MATHEMATICS

FUNCTION

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FUNCTION

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IMPORTANT FUNCTIONS (MODULUS, G.I.F., F.P.F. ETC.)

1. Let $f(x) = |x - 1|$, then :

(A) $f(x^2) = (f(x))^2$ (B) $f(x+y) = f(x) + f(y)$ (C) $f(|x|) = |f(x)|$ (D) None of these

यदि $f(x) = |x - 1|$, तो :

(A) $f(x^2) = (f(x))^2$ (B) $f(x+y) = f(x) + f(y)$ (C) $f(|x|) = |f(x)|$ (D) इनमें से कोई नहीं

Ans. D

Sol. $f(x) = |x - 1|$

$$\Rightarrow f(|x|) = ||x| - 1| = \begin{cases} |x| - 1 & \text{if } |x| \geq 1 \\ 1 - |x| & |x| < 1 \end{cases}$$

$$\text{but } |f(x)| = \|x - 1\| = \begin{cases} x - 1 & \text{if } x \geq 1 \\ 1 - x & x < 1 \end{cases}$$

$$\Rightarrow f(|x|) \neq |f(x)|$$

$f(x + y) = |x + y - 1| \neq |x - 1| + |y - 1|$

2. If $f(x) = \cos\left[\frac{\pi^2}{2}\right]x + \sin\left[-\frac{\pi^2}{2}\right]x$, where [x] denotes the greatest integer function, then which of the following is not correct :

यदि $f(x) = \cos\left[\frac{\pi^2}{2}\right]x + \sin\left[-\frac{\pi^2}{2}\right]x$, जहाँ [x] महत्तम पूर्णांक फलन को प्रदर्शित करता है, तो निम्नलिखित में से कौन–सा सही नहीं है :

(A) $f(0) = 1$ (B) $f\left(\frac{\pi}{3}\right) = \frac{1}{\sqrt{3}+1}$ (C) $f\left(\frac{\pi}{2}\right) = 0$ (D) $f(\pi) = 0$

Ans. D

Sol. $f(x) = \cos\left[\frac{\pi^2}{2}\right]x + \sin\left[-\frac{\pi^2}{2}\right]x$

$\Rightarrow f(x) = \cos 4x + \sin(-5x)$

$\Rightarrow f(0) = 1$

$$f\left(\frac{\pi}{3}\right)=\cos\frac{4\pi}{3}-\sin\frac{5\pi}{3}=-\cos\frac{\pi}{3}+\sin\frac{\pi}{3}$$

$$=\frac{\sqrt{3}-1}{2}=\frac{1}{\sqrt{3}+1}$$

$$f\left(\frac{\pi}{2}\right)=\cos 2\pi-\sin\left(\frac{5\pi}{2}\right)=0$$

$f(\pi)=\cos 4\pi-\sin(5\pi)=1$

DOMAIN OF FUNCTIONS

3. Let f(x) be a function whose domain is [– 5, 7]. Let g(x) = |2x + 5|. Then domain of (fog) (x) is :
(A) [– 4, 1] (B) [– 5, 1] (C) [– 6, 1] (D) None of these

माना f(x) एक फलन है जिसका प्रान्त [– 5, 7] है। यदि g(x) = |2x + 5| हो, तो (fog) (x) का प्रान्त है :
(A) [– 4, 1] (B) [– 5, 1] (C) [– 6, 1] (D) इनमें से कोई नहीं

Ans. C

Sol. Domain of f(g(x)). Range of g(x) ≡ Domain of f(x)
$\Rightarrow \quad -5\le|2x+5|\le 7 \quad \Rightarrow \quad 0\le|2x+5|\le 7 \quad \Rightarrow \quad -7\le 2x+5\le 7$
$\Rightarrow \quad -12\le 2x\le 2 \quad \Rightarrow \quad -6\le x\le 1$
Hindi f(g(x)) के प्रान्त के लिए . g(x) परिसर ≡ f(x) प्रान्त
$\Rightarrow \quad -5\le|2x+5|\le 7 \quad \Rightarrow \quad 0\le|2x+5|\le 7 \quad \Rightarrow \quad -7\le 2x+5\le 7$
$\Rightarrow \quad -12\le 2x\le 2 \quad \Rightarrow \quad -6\le x\le 1$

4. The domain of the function $f(x)=\log_{1/2}\left(-\log_2\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right)$ is :

(A) $0<x<1$ (B) $0<x\le 1$ (C) $x\ge 1$ (D) null set

फलन $f(x)=\log_{1/2}\left(-\log_2\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right)$ का प्रान्त है :

(A) $0<x<1$ (B) $0<x\le 1$ (C) $x\ge 1$ (D) रिक्त समुच्चय

Ans. D

Sol. $f(x)=\log_{1/2}\left(-\log_2\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right) \Rightarrow -\log_2\left(1+\frac{1}{x^{1/4}}\right)-1>0$

$\Rightarrow \quad -\infty < \log_2\left(1+\frac{1}{x^{1/4}}\right) < -1 \Rightarrow 0 < 1 + \frac{1}{x^{1/4}} < \frac{1}{2} \quad \Rightarrow \quad -1 < \frac{1}{x^{1/4}} < -\frac{1}{2}$

$\Rightarrow \quad x \in \phi$ (null set) $\quad \Rightarrow \quad x \in \phi$ (रिक्त समुच्चय)

5. The domain of the function, $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is :

फलन $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ का प्रान्त है :

(A) [1 , 2] (B) [2 , 3) (C) [2 , 3] (D) [1 , 2)

Ans. B

Sol. To define $f(x)$, $9 - x^2 > 0 \Rightarrow -3 < x < 3$ ... (1)

$-1 \le (x-3) \le 1 \Rightarrow 2 \le x \le 4$ ... (2)

From (i) and (ii), $2 \le x < 3$ i.e., [2, 3).

6. The domain of the function $f(x) = \frac{1}{\sqrt{|x|-x}}$ is : **[AIEEE-2011]**

फलन $f(x) = \frac{1}{\sqrt{|x|-x}}$ का प्रान्त है : **[AIEEE-2011]**

(A) $(-\infty, \infty)$ (B) $(0, \infty)$ (C) $(-\infty, 0)$ (D) $(-\infty, \infty) - \{0\}$

Ans. C

Sol. $f(x) = \frac{1}{\sqrt{|x|-x}}$

$\Rightarrow |x| - x > 0$

$\Rightarrow |x| > x$

$\Rightarrow x \in (-\infty, 0)$

RANGE OF FUNCTIONS

7. If $[2\cos x]+[\sin x]=-3$, then the range of the function, $f(x)=\sin x+\sqrt{3}\cos x$ in $[0, 2\pi]$ is : (where [ ] denotes greatest integer function)

यदि $[2\cos x]+[\sin x]=-3$, हो तो फलन $f(x)=\sin x+\sqrt{3}\cos x$ का $[0, 2\pi]$ में परिसर है : (जहाँ [ ] महत्तम पूर्णांक फलन को प्रदर्शित करता है।)

(A) $[-2, -1)$ (B) $(-2, -1]$ (C) $(-2, -1)$ (D) $[-2, -\sqrt{3})$

Ans. D

Sol. $[2\cos x]+[\sin x]=-3$

$-2 \le 2\cos x \le 2 \quad \Rightarrow \quad [2\cos x]=2, 1, -1, -2$

$-1 \le \sin x \le 1 \quad \Rightarrow \quad [\sin x]=1, 0, -1$

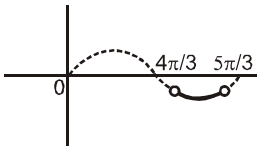
Equation holds true for $[2\cos x]=-2$ and $[\sin x]=-1$

$\Rightarrow \quad -1 \le \cos x < -\frac{1}{2}$ and $-1 \le \sin x < 0$

$\Rightarrow \quad x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$ and $x \in (\pi, 2\pi) \quad \Rightarrow \quad x \in \left(\pi, \frac{4\pi}{3}\right)$

$f(x)=\sin x+\sqrt{3}\cos x$

$f(x)=2\sin\left(x+\frac{\pi}{3}\right)$



$\frac{4\pi}{3} < x+\frac{\pi}{3} < \frac{5\pi}{3} \quad \Rightarrow \quad -1 \le \sin\left(x+\frac{\pi}{3}\right) < -\frac{\sqrt{3}}{2} \quad \Rightarrow \quad -2 \le 2\sin\left(x+\frac{\pi}{3}\right) < -\sqrt{3}$

Hence range is $\left[-2, -\sqrt{3}\right]$

Hindi $[2\cos x]+[\sin x]=-3$

$-2 \le 2\cos x \le 2 \quad \Rightarrow \quad [2\cos x]=2, 1, -1, -2$

$-1 \le \sin x \le 1 \quad \Rightarrow \quad [\sin x]=1, 0, -1$

समीकरण सत्य होगी यदि $[2\cos x]=-2$ और $[\sin x]=-1$

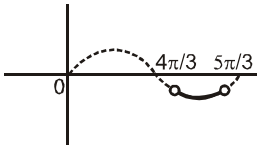
$\Rightarrow \quad -1 \le \cos x < -\frac{1}{2}$ और $-1 \le \sin x < 0$

$\Rightarrow \quad x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$ और $x \in (\pi, 2\pi)$

$\Rightarrow \quad x \in \left(\pi, \frac{4\pi}{3}\right)$

$f(x)=\sin x+\sqrt{3}\cos x$

$f(x)=2\sin\left(x+\frac{\pi}{3}\right)$



$\frac{4\pi}{3} < x + \frac{\pi}{3} < \frac{5\pi}{3} \quad \Rightarrow \quad -1 \le \sin\left(x+\frac{\pi}{3}\right) < -\frac{\sqrt{3}}{2} \quad \Rightarrow \quad -2 \le 2\sin\left(x+\frac{\pi}{3}\right) < -\sqrt{3}$

अतः परिसर है $y \in \left[-2, -\sqrt{3}\right]$.

8. Range of $f(x) = \log(3x^2 - 4x + 5)$ is :

फलन $f(x) = \log(3x^2 - 4x + 5)$ का परिसर है :

(A) $\left[\log\frac{11}{3}, \infty\right)$ (B) $[\log 10, \infty)$ (C) $\left[\log\frac{11}{6}, \infty\right)$ (D) $\left[\log\frac{11}{12}, \infty\right)$

Ans. A

Sol. $f(x) = \log_e(3x^2 - 4x + 5)$

$3x^2 - 4x + 5 \ge \frac{11}{3} \quad \Rightarrow \quad \ell n(3x^2 - 4x + 5) \ge \ell n\frac{11}{3}$ [$\because$ ℓn is an increasing function]

$\therefore$ Range is $\left[\ell n\frac{11}{3}, \infty\right)$

Hindi $f(x) = \log_e(3x^2 - 4x + 5)$

$3x^2 - 4x + 5 \ge \frac{11}{3} \quad \Rightarrow \quad \ell n(3x^2 - 4x + 5) \ge \ell n\frac{11}{3}$ [$\because$ ℓn एक निरन्तर वर्द्धमान फलन है]

$\therefore$ परिसर $\left[\ell n\frac{11}{3}, \infty\right)$

9. Range of $f(x) = 4^x + 2^x + 1$ is :

फलन $f(x) = 4^x + 2^x + 1$ का परिसर है :

(A) $(0, \infty)$ (B) $(1, \infty)$ (C) $(2, \infty)$ (D) $(3, \infty)$

Ans. B

Sol. $f(x) = 4^x + 2^x + 1$

Let $2^x = t > 0, \forall\, x \in R$

$\therefore$ $f(x) = g(t) = t^2 + t + 1, \ t > 0$

$g(t) = \left(t+\frac{1}{2}\right)^2 + \frac{3}{4} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right) > \frac{1}{2} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right)^2 > \frac{1}{4} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right)^2 + \frac{3}{4} > 1$

Range is $(1, \infty)$

Hindi $f(x) = 4^x + 2^x + 1$

माना $2^x = t > 0, \forall\, x \in R$

$\therefore$ $f(x) = g(t) = t^2 + t + 1, \ t > 0$

$g(t)=\left(t+\frac{1}{2}\right)^2+\frac{3}{4} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right)>\frac{1}{2} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right)^2>\frac{1}{4} \quad \Rightarrow \quad \left(t+\frac{1}{2}\right)^2+\frac{3}{4}>1$

परिसर = $(1, \infty)$

10. The range of the functions $f(x)=\log_{\sqrt{2}}\left(2-\log_2\left(16\sin^2 x+1\right)\right)$ is :

फलन $f(x)=\log_{\sqrt{2}}\left(2-\log_2\left(16\sin^2 x+1\right)\right)$ का परिसर है :

(A) $(-\infty, 1)$ (B) $(-\infty, 2)$ (C) $(-\infty, 1]$ (D) $(-\infty, 2]$

Ans. D

Sol. Here $(2-\log_2(16\sin^2x+1)>0 \quad \Rightarrow \quad 0<16\sin^2x+1<4$

$\Rightarrow \quad \frac{-1}{16}\le\sin^2x<\frac{3}{16} \quad \Rightarrow \quad 1\le 16\sin^2x+1\le 4 \quad \Rightarrow \quad 0\le\log_2(16\sin^2x+1)<2$

$\Rightarrow \quad 2\ge 2-\log_2(16\sin^2x+1)>0$

$\Rightarrow \quad \log_{\sqrt{2}}2\ge\log_{\sqrt{2}}(2-\log_2(16\sin^2x+1))>-\infty \Rightarrow \quad 2\ge y>-\infty$

Hence range is $y\in(-\infty, 2]$

Hindi यहाँ $(2-\log_2(16\sin^2x+1)>0 \quad \Rightarrow \quad 0<16\sin^2x+1<4$

$\Rightarrow \quad \frac{-1}{16}\le\sin^2x<\frac{3}{16} \quad \Rightarrow \quad 1\le 16\sin^2x+1\le 4$

$\Rightarrow \quad 0\le\log_2(16\sin^2x+1)<2$

$\Rightarrow \quad 2\ge 2-\log_2(16\sin^2x+1)>0$

$\Rightarrow \quad \log_{\sqrt{2}}2\ge\log_{\sqrt{2}}(2-\log_2(16\sin^2x+1))>-\infty$

$\Rightarrow \quad 2\ge y>-\infty$

अतः परिसर है $y\in(-\infty, 2]$

11. The range of the function, $f(x)=\frac{2+x}{2-x}$, $x\neq 2$ is :

फलन $f(x)=\frac{2+x}{2-x}$, $x\neq 2$ का परिसर होगा :

(A) R (B) $R-\{-1\}$ (C) $R-\{1\}$ (D) $R-\{2\}$

Ans. B

Sol. $f(x)=\frac{x+2}{2-x}; x\neq 2$

Range = R – {–1}

12. The range of the function, $f(x) = {}^{7-x}P_{x-3}$ is :

फलन $f(x) = {}^{7-x}P_{x-3}$ का परिसर है :

(A) {1, 2, 3, 4} (B) {1, 2, 3, 4, 5, 6} (C) {1, 2, 3, 4, 5} (D) {1, 2, 3}

Ans. D

Sol. $f(3) = {}^{7-3}P_0 = 1$, $f(4) = {}^3P_1 = 3$ and $f(5) = {}^2P_2 = 2$

Hence, range of $f = \{1, 2, 3\}$.

TRANSFORMATION OF GRAPHS

13. The graph of the function, $y = f(x)$ is symmetrical about the line $x = 2$, then :

यदि $y = f(x)$ का ग्राफ रेखा $x = 2$ के सापेक्ष सममित हो, तो :

(A) $f(x) = f(-x)$ (B) $f(2 + x) = f(2 - x)$ (C) $f(x + 2) = f(x - 2)$ (D) $f(x) = -f(-x)$

Ans. B

Sol. $y = f(x)$ is symmetrical about the line $x = 2$

$\Rightarrow f(2 + x) = f(2 - x)$

CLASSIFICATION OF FUNCTIONS (ONE-ONE, ONTO ETC.)

14. If $f(x) = \frac{4a-7}{3} x^3 + (a-3)x^2 + x + 5$ is a one-one function, then :

(A) $2 \le a \le 8$ (B) $1 \le a \le 2$ (C) $0 \le a \le 1$ (D) None of these

यदि $f(x) = \frac{4a-7}{3} x^3 + (a-3)x^2 + x + 5$ एकैकी फलन हो, तो :

(A) $2 \le a \le 8$ (B) $1 \le a \le 2$ (C) $0 \le a \le 1$ (D) इनमें से कोई नहीं

Ans. A

Sol. $f(x) = \frac{4a-7}{3} x^3 + (a-3)x^2 + x + 5$

$\frac{df}{dx} = \frac{4a-7}{3} x^2 + 2(a-3)x + 1$

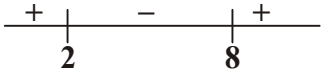
$4a - 7 > 0$ $D = 4(a-3)^2 - 4(4a-7)$

$4a > 7$ $= 4(a^2 - 6a + 9 - 4a + 7)$

$\Rightarrow a > \frac{7}{4}$

$= 4(a^2 - 10a + 16)$

$= 4(a-8)(a-2)$



$\Rightarrow a \in [2,8]$

15. A function 'f' from the set of natural numbers to integers defined by, $f(x) = \begin{cases} \frac{n-1}{2} & , \text{ when 'n' is odd} \\ -\frac{n}{2} & , \text{ when 'n' is even} \end{cases}$ is :

(A) onto but not one-one
(B) one-one and onto both
(C) one-one but not onto
(D) neither one-one nor onto

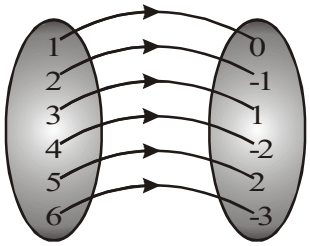
एक फलन 'f' प्राकृत संख्याओं के समुच्चय से पूर्णांको के समुच्चय, $f(x) = \begin{cases} \frac{n-1}{2} & , \text{ t gka 'n' fo"ke gS} \\ -\frac{n}{2} & , \text{ t gka 'n' l e gS} \end{cases}$ में इस प्रकार है कि :

(A) आच्छादक लेकिन एकैकी नहीं है
(B) एकैकी एवं आच्छादक दोनों है
(C) एकैकी लेकिन आच्छादक नहीं है
(D) न एकैकी न आच्छादक है

Ans. B

Sol. $f : N \to I$

$f(1) = 0$, $f(2) = -1$, $f(3) = 1$, $f(4) = -2$, $f(5) = 2$ and $f(6) = -3$ so on.



In this type of function every element of set A has unique image in set B and there is no element left in set B. Hence f is one-one and onto function.

16. If $f : R \to S$, defined by $f(x) = \sin x - \sqrt{3}\cos x + 1$, is onto , then the interval of 'S' is :

यदि $f : R \to S$ में परिभाषित फलन $f(x) = \sin x - \sqrt{3}\cos x + 1$ आच्छादक हो, तो 'S' का अन्तराल है :

(A) $[0, 1]$ (B) $[-1, 1]$ (C) $[0, 3]$ (D) $[-1, 3]$

Ans. D

Sol. Using $-\sqrt{a^2+b^2} \le (a\sin x + b\cos x) \le \sqrt{a^2+b^2}$

$-\sqrt{1+(-\sqrt{3})^2} \le (\sin x - \sqrt{3}\cos x) \le \sqrt{1+(-\sqrt{3})^2}$

$-2 \le (\sin x - \sqrt{3}\cos x) \le 2$

$-2+1 \le (\sin x - \sqrt{3}\cos x + 1) \le 2+1$

$-1 \le (\sin x - \sqrt{3}\cos x + 1) \le 3$ i.e., range = $[-1,3]$

$\therefore$ For f to be onto S = $[-1, 3]$.

17. For real x, let $f(x) = x^3 + 5x + 1$, then : **[AIEEE-2009]**

(A) f is one – one but not onto R (B) f is onto R but not one – one

(C) f is one – one and onto R (D) f is neither one – one nor onto R

वास्तविक x के लिए $f(x) = x^3 + 5x + 1$ तो : **[AIEEE-2009]**

(A) f समुच्चय R में एकेकी लेकिन आच्छादक नहीं (B) f समुच्चय R में आच्छादक लेकिन एकेकी नहीं

(C) f समुच्चय R में एकेकी तथा आच्छादक (D) f समुच्चय R में ना तो एकेकी ना ही आच्छादक

Ans. C

Sol. For real x, $f(x) = x^3 + 5x + 1$

$\lim\limits_{x\to\infty} f(x) = +\infty$

and $\lim\limits_{x\to\infty} f(x) = -\infty$

$\therefore$ Range is R – f(x) is on to

Now $f'(x) = 3x^2 + 5 > 0$

$\therefore$ f(x) is one-one

f(x) is one-one onto.

18. The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function, $\left[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)\right]$ is defined as :

$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में उपस्थित अधिकतम अन्तराल जिसके लिए फलन $\left[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)\right]$ परिभाषित है :

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) $\left[0 \ \frac{\pi}{2}\right)$ (D) $[0, \pi]$

Ans. C

Sol. $f(x) = 4^{-x} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log \cos x$

$\cos x > 0$ & $\frac{x}{2} - 1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

In $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ & $x \in [2 - \pi, 2 + \pi]$

$x \in \left[0, \frac{\pi}{2}\right]$

$\Rightarrow x \in \left[0, \frac{\pi}{2}\right)$

FUNCTIONAL EQUATION

19. If $f(x) = \cos(\ell n\, x)$ then, $f(x)\, f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value :

(A) –1 (B) $\frac{1}{2}$ (C) –2 (D) None of these

यदि $f(x) = \cos(\ell n\, x)$ हो, तो $f(x)\, f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ का मान है :

(A) –1 (B) $\frac{1}{2}$ (C) –2 (D) इनमें से कोई नहीं

Ans. D

Sol. $f(x) = \cos(\ln x)$

$f(x) \cdot f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$

$= \cos(\ln x)\cos(\ln y) - \frac{1}{2}\left[\cos(\ln x - \ln y) + \cos(\ln x + \ln y)\right]$

$= 0$

20. If $y = f(x)$ satisfies the condition $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$ $(x \neq 0)$ then $f(x) =$

यदि $y = f(x)$ प्रतिबन्ध $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$ $(x \neq 0)$ को सन्तुष्ट करता है, तो $f(x) =$

(A) $-x^2 + 2$ (B) $-x^2 - 2$ (C) $x^2 - 2, x \in R - \{0\}$ (D) $x^2 - 2, |x| \in [2, \infty)$

Ans. D

Sol. $f\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^2 - 2$ $(x \neq 0)$

$x + \frac{1}{x} \leftrightarrow x$

$f(y) = x^2 - 2; x \in (-\infty, -2] \cup [2, \infty)$

21. If $f(1) = 1$ and $f(n + 1) = 2f(n) + 1$ if $n \geq 1$, then $f(n)$ is equal to :

यदि $f(1) = 1$ एवं $f(n + 1) = 2f(n) + 1$ यदि $n \geq 1$, तो $f(n) =$

(A) $2^n + 1$ (B) 2^n (C) $2^n - 1$ (D) $2^{n-1} - 1$

Ans. C

Sol. $f(1) = 1$

$f(n + 1) = 2f(n) + 1$

$\Rightarrow f(2) = 2f(1) + 1 = 3 = 2^2 - 1$

$f(3) = 2f(2) + 1 = 7 = 2^3 - 1$

$\vdots$

$\Rightarrow f(n) = 2^n - 1$

22. If $f: R \to R$ satisfies $f(x + y) = f(x) + f(y)$, for all $x, y \in R$ and $f(1) = 7$, then $\sum_{r=1}^{n} f(r)$ is :

यदि $f: R \to R$, $f(x + y) = f(x) + f(y)$, $\forall\, x, y \in R$ को संतुष्ट करता है एवं $f(1) = 7$ हो, तो $\sum_{r=1}^{n} f(r)$ है :

(A) $\frac{7(n + 1)}{2}$ (B) $7n(n + 1)$ (C) $\frac{7n(n + 1)}{2}$ (D) $\frac{7n}{2}$

Ans. C

Sol. $f(x+y) = f(x) + f(y) \quad \forall x, y \in R$

$\Rightarrow f(x) = kx$

Given $f(1) = 7$

$\Rightarrow k = 7$

$$\Rightarrow \sum_{r=1}^{n} f(r) = \sum_{r=1}^{n} 7r = \frac{7 \cdot n(n+1)}{2}$$

23. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$ and $S = \{x \in R : f(x) = f(-x)\}$; then S : **[JEE(Main)-2016]**

(A) contains exactly two elements
(B) contains more than two elements
(C) is an empty set
(D) contains exactly one element

यदि $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$ है, तथा $S = \{x \in R : f(x) = f(-x)\}$ है; तो S : **[JEE(Main)-2016]**

(A) में तथ्यतः दो अवयव है
(B) में दो से अधिक अवयव हैं
(C) एक रिक्त समुच्चय है
(D) में केवल एक अवयव है

Ans. A

Sol. $f(x) + 2f(1/x) = 3x \quad (1) \qquad (x \neq 0)$

$f(1/x) + 2f(x) = \frac{3}{x} \quad (2)$

From (1) & (2)

$f(x) = \frac{2 - x^2}{x}$

$f(x) = f(-x)$

$x^2 = 2$

$x = \pm \sqrt{2}$

S contains exactly two elements

24. Let $a, b, c \in R$. If $f(x) = ax^2 + bx + c$ is such that $a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy$, $\forall x, y \in R$, then $\sum_{n=1}^{10} f(n)$ is equal to : **[JEE(Main)-2017]**

माना a, b, c ∈ R । यदि $f(x) = ax^2 + bx + c$ ऐसा है कि a + b + c = 3 है तथा सभी x, y ∈ R के लिए $f(x + y) = f(x) + f(y) + xy$ है, तो $\sum_{n=1}^{10} f(n)$ बराबर है : **[JEE(Main)-2017]**

(A) 190 (B) 255 (C) 330 (D) 165

Ans. C

Sol. $f(1) = 3$

Put y = 1 $f(x + 1) - f(x) = 3 + x$

$(a(x + 1)^2 + b(x + 1) + c) - (ax^2 + bx + c) = 3 + x$

Comparing coefficients

$$a = \frac{1}{2} \qquad b = \frac{5}{2} \qquad c = 0$$

$$f(n) = \frac{n^2 + 5n}{2}$$

$$\sum_{n=1}^{10} \frac{n^2 + 5n}{2} = 330$$

Alternate Solution :

f(1) = 3

f(2) = f(1) + f(1) + 1 ⇒ 3 + 3 + 1 = 7

f(3) = f(2) + f(1) + 2 ⇒ 7 + 3 + 2 = 12

f(4) = 18

f(5) = 25

f(6) = 33

f(7) = 42

f(8) = 52

f(9) = 63

f(10) = 75

---

add = 330

COMPOSITE FUNCTIONS

25. If $f(x) = \frac{ax + b}{cx + d}$, then (fof) (x) = x, provided that :

यदि $f(x) = \frac{ax + b}{cx + d}$ हो, तो (fof) (x) = x, यह इस शर्त पर है कि :

(A) d + a = 0 (B) d – a = 0 (C) a = b = c = d = 1 (D) a = b = 1

Ans. A

Sol. $f(x)=\dfrac{ax+b}{cx+d} \Rightarrow fof(x)=\dfrac{a\left(\dfrac{ax+b}{cx+d}\right)+b}{c\left(\dfrac{ax+b}{cx+d}\right)+d} \Rightarrow fof(x)=\dfrac{a^2x+ab+bcx+bd}{acx+bc+cdx+d^2}$

$fof(x)=\dfrac{(a^2+bc)x+(ab+bd)}{(ac+cd)x+(bc+d^2)}=x$

on comparing coefficient of both side $(a^2 + bc)\, x + (ab + bd) = (ac + cd)\, x^2 + (bc + d^2)\, x$

$a^2 + bc = bc + d^2 \quad \Rightarrow \quad a = d$ or $a = -d$

and $ab + bd = 0 \quad \Rightarrow \quad b = 0$ or $a = -d$

and $ac + cd = 0 \quad \Rightarrow \quad c = 0$ or $a = -d$

which can be simultaneously true for $a = -d$

Hindi $f(x)=\dfrac{ax+b}{cx+d} \Rightarrow fof(x)=\dfrac{a\left(\dfrac{ax+b}{cx+d}\right)+b}{c\left(\dfrac{ax+b}{cx+d}\right)+d} \Rightarrow fof(x)=\dfrac{a^2x+ab+bcx+bd}{acx+bc+cdx+d^2}$

$fof(x)=\dfrac{(a^2+bc)x+(ab+bd)}{(ac+cd)x+(bc+d^2)}=x \quad \Rightarrow \quad (a^2 + bc)\, x + (ab + bd) = (ac + cd)\, x^2 + (bc + d^2)\, x$

समीकरण के दोनों पक्षों में गुणांकों की तुलना करने पर

$a^2 + bc = bc + d^2 \quad \Rightarrow \quad a = d$ या $a = -d$

तथा $ab + bd = 0 \quad \Rightarrow \quad b = 0$ या $a = -d$

तथा $ac + cd = 0 \quad \Rightarrow \quad c = 0$ या $a = -d$

ये सभी एक साथ सत्य होगी यदि $a = -d$

26. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is: **[JEE-2011]**

(A) $\pm\sqrt{n\pi}, n \in \{0,1,2,....\}$ (B) $\pm\sqrt{n\pi}, n \in \{1,2,....\}$

(C) $\dfrac{\pi}{2}+2n\pi, n \in \{....-2,-1,0,1,2,....\}$ (D) $2n\pi, n \in \{...,-2,-1,0,1,2,...\}$

माना $f(x) = x^2$ तथा $g(x) = \sin x$ सभी $x \in R$ के लिए तो x के मान ज्ञात कीजिए जो $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$ को संतुष्ट करें, जहाँ $(f \circ g)(x) = f(g(x))$: **[JEE-2011]**

(A) $\pm\sqrt{n\pi}, n \in \{0,1,2,....\}$ (B) $\pm\sqrt{n\pi}, n \in \{1,2,....\}$

(C) $\dfrac{\pi}{2}+2n\pi, n \in \{....-2,-1,0,1,2,....\}$ (D) $2n\pi, n \in \{...,-2,-1,0,1,2,...\}$

Ans. A

Sol. Given $f(x) = x^2$; $g(x) = \sin x$

$fogogof(x) = \sin^2(\sin x^2)$

and $gogof(x) = \sin(\sin x^2)$

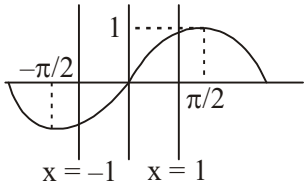
given $fogogof(x) = gogof(x)$

$\Rightarrow \sin^2(\sin x^2) = \sin(\sin x^2)$

$\Rightarrow \sin(\sin x^2) = 0$ or 1 (rejected)

$\sin(\sin x^2) = 0 \quad \Rightarrow \quad x^2 = n\pi$

$\Rightarrow \quad x = \pm\sqrt{n\pi};\ x \in \{0, 1, 2, 3, ...\}$



EVEN AND ODD FUNCTIONS

27. The function $f(x) = \log\left(\frac{1+\sin x}{1-\sin x}\right)$ is :

(A) Even (B) Odd

(C) Neither even nor odd (D) Both even and odd

फलन $f(x) = \log\left(\frac{1+\sin x}{1-\sin x}\right)$ है :

(A) सम फलन (B) विषम फलन

(C) न तो सम न ही विषम (D) सम एवं विषम दोनों

Ans. B

Sol. $f(x) = \log\left(\frac{1+\sin x}{1-\sin x}\right) \quad \Rightarrow \quad f(-x) = \log\left(\frac{1-\sin x}{1+\sin x}\right) = -\log\left(\frac{1+\sin x}{1-\sin x}\right) = -f(x)$ odd function विषम फलन

28. The function $f(x) = [x] + \frac{1}{2}$, $x \notin I$ is a/an :

(A) Even (B) Odd

(C) Neither even nor odd (D) None of these

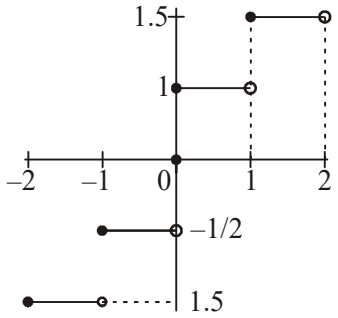
फलन $f(x) = [x] + \frac{1}{2}$, $x \notin I$ है :

(A) सम (B) विषम

(C) न सम न विषम (D) इनमें से कोई नहीं

Ans. B

Sol. $f(x) = [x] + \frac{1}{2}, x \notin I \Rightarrow f(-x) = [-x] + \frac{1}{2} = -[x] - 1 + \frac{1}{2} = -\left([x] + \frac{1}{2}\right) = -f(x)$ odd function



INVERSE OF A FUNCTION

29. The inverse of the function $f(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is :

फलन $f(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ का प्रतिलोम है :

(A) $\frac{1}{2}\log\frac{1+x}{1-x}$ (B) $\frac{1}{2}\log\frac{2+x}{2-x}$ (C) $\frac{1}{2}\log\frac{1-x}{1+x}$ (D) $2\log(1+x)$

Ans. A

Sol. $\frac{y}{1} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

By compnendo and dividendo योगान्तरानुपात से

$\frac{1+y}{1-y} = \frac{2e^x}{2e^{-x}} \Rightarrow 2x = \ell n\left(\frac{1+y}{1-y}\right) \quad \therefore \quad f^{-1}(x) = \frac{1}{2}\ell n\left(\frac{1+x}{1-x}\right)$

PERIODIC FUNCTIONS

30. Fundamental period of $f(x) = \sec(\sin x)$ is :

(A) $\frac{\pi}{2}$ (B) 2π (C) π (D) aperiodic

$f(x) = \sec(\sin x)$ का मूलभूत आवर्तकाल है :

(A) $\frac{\pi}{2}$ (B) 2π (C) π (D) अनावर्ती

Ans. C

Sol. f(x) = sec (sin x). Since sin x is a periodic function with fundamental period 2π. f(x) has a period 2π for fundamental period $f(x+\pi) = \sec(\sin(\pi+x)) = \sec(-\sin x) = \sec(\sin x) = f(x)$

$f\left(x+\frac{\pi}{2}\right) \neq f(x)$ hence fundamental period is π

Hindi. f(x) = sec (sin x) ∵ sin x, मूलभुत आवर्तकाल 2π का आवर्ती फलन है, इसलिए f(x) आवर्तकाल 2π है
अब मूलभुत आवर्तकाल के लिए $f(x+\pi) = \sec(\sin(\pi+x)) = \sec(-\sin x) = \sec(\sin x) = f(x)$

$f\left(x+\frac{\pi}{2}\right) \neq f(x)$ अतः मूलभुत आवर्तकाल π है।

31. The fundamental period of function $f(x) = [x] + \left[x+\frac{1}{3}\right] + \left[x+\frac{2}{3}\right] - 3x + 15$:

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 1 (D) Non-periodic

फलन $f(x) = [x] + \left[x+\frac{1}{3}\right] + \left[x+\frac{2}{3}\right] - 3x + 15$ का आवर्त काल होगा :

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 1 (D) आवर्ती फलन नहीं है

Ans. A

Sol. $f(x) = [x] + \left[x+\frac{1}{3}\right] + \left[x+\frac{2}{3}\right] - 3x + 15$

$= \left\{[x] + \left[x+\frac{1}{n}\right] + \left[x+\frac{2}{n}\right] + ... + \left[x+\frac{n-1}{n}\right] = [nx]\right\}$ $n \in N, x \in R$

$\Rightarrow f(x) = [x] + \left[x+\frac{1}{3}\right] + \left[x+\frac{2}{3}\right] = [3x]$

$\Rightarrow f(x) = [3x] - 3x + 15$

$= -(3x - [3x]) + 15$

$= -\{3x\} + 15$

Periodic function with period $= \frac{1}{3}$

Exercise J-2

IMPORTANT FUNCTIONS (MODULUS, SIGNUM, G.I.F., F.P.F. ETC.)

1. Read the symbols [ ] and { } as greatest integer function less than or equal to x and fractional part function :

(i) Find the number of real values of x, satisfying the equation $(x-2)[x] = \{x\} - 1$.

(ii) Find the number of solutions of the equation, $x^2 - 3x + [x] = 0$ in the interval $[0, 3]$.

(iii) If $[x]^2 + 3[x] - 10 \geq 0$, then find the range of x.

(iv) If $[x]^2 - 5[x] + 6 = 0$, then find the range of x.

[ ] व {} महत्तम पूर्णांक फलन व भिन्नात्मक फलन है :

(i) समीकरण $(x-2)[x] = \{x\} - 1$ को सन्तुष्ट करने वाले x के वास्तविक मानों की संख्या ज्ञात कीजिए।

(ii) अन्तराल $[0, 3]$ में समीकरण $x^2 - 3x + [x] = 0$ के हलों की संख्या होगी।

(iii) यदि $[x]^2 + 3[x] - 10 \geq 0$ हो, तो x का परिसर ज्ञात कीजिए।

(iv) यदि $[x]^2 - 5[x] + 6 = 0$ हो, तो x का परिसर ज्ञात कीजिए।

Ans. (i) infinite (ii) 2 (iii) $x \in (-\infty, -4) \cup [2, \infty)$ (iv) $x \in [2, 4)$

Sol. (i) $(x-2)[x] = \{x\} - 1$

$\because \{x\} = x - [x]$

$\Rightarrow (x-2)[x] = x - [x] - 1$

$\Rightarrow x[x] - 2[x] = x - [x] - 1$

$\Rightarrow [x](x-1) = x - 1$

$\Rightarrow (x-1)([x]-1) = 0$

$x = 1$ or $[x] = 1$

$x = 1$ or $0 \leq x < 1$

$\Rightarrow 0 \leq x \leq 1$

Number of x satisfying given equation is infinite.

(ii) $x^2 - 3x + [x] = 0; \quad x \in [0,3]$

If $0 \leq x < 1$ then $x^2 - 3x = 0 \Rightarrow x = 0$ or $x = 3$

$\Rightarrow x = 0$

If $1 \leq x < 2$ then $x^2 - 3x + 1 = 0 \Rightarrow x = \dfrac{3 \pm \sqrt{5}}{2}$

$\Rightarrow x \in \phi$

If $2 \leq x < 3$ then $x^2 - 3x + 2 = 0 \Rightarrow x = 2$ or 1

$\Rightarrow x = 2$

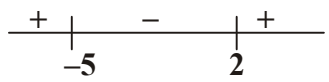
$\Rightarrow$ No. of values of x is 2.

(iii) $[x]^2 + 3[x] - 10 \geq 0$

let $[x] = t$

$t^2 + 3t - 10 \geq 0$

$\Rightarrow (t + 5)(t - 2) \geq 0$

$\Rightarrow$ 

$\Rightarrow t \in (-\infty, -5] \cup [2, \infty)$

$\Rightarrow [x] \leq -5$ or $[x] \geq 2$

$\Rightarrow x < -4$ or $x \geq 2$

$\Rightarrow x \in (-\infty, -4) \cup [2, \infty)$

(iv) $[x]^2 - 5[x] + 6 = 0$

let $[x] = t$

$\Rightarrow t^2 - 5t + 6 = 0$

$\Rightarrow (t - 2)(t - 3) = 0$

$\Rightarrow [x] = 2$ or $[x] = 3$

$\Rightarrow 2 \leq x < 3$ or $3 \leq x < 4$

$\Rightarrow x \in [2, 4)$

DOMAIN AND RANGE OF FUNCTIONS

2. Find the domains of definitions of the following functions : (Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

निम्न फलनों का प्रान्त ज्ञात कीजिये ।([] व {} महत्तम पूर्णांक फलन व भिन्नात्मक फलन है ।)

(i) $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$

(ii) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

(iii) $f(x) = \ln\left(\sqrt{x^2 - 5x - 24} - x - 2\right)$

(iv) $f(x) = \sqrt{\dfrac{1 - 5^x}{7^{-x} - 7}}$

(v) $y = \log_{10} \sin(x - 3) + \sqrt{16 - x^2}$

(vi) $f(x) = \log_{100x}\left(\dfrac{2\log_{10} x + 1}{-x}\right)$

(vii) $f(x) = \dfrac{1}{\sqrt{4x^2 - 1}} + \ln x(x^2 - 1)$

(viii) $f(x) = \sqrt{\log_{\frac{1}{2}} \dfrac{x}{x^2 - 1}}$

(ix) $f(x)=\sqrt{x^2-|x|}+\dfrac{1}{\sqrt{9-x^2}}$ (x) $f(x)=\sqrt{(x^2-3x-10).ln^2(x-3)}$

(xi) $f(x)=\sqrt{\log_x(\cos 2\pi x)}$ (xii) $f(x)=\dfrac{\sqrt{\cos x-(1/2)}}{\sqrt{6+35x-6x^2}}$

(xiii) $f(x)=\sqrt{\log_{1/3}\left(\log_4\left([x]^2-5\right)\right)}$ (xiv) $f(x)=\dfrac{[x]}{2x-[x]}$ (xv) $f(x)=\log_x \sin x$

(xvi) $f(x)=\log_2\left(-\log_{1/2}\left(1+\dfrac{1}{\sin\left(\frac{x^\circ}{100}\right)}\right)\right)+\sqrt{\log_{10}\left(\log_{10}x\right)-\log_{10}\left(4-\log_{10}x\right)-\log_{10}3}$

(xvii) $f(x)=\dfrac{1}{[x]}+\log_{1-\{x\}}(x^2-3x+10)+\dfrac{1}{\sqrt{2-|x|}}+\dfrac{1}{\sqrt{\sec(\sin x)}}$

(xviii) $f(x)=\sqrt{(5x-6-x^2)\left[\{ln\{x\}\}\right]}+\sqrt{(7x-5-2x^2)}+\left(ln\left(\dfrac{7}{2}-x\right)\right)^{-1}$

(xix) $f(x)=\log_{\left[x+\frac{1}{x}\right]}\left|x^2-x-6\right|+{}^{16-x}C_{2x-1}+{}^{20-3x}P_{2x-5}$

(xx) $f(x)=\sqrt{\dfrac{\log_{0.3}(x-1)}{x^2-2x-8}}$, $g(x)=\sqrt{{}^{x^2+4x}C_{2x^2+3}}$ and $h(x)=\sqrt{1-\log_3(x^2-6x+11)}$.

Ans. (i) $\left[-\dfrac{5\pi}{4},\dfrac{-3\pi}{4}\right]\cup\left[-\dfrac{\pi}{4},\dfrac{\pi}{4}\right]\cup\left[\dfrac{3\pi}{4},\dfrac{5\pi}{4}\right]$ (ii) $\left(-4,-\dfrac{1}{2}\right)\cup(2,\infty)$ (iii) $(-\infty,-3]$

(iv) $(-\infty,-1)\cup[0,\infty)$ (v) $(3-2\pi<x<3-\pi)\cup(3<x\le 4)$

(vi) $\left(0,\dfrac{1}{100}\right)\cup\left(\dfrac{1}{100},\dfrac{1}{\sqrt{10}}\right)$ (vii) $(-1<x<-1/2)\ \cup\ (x>1)$ (viii) $\left[\dfrac{1-\sqrt{5}}{2},0\right)\cup\left[\dfrac{1+\sqrt{5}}{2},\infty\right)$

(ix) $(-3,-1]\cup\{0\}\cup[1,3)$ (x) $\{4\}\cup[5,\infty)$ (xi) $(0,1/4)\cup(3/4,1)\cup\{x: x\in N, x\ge 2\}$

(xii) $\left(-\dfrac{1}{6},\dfrac{\pi}{3}\right]\cup\left[\dfrac{5\pi}{3},6\right)$ (xiii) $[-3,-2)\cup[3,4)$ (xiv) $R-\left\{-\dfrac{1}{2},0\right\}$

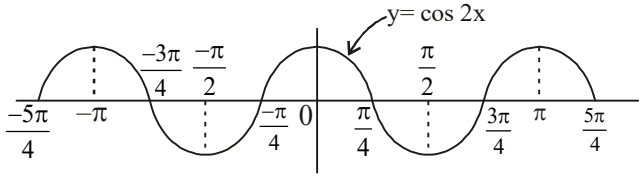
(xv) $2K\pi<x<(2K+1)\pi$ but $x\ne 1$ where K is non–negative integer

(xvi) $\{x\mid 1000\le x<10000\}$ (xvii) $(-2,-1)\cup(-1,0)\cup(1,2)$ (xviii) $(1,2)\cup(2,5/2)$;

(xix) $x\in\{4,5\}$ (xx) $[2,4)$; $\{1,2,3\}$; $2\le x\le 4$

Sol. (i) $f(x)=\sqrt{\cos 2x}+\sqrt{16-x^2}$

Domain $16-x^2\ge 0$ & $\cos x\ge 0$



$\Rightarrow x \in \left[-\frac{5\pi}{4}, -\frac{3\pi}{4}\right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$

(ii) $f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$

$\Rightarrow \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x) > 0$

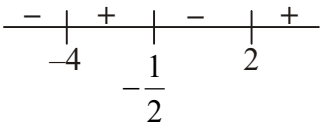
$\Rightarrow \log_3 \log_2 (2x^3 + 5x^2 - 14x) > 1$

$\Rightarrow 2x^3 + 5x^2 - 14x > 8$

$\Rightarrow 2x^3 + 5x^2 - 14x - 8 > 0$

$\Rightarrow (x - 2)(2x^2 + 9x + 4) > 0$

$\Rightarrow (x-2)2\left(x+\frac{1}{2}\right)(x+4) > 0$

$\Rightarrow$ 

$\Rightarrow x \in \left(-4, -\frac{1}{2}\right) \cup (2, \infty)$

(iii) $f(x) = \ln\left(\sqrt{x^2 - 5x - 24} - x - 2\right)$ & $\sqrt{x^2 - 5x - 24} > x + 2$

$\sqrt{x^2 - 5x - 24} > x + 2$

$x^2 - 5x - 24x > x^2 + 4x + 7$

$9x < -28$

$x < -\frac{28}{9}x - 30$ ...(1)

Also, $x^2 - 5x - 24 \geq 0$

$(x - 8)(x + 3) \geq 0$

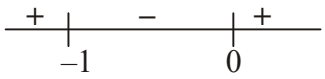
$x \in (-\infty, -3] \cup [8, \infty)$ ...(2)

Intersection of (1) & (2)

$x \in (-\infty, 3]$

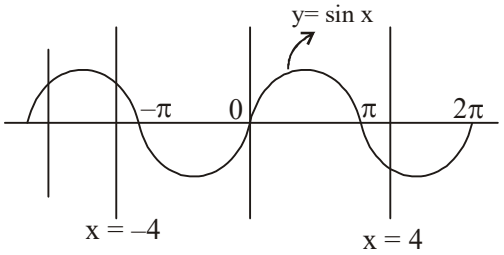
(iv) $f(x) = \sqrt{\frac{1 - 5^x}{7^{-x} - 7}}$

$\Rightarrow \dfrac{1-5^x}{7^{-x}-7} \geq 0$ $\qquad\qquad$ $\Rightarrow \dfrac{\left(5^x-1\right)7^x}{7^{x+1}-1} \geq 0$



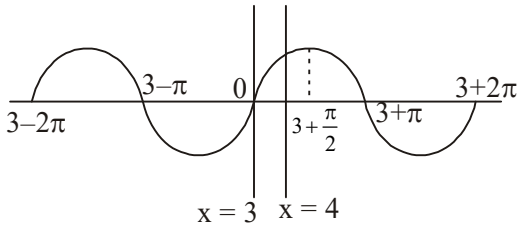
$\Rightarrow x \in (-\infty,-1) \cup [0,\infty)$

(v) $y = \log_{10} \sin(x-3) + \sqrt{16-x^2}$



$\sin(x-3) > 0$ & $16 - x^2 \geq 0$ & $x \in [-4, 4]$

$\Rightarrow x \in (3-2\pi, 3-\pi) \cup (3, 4]$

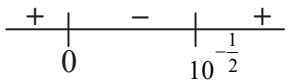


(vi) $f(x) = \log_{100x}\left(\dfrac{2\log_{10} x + 1}{-x}\right)$

$\Rightarrow 100x > 0$ & $100x \neq 1$ & $\dfrac{2\log_{10} x + 1}{-x} > 0$

$\Rightarrow x > 0$ & $x \neq \dfrac{1}{100}$

$\Rightarrow x > 0$ & $x \neq \dfrac{1}{100}$ & $\dfrac{2\log_{10} x + 1}{x} < 0$



$\Rightarrow x > 0$ & $x \neq \dfrac{1}{100}$ & $x \in \left(0, 10^{-\frac{1}{2}}\right)$

$\Rightarrow x \in \left(0, \dfrac{1}{100}\right) \cup \left(\dfrac{1}{100}, \dfrac{1}{\sqrt{10}}\right)$

(vii) $f(x)=\frac{1}{\sqrt{4x^2-1}}+\ln x\left(x^2-1\right)$

$x(x^2-1)>0$ & $4x^2-1>0$

$\Rightarrow x(x-1)(x+1)>0$ & $x^2-\frac{1}{4}>0$

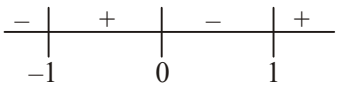
$\Rightarrow x\in(-1,0)\cup(1,\infty)$ & $x\in\left(-\infty,-\frac{1}{2}\right)\cup\left(\frac{1}{2},\infty\right)$

$\Rightarrow x\in\left(-1,-\frac{1}{2}\right)\cup(1,\infty)$

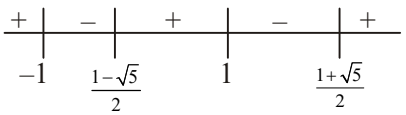
(viii) $f(x)=\sqrt{\log_{\frac{1}{2}}\frac{x}{x^2-1}}$

$\Rightarrow \log_{\frac{1}{2}}\frac{x}{x^2-1}\geq 0$

$\Rightarrow \frac{x}{x^2-1}\leq 1$ & $\frac{x}{x^2-1}>0$



$\frac{x-x^2+1}{x^2-1}\leq 0$ $\Rightarrow x\in(-1,0)\cup(1,\infty)$



$\frac{x^2-x-1}{x^2-1}\geq 0$

$\Rightarrow x\in(-\infty,-1)\cup\left[\frac{1-\sqrt{5}}{2},1\right)\cup\left[\frac{1+\sqrt{5}}{2},\infty\right)$

$\Rightarrow x\in\left[\frac{1-\sqrt{5}}{2},0\right)\cup\left[\frac{1+\sqrt{5}}{2},\infty\right)$

(ix) $f(x)=\sqrt{x^2-|x|}+\frac{1}{\sqrt{9-x^2}}$

$x^2 - |x| \geq 0$ & $9 - x^2 > 0$

$x^2 \leq |x|$ & $x^2 - 9 > 0$

$x \in (-\infty, -1] \cup [1, \infty) \cup \{0\}$ & $x \in (-3, 3)$ $\quad x \in (-3, -1] \cup [1, 3) \cup \{0\}$

(x) $f(x) = \sqrt{(x^2 - 3x - 10)\ln^2(x-3)}$

$(x^2 - 3x - 10)\ln^2(x-3) \geq 0$

For $\ln^2(x-3)$; $x - 3 > 0$ i.e. $x > 3$

$\Rightarrow x^2 - 3x - 10 \geq 0$ or $\ln^2(x-3) = 0$ $\rightarrow x \in \{4\}$

$\Rightarrow (x-5)(x+2) \geq 0$ $\Rightarrow x = 4$

$\Rightarrow x \in (-\infty, -2] \cup [5, \infty) \rightarrow x \in [5, \infty)$ $\rightarrow x \in [5, \infty) \cup \{4\}$

(xi) $f(x) = \sqrt{\log_x(\cos 2\pi x)}$

$\log_x(\cos 2\pi x) \geq 0$ & $x > 0$ & $x \neq 1$ & $\cos 2\pi x > 0$

$\Rightarrow \cos 2\pi x \geq 1$ If $x > 1$ $\cos 2\pi x \leq 1$ if $0 < x < 1$

$\cos(2\pi x) = 1$

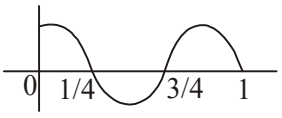
$x \in Z$

Now $\cos(2\pi x) > 0$

If $0 < x < 1$ then $\cos 2\pi x > 0$ $x \in \left(0, \frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right)$

final answer

$x \in \left(0, \frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right) \cup \{x; x \in N, x \geq 2\}$



(xii) $f(x) = \frac{\sqrt{\cos x - \frac{1}{2}}}{\sqrt{6 + 35x - 6x^2}}$

Domain

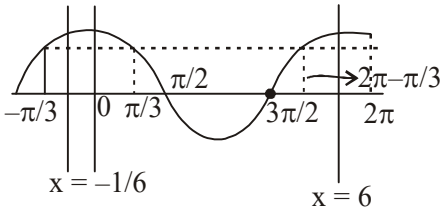
$\cos x \geq \frac{1}{2}$ & $-6x^2 + 35x + 6 < 0$

$\cos x \geq \frac{1}{2}$ & $6x^2 - 36x + x - 6 < 0$

$(x-6)(6x+1) < 0$

$\Rightarrow x \in \left(-\frac{1}{6}, 6\right)$

final answer



$x \in \left(-\frac{1}{6}, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 6\right)$

(xiii) $f(x) = \sqrt{\log_{\frac{1}{3}}\left(\log_4\left([x]^2 - 5\right)\right)}$

$\log_{\frac{1}{3}}\left(\log_4\left([x]^2 - 5\right)\right) \geq 0$

$\log_4\left([x]^2 - 5\right) > 0$

$\log_4([x]^2 - 5) \leq 1$

$[x]^2 > 6$

$[x]^2 - 5 \leq 4$

$\Rightarrow [x]^2 \leq 9$

$[x]^2 - 5 > 0$

$\Rightarrow 6 < [x]^2 \leq 9$

$\because [x] \in \mathbb{Z}$

$\Rightarrow [x]^2 = 9$

$\Rightarrow [x] = 3$ or $[x] = -3$

$\Rightarrow x \in [-3, -2) \cup [3, 4)$

(xiv) $f(x) = \frac{[x]}{2x - [x]}$

Domain $2x - [x] \neq 0$

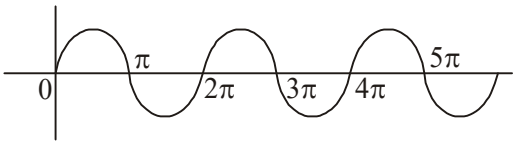
$\Rightarrow 2x \neq [x]$

$2x = [x]$

$x \in R - \left(\{0\} \cup \left\{\frac{-1}{2}\right\}\right)$

$\Rightarrow x = 0$ or $x = -\frac{1}{2}$

$\therefore$ Domain $= R - \left\{0, \frac{-1}{2}\right\}$



(xv) $f(x) = \log_x \sin x$

$x > 0$ & $x \neq 1$ & $\sin x > 0$

$\Rightarrow x \in (0,\pi) - \{1\} \cup (2\pi, 3\pi) \cup (4\pi, 5\pi) \cup .....$

$x \in \left(2k\pi, (2k+1)\pi - \{1\}; k \in \mathbb{N} \cup \{0\}\right)$

(xvi) $f(x) = \log_2\left(-\log_{\frac{1}{2}}\left(1 + \frac{1}{\sin\left(\frac{x^o}{100}\right)}\right)\right) + \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x)} - \log_{10} 3$

For $\log_2\left(-\log_{\frac{1}{2}}\left(1 + \frac{1}{\sin\left(\frac{x^o}{100}\right)}\right)\right)$

$\sin\left(\frac{x^o}{100}\right) \neq 0$ & $1 + \frac{1}{\sin\left(\frac{x^o}{100}\right)} > 0$ & $-\log_{\frac{1}{2}}\left(1 + \frac{1}{\sin\left(\frac{x^o}{100}\right)}\right) > 0$

$\Rightarrow \sin\left(\frac{x^o}{100}\right) \neq 0$ & $\frac{1}{\sin\left(\frac{x^o}{100}\right)} > -1$ & $1 + \frac{1}{\sin\left(\frac{x^o}{100}\right)} > 1$

$\Rightarrow \frac{1}{\sin\left(\frac{x^o}{100}\right)} > 0$ & $\sin\left(\frac{x^o}{100}\right) \neq 1$

$\Rightarrow \frac{x^o}{100} \in \left(2k\pi, (2k+1)\pi\right) \Rightarrow \frac{x}{1800}\left(2k, (2k+1)\right)$

Now $\sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$

$\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3 \geq 0$

& $\log_{10} x > 0$ & $x > 0$ & $4 - \log_{10} x > 0$

$\Rightarrow \log_{10}\left(\frac{\log_{10} x}{3 \cdot 4 - 3\log_{10} x}\right) \geq 0$

$\Rightarrow \frac{4\log_{10} x - 3 \cdot 4}{3 \cdot 4 - 3\log_{10} x} \geq 0$

let $\log_{10}x = t$

$$\frac{4t-12}{12-3t} \geq 0$$

$$\frac{4t-12}{3t-12} \leq 0$$

$$\Rightarrow t \in [3,4)$$

$$\Rightarrow \log_{10} x \in [3,4)$$

$0 < \log_{10}x < 4$

$$\Rightarrow \log_{10} x \in [3,4)$$

$$\Rightarrow x \in [1000,10000)$$

$$\Rightarrow x \in [1000,10000)$$

(xvii) $f(x) = \frac{1}{[x]} + \log_{1-\{x\}}\left(x^2 - 3x + 10\right)$

$$+\frac{1}{\sqrt{2-|x|}} + \frac{1}{\sqrt{\sec(\sin x)}}$$

$[x] \neq 0$ $\quad$ $1 - \{x\} > 0$ & $x^2 - 3x + 10 > 0$ & $1 - \{x\} \neq 1$

$\Rightarrow x \in R - [0,1)$ $\quad$ $\Rightarrow \{x\} < 1$ & $x \in R$ & $\{x\} \neq 0 \rightarrow x \notin Z$

$2 - |x| > 0$ $\quad$ $\sec(\sin x) > 0$

$\Rightarrow |x| < 2$ $\quad$ $x \in R$

$\Rightarrow -2 < x < 2$

$\Rightarrow x \in (-2,-1) \cup (-1,0) \cup (1,2)$

(xviii) $f(x) = \sqrt{\left(5x - 6 - x^2\right)\left[\{\ln\{x\}\}\right]} + \sqrt{\left(7x - 5 - 2x^2\right)} + \left(\ln\left(\frac{7}{2} - x\right)\right)^{-1}$

For domain

$7x - 5 - 2x^2 \geq 0$ $\quad$ $\frac{7}{2} - x > 0$ & $\ln\left(\frac{7}{2} - x\right) \neq 0$

$2x^2 - 7x + 5 \leq 0$ $\quad$ $\Rightarrow x < \frac{7}{2}$ & $\frac{7}{2} - x \neq 1$

$x < \frac{7}{2}$ & $\frac{5}{2} \neq x$

$\Rightarrow x \in \left[1, \frac{5}{2}\right]$ $\quad$ $x \in \left(-\infty, \frac{7}{2}\right) - \left\{\frac{5}{2}\right\}$

$$x \in \left[1, \frac{5}{2}\right) \ \& \ \{x\} > 0$$

$(5x - 6 - x^2)$ $$\underbrace{\left[\left\{\ln\{x\}\right\}\right]}_{=0} \geq 0 \ \& \ x \notin \mathbb{Z}$$

$$\Rightarrow x \in (1,2) \cup \left(2, \frac{5}{2}\right) \text{ (final answer)}$$

(xix) $f(x) = \log_{\left[x+\frac{1}{x}\right]} \left|x^2 - x - 6\right| + {}^{16-x}C_{2x-1} + {}^{20-3x}P_{2x-5}$

$$\underbrace{x + \frac{1}{x} > 0}_{\substack{\downarrow \\ x>0}} \ \& \ \left[x + \frac{1}{x}\right] \neq 1 \ \& \ |x^2 - x - 6| \neq 0$$

$x \neq -2$ $\quad$ $x \neq 3$

$\Rightarrow x \in (0, \infty) - \{3\}$ $\quad$ ...(1)

$16 - x \in \mathbb{N}$ & $2x - 1 \in \mathbb{N} \cup \{0\}$

$16 - x \geq 1$ & $2x - 1 \geq 0$ & $16 - x \geq 2x - 1$

$$\Rightarrow \frac{1}{2} \leq x \leq \frac{17}{3} \quad \text{...(2)}$$

$20 - 3x \in N$ & $2x - 5 \in N \cup \{0\}$

$20 - 3x \geq 1$ & $2x - 5 \geq 0$ & $20 - 3x \geq 2x - 5$

$$\frac{5}{2} \leq x \leq 5$$

$$x \in \{3,4,5\} \quad \text{...(3)}$$

from (1), (2) & (3)

$$\Rightarrow x \in \{4,5\}$$

(xx) $$f(x) = \sqrt{\frac{\log_{0.3}(x-1)}{x^2 - 2x - 8}}$$

$$\Rightarrow \frac{\log_{0.3}(x-1)}{x^2 - 2x - 8} \geq 0 \ \& \ x - 1 > 0$$

$$\Rightarrow x \in [2,4)$$

$$g(x) = \sqrt{{}^{x^2+4x}C_{2x^2+3}}$$

$x^2 + 4x \in N$ & $2x^2 + 3 \in N \cup \{0\}$

$\Rightarrow x^2+4x \geq 1$ & $\underbrace{2x^2+3 \geq 0}_{x \in R}$ & $\underbrace{x^2+4x \geq 2x^2+3}_{x^2-4x+3 \leq 0}$

$\Rightarrow x^2+4x-1 \geq 0$

$x \in \left(-\infty, -2-\sqrt{5}\right] \cup \left[-2+\sqrt{5}, \infty\right)$ $\qquad$ $x \in [1,3]$

$x \in \{1,2,3\}$

$h(x) = \sqrt{1-\log_3\left(x^2-6x+11\right)}$

$1-\log_3\left(x^2-6x+11\right) \geq 0$ & $\underbrace{x^2-6x+11>0}_{x \in R}$

$\Rightarrow \log_3\left(x^2-6x+11\right) \leq 1$

$\Rightarrow x^2-6x+11 \leq 3$

$\Rightarrow x^2-6x+8 \leq 0$

$\Rightarrow x \in [2,4]$

3. Find the domain & range of the following functions. (Read the symbols [] and {} as greatest integers and fractional part functions respectively.)

निम्न फलनो का प्रान्त व परिसर ज्ञात कीजिये। ([] व {} महत्तम पूर्णांक फलन व भिन्नात्मक फलन है।)

(i) $y = \log_{\sqrt{5}}\left(\sqrt{2}(\sin x - \cos x)+3\right)$ $\qquad$ (ii) $y = \dfrac{2x}{1+x^2}$ $\qquad$ (iii) $f(x) = \dfrac{x^2-3x+2}{x^2+x-6}$

(iv) $f(x) = \dfrac{x}{1+|x|}$ $\qquad$ (v) $y = \sqrt{2-x} + \sqrt{1+x}$

(vi) $f(x) = \log_{(\operatorname{cosec} x - 1)}(2-[\sin x]-[\sin x]^2)$ $\qquad$ (vii) $f(x) = \dfrac{\sqrt{x+4}-3}{x-5}$

Ans. (i) $D: x \in R$ $R: [0,2]$

(ii) $D = R$; range $[-1, 1]$

(iii) $D: \{x \mid x \in R;\ x \neq -3;\ x \neq 2\}$ $R: \{f(x) \mid f(x) \in R,\ f(x) \neq 1/5;\ f(x) \neq 1\}$

(iv) $D: R$; $R: (-1, 1)$

(v) $D: -1 \leq x \leq 2$ $\quad R: \left[\sqrt{3}, \sqrt{6}\right]$

(vi) $D:\ x \in (2n\pi, (2n+1)\pi) - \left\{2n\pi+\frac{\pi}{6}, 2n\pi+\frac{\pi}{2}, 2n\pi+\frac{5\pi}{6}, n \in I\right\}$ and
$R:\ \log_a 2;\ a \in (0,\infty) - \{1\} \Rightarrow$ Range is $(-\infty, \infty) - \{0\}$

(vii) $D: [-4, \infty) - \{5\}$; $R: \left(0, \dfrac{1}{6}\right) \cup \left(\dfrac{1}{6}, \dfrac{1}{3}\right]$

Sol. (i) $y = \log_{\sqrt{5}}\left(\sqrt{2}\left(\sin x - \cos x\right) + 3\right)$

for domain

$\sqrt{2}\left(\sin x - \cos x\right) + 3 > 0$

$\Rightarrow \sin x - \cos x > \dfrac{-3}{\sqrt{2}}$

$\Rightarrow \sin x \cdot \dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{2}} \cos x > -\dfrac{3}{2}$

$\Rightarrow \sin\left(x - \dfrac{\pi}{4}\right) > -\dfrac{3}{2}$

$x \in R$

Range

$\sin x - \cos x \in \left[-\sqrt{2}, \sqrt{2}\right]$

$\Rightarrow \sqrt{2}\left(\sin x - \cos x\right) \in \left[-2, 2\right]$

$\Rightarrow 3 + \sqrt{2}\left(\sin x - \cos x\right) \in \left[1, 5\right]$

$\Rightarrow 1 \le 3 + \sqrt{2}\left(\sin x - \cos x\right) \le 5$

$\Rightarrow \log_{\sqrt{5}} 5 \ge y \ge 0$

$\Rightarrow 0 \le y \le 2$

(ii) $y = \dfrac{2x}{1 + x^2}$

Domain : $x \in R$

For Range : $y + yx^2 - 2x = 0$

$\Rightarrow x^2 \cdot y - 2x + y = 0$

Case I : $y \neq 0$ $D \ge 0$

$4 - 4y^2 \ge 0$

$\Rightarrow y^2 - 1 \le 0$

$\Rightarrow y \in [-1,1] - \{0\}$

Case II : $y = 0$

$\Rightarrow x = 0$ (Real)

$\Rightarrow y \in [-1,1]$

(iii) $f\left(x\right) = \dfrac{x^2 - 3x + 2}{x^2 + x - 6} = \dfrac{\left(x-1\right)\left(x-2\right)}{\left(x+3\right)\left(x-2\right)}$

Domain : $R - \{2, -3\}$

Range : $R-\left\{1,\frac{1}{5}\right\}$

(iv) $f(x)=\frac{x}{1+|x|}=\begin{cases}\frac{x}{1-x} & x<0\\ \frac{x}{1+x} & x\geq 0\end{cases}$

Domain = R

$f(x)=\begin{cases}\frac{-x}{x-1}\equiv\frac{1-x-1}{x-1}=-1-\frac{1}{x-1} & x<0\\ 1-\frac{1}{1+x} & x\geq 0\end{cases}$

$f(x)\in(-1,1)$

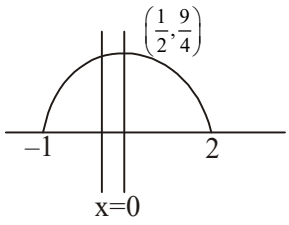
(v) $y=\sqrt{2-x}+\sqrt{1+x}$

domain $2-x\geq 0$ & $1+x\geq 0$

$\Rightarrow x\leq 2$ & $x\geq -1$

$\Rightarrow x\in[-1,2]\Rightarrow y^2=3+2\sqrt{2+2x-x-x^2}$

$=3+2\sqrt{x-x^2+2}$



$\Rightarrow -x^2+x+2\in\left[0,\frac{9}{4}\right]$

$\Rightarrow y^2\in 3+2\left[0,\frac{3}{2}\right]$

$\Rightarrow y^2\in[3,6]$

$\Rightarrow y\in\left[\sqrt{3},\sqrt{6}\right]$ as $y>0$

(vi) $f(x)=\log_{\operatorname{cosec}x-1}\left(2-[\sin x]-[\sin x]^2\right)$

Domain

cosec x – 1 > 0 & cosec x ≠ 2

cosec x > 1 & cosex x ≠ 2 $\rightarrow x \neq 2n\pi + \frac{\pi}{6}; n \in I$

& 2 – [sin x] – [sin x]^2 > 0

$x \in (2n\pi, (2n+1)\pi) - \left\{2n\pi + \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right\}$

$\therefore$ Domain = $(2n\pi, (2n+1)\pi) - \left\{2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{2}, 2n\pi + \frac{5\pi}{6}\right\}; n \in I$

Range

In this doamin

sin x $\in$ (0,1)

$\Rightarrow$ [sin x] = 0

$\Rightarrow$ f(x) = $\log_{\text{cosecx}-1}2$ $\Rightarrow$ Range is R – {0}.

cosec x – 1 > 0 & cosec x ≠ 2

(vii) $f(x) = \frac{\sqrt{x+4}-3}{x-5}$

Domain x + 4 ≥ 0 & x ≠ 5

$\Rightarrow x \in [-4,5) \cup (5, \infty)$

$y = \frac{\sqrt{x+4}-3}{x-5}$

$y = \frac{\sqrt{x+4}-3}{(x+4)-9}$

let $\sqrt{x+4} = t; t \geq 0$

$y = \frac{t-3}{t^2-9}$

$y = \frac{t-3}{(t-3)(t+3)}$

$y \in (0,\infty) - \left\{-3, \frac{1}{6}\right\}$ and $t+3 = \frac{1}{y}$

$t+3 \geq 3$

$$\Rightarrow \frac{1}{y} \geq 3 \Rightarrow y \leq \frac{1}{3}$$

$$y \in \left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$$

4. Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in N$ then find the value of $(a + b)$.

$f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ जहाँ $x \in [-6, 6]$ यदि $f(x)$ का परिसर $[a, b]$ है $a, b \in N$ तो $(a + b)$ का मान ज्ञात कीजिए।

Ans. 5049

Sol. $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$

$x \in [-6, 6]$

$= (x^2 + 5x + 4)(x^2 + 5x + 6) + 5$

Let $t = x^2 + 5x$

$(t + 4)(t + 6) + 5 = t^2 + 10t + 24 + 5$

$= t^2 + 10t + 29$

$= (t + 5)^2 + 4$

$f(x) = (x^2 + 5x + 5)^2 + 4$

$4 \leq f(x) \leq (36 + 30 + 5)^2 + 4$

$4 \leq f(x) \leq 5041 + 4 = 5045$

$a + b = 5049.$

EQUAL OR IDENTICAL FUNCTIONS

5. Identify the pair(s) of functions which are identical? (where [x] denotes greatest integer and {x} denotes fractional part function)

(i) $f(x) = \text{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$

(ii) $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$ and $g(x) = \tan x$

(iii) $f(x) = \ln(1 + x) + \ln(1 - x)$ and $g(x) = \ln(1 - x^2)$

(iv) $f(x) = \frac{\cos x}{1 - \sin x}$ and $g(x) = \frac{1 + \sin x}{\cos x}$

निम्न युग्मों में समरूप फलन ज्ञात करें? ([] व {} महत्तम पूर्णांक फलन व भिन्नात्मक फलन है।)

(i) $f(x) = \text{sgn}(x^2 - 3x + 4)$ तथा $g(x) = e^{[\{x\}]}$

(ii) $f(x) = \sqrt{\frac{1-\cos 2x}{1+\cos 2x}}$ तथा $g(x) = \tan x$

(iii) $f(x) = \ln(1+x) + \ln(1-x)$ तथा $g(x) = \ln(1-x^2)$

(iv) $f(x) = \frac{\cos x}{1-\sin x}$ तथा $g(x) = \frac{1+\sin x}{\cos x}$

Ans. (i), (iii) are identical

Sol. (i) $f(x) = \text{sgn}(x^2 - 3x + 4) \equiv 1$ $\quad a > 0$ & $D < 0$

$g(x) = e^{[\{x\}]} \equiv 1$ $\quad \Rightarrow f(x) = g(x)$

$D_f = R$ & $D_g = R$

$\Rightarrow$ f & g are identical

Range of g(x) and f(x) = {1}

(ii) $f(x) = \sqrt{\frac{1-\cos 2x}{1+\cos 2x}}$ and $g(x) = \tan x$

$D_g = R - \left\{(2n+1)\frac{\pi}{2}; n \in I\right\}$ $\qquad D_f(x) = R - \left\{(2n+1)\frac{\pi}{2}; n \in I\right\}$

$R_g = R = (-\infty, \infty)$ $\qquad R_g = (0, \infty)$

(iii) $f(x) = \ln(1+x) + \ln(1-x)$

$g(x) = \ln(1-x^2)$

$D_f = (-1,1)$ $\qquad f(x) = \ln(1-x^2)$; $\quad g(x) = \ln(1-x^2)$

$D_g = (-1,1)$ $\qquad \Rightarrow f(x) = g(x)\ \forall x \in (-1,1)$

$\Rightarrow$ f & g are identical.

(iv) $f(x) = \frac{\cos x}{1-\sin x}$ $\qquad g(x) = \frac{1+\sin x}{\cos x}$

$D_f = R - \left\{(4n+1)\frac{\pi}{2}; n \in I\right\}$

$D_g = R - \left\{(2n+1)\frac{\pi}{2}; n \in I\right\}$

$\Rightarrow D_f \neq D_g$

$\Rightarrow$ f & g are not identical.

IMPLICIT AND EXPLICIT FUNCTIONS

6. (i) Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :

y को x के पदों में लिखे। फलन का प्रान्त भी ज्ञात कीजिये। :

(a) $10^x + 10^y = 10$ (b) $x + |y| = 2y$

(ii) The function f(x) is defined on the interval [0,1]. Find the domain of definition of the functions.
फलन f(x) [0,1] में परिभाषित है। निम्न फलनों के प्रान्त ज्ञात कीजिये।

(a) $f(\sin x)$ (b) $f(2x+3)$

(iii) Given that y = f(x) is a function whose domain is [4, 7] and range is [–1, 9]. Find the range and domain of :
दिया है y = f(x) एक ऐसा फलन है जिसका प्रान्त [4, 7] तथा परिसर [–1, 9] है। निम्न फलनों के प्रान्त व परिसर ज्ञात कीजिये:

(a) $g(x) = \frac{1}{3} f(x)$ (b) $h(x) = f(x-7)$

Ans. (i) (a) $y = \log(10 - 10^x)$, $-\infty < x < 1$ (b) $y = x/3$ when $-\infty < x < 0$ and $y = x$ when $0 \le x < +\infty$
(ii) (a) $2K\pi \le x \le 2K\pi + \pi$ where () $K \in I$ (b) $[-3/2, -1]$
(iii) (a) Range : $[-1/3, 3]$, Domain = [4, 7] (b) Range [–1, 9] and domain [11, 14]

Sol. (i) (a) $10^x + 10^y = 10$

$10^y = 10 - 10^x$

$y = \log_{10}(10 - 10^x)$

$10 - 10^x > 0$

$\Rightarrow 10^x < 10$

$\Rightarrow x < 1$

$\therefore$ Domain = $(-\infty, 1)$

(b) $x + |y| = 2y$

$y \ge 0$: $x = y$

$y < 0$: $x = 3y$

$\therefore y = x$ if $0 \le x < \infty$

$y = x$ if $-\infty < x < 0$

(ii) f(x); $x \in [0,1]$

(a) $f(\sin x)$

$\Rightarrow \sin x \in [0,1]$

$\Rightarrow x \in [2k\pi, (2k+1)\pi]$; $k \in I$

(b) $f(2x + 3)$

$\Rightarrow 2x + 3 \in [0,1]$ $\Rightarrow 2x \in [-3,-2]$

$\Rightarrow x \in \left[-\frac{3}{2}, -1\right]$

(iii) f(x)

$D_f \equiv [4,7]$

$R_f = [-1,9]$

$\Rightarrow g(x) = \frac{1}{3} f(x)$ $h(x) = f(x-7)$

$\Rightarrow D_g = [4,7]$ $D_h = [11, 14]$

$R_g = \left[-\frac{1}{3}, 3\right]$ $R_h = [-1, 9]$

CLASSIFICATION OF FUNCTIONS (ONE-ONE, ONTO ETC.)

7. Classify the following functions f(x) defined in $R \to R$ as injective, surjective, both or none :

निम्न फलनों मे एकैकी, आच्छादक, दोनों या कोई भी नहीं ज्ञात करों। $f : R \to R$:

(a) $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$ (b) $f(x) = x^3 - 6x^2 + 11x - 6$

(c) $f(x) = (x^2+x+5)(x^2+x-3)$

Ans. (a) neither surjective nor injective (b) surjective but not injective (c) neither injective nor surjective

Sol. (a) $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$

Neither injective nor surjective.

(b) $f(x) = x^3 - 6x^2 + \underbrace{11x - 6}_{\text{surjective}}$

Neither injective as $f'(x) = 3x^2 - 12x + 11$

(c) $f(x) = (x^2 + x + 5)(x^2 + x - 3)$

Neither injective nor surjective

FUNCTIONAL EQUATION

8. Solve the following problems from (a) to (e) on functional equation.

फलनो के समीकरण पर आधारित प्रश्न (a) से (e) तक हल कीजिए।

(a) The function f(x) defined on the real numbers has the property that $f\left(f(x)\right)\cdot\left(1+f(x)\right) = -f(x)$ for all x in the domain of f. If the number 3 is in the domain and range of f, compute the value of f(3).

फलन f(x) यह गुणधर्म है कि $f\left(f(x)\right)\cdot\left(1+f(x)\right) = -f(x)$ जहाँ x प्रात का अवयव है। यदि 3 प्रान्त व परिसर दोनों में है तो f(3) ज्ञात कीजिए।

(b) Suppose f is a real function satisfying f(x + f(x)) = 4 f(x) and f(1) = 4. Find the value of f(21).
फलन f सम्बन्ध f(x + f(x)) = 4 f(x) को संतुष्ट करता है तथा f(1) = 4 । f(21) का मान ज्ञात कीजिए।

(c) Let 'f' be a function defined from $R^+ \to R^+$. If $[f(xy)]^2 = x\left(f(y)\right)^2$ for all positive numbers x and y and f(2) = 6, find the value of f(50).

फलन 'f' $R^+ \to R^+$ यदि $[f(xy)]^2 = x\left(f(y)\right)^2$ x व y धनात्मक है तथा f(2) = 6 तो f(50) का मान ज्ञात कीजिए।

(d) Let f(x) be a function with two properties
फलन f(x) में दो निम्न गुणधर्म है

(i) for any two real number x and y, f(x + y) = x + f(y) and (ii) f(0) = 2.
Let f(x) be a function with two properties
वास्तविक संख्या x व y के लिये f(x + y) = x + f(y) तथा (ii) f(0) = 2.
Find the value of f(100).
f(100) का मान ज्ञात करें।

(e) Let f be a function such that f(3) = 1 and f(3x) = x + f(3x – 3) for all x. Then find the value of f(300).
फलन f ऐसा है कि f(3) = 1 तथा f(3x) = x + f(3x – 3), x के सभी वास्तविक मानों के लिये f(300) का मान ज्ञात कीजिए।

(f) Suppose that f(x) is a function of the form $f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$ $(x \neq 0)$. If f(5) = 2 then find the value of f(– 5).

यदि फलन f(x) ऐसा है कि $f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$ $(x \neq 0)$. व f(5) = 2 तो f(– 5) का मान ज्ञात कीजिए।

Ans. (a) – 3/4 (b) 64 (c) 30 (d) 102 (e) 5050 (f) 28

Sol. (a) f(f(x) · (1 + f(x)) = – f(x)

$$\Rightarrow f\left(f(x)\right) = -\frac{f(x)}{1+f(x)}$$

$$f(x) \leftrightarrow x$$

$$\Rightarrow f(x) = -\frac{x}{1+x}$$

$$f(3) = -\frac{3}{4}$$

(b) $f(x + f(x)) = 4\, f(x)$ & $f(1) = 4$

$\Rightarrow f(5) = 16$

$\because\ f(x) = 3x + 1$

$\Rightarrow f(21) = 63 + 1 = 64$

(c) $[f(xy)]^2 = x(f(y))^2$

$y = 2$

$\Rightarrow (f(2x))^2 = x \cdot 36$

$\Rightarrow (f(2x))^2 = 36x$

put $x = 25$

$\Rightarrow f(50) = 30$

(d) $f(x + y) = x + f(y)$

put $y = 0$

$f(x) = x + 2$

$\Rightarrow f(100) = 102$

(e) $f(3x) = x + f(3x - 3)$

$\Rightarrow f(3x) - f(3x - 3) = x \Rightarrow f(0) = 0$

$$\Rightarrow \sum_{x=1}^{100} f\left(3x\right) - f\left(3x-3\right) = \sum_{x=1}^{100} x = 5050$$

$\Rightarrow -f(0) + f(300) = 5050$

$\Rightarrow f(300) = 5050$

(f) $f\left(x\right) = \dfrac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$

$$f\left(5\right) + f\left(-5\right) = \frac{15\left(5+5\right)}{5}$$

$2 + f(-5) = 30$

$\Rightarrow f(-5) = 28$

COMPOSITE FUNCTIONS

9. Suppose $f(x) = \sin x$ and $g(x) = 1 - \sqrt{x}$. Then find the domain and range of the following functions :

$f(x) = \sin x$ व $g(x) = 1 - \sqrt{x}$ तो निम्न फलनों का प्रान्त व परिसर ज्ञात कीजिए :

(a) fog (b) gof (c) fof (d) gog

Ans. (a) Domain is $x \geq 0$; range $[-1, 1]$ (b) Domain $2k\pi \leq x \leq 2k\pi + \pi$; range $[0, 1]$
(c) Domain $x \in R$; range $[-\sin 1, \sin 1]$ (d) Domain is $0 \leq x \leq 1$; range is $[0, 1]$

Sol. $f(x) = \sin x$ $g(x) = 1 - \sqrt{x}$

(a) $fog(x) = \sin\left(1 - \sqrt{x}\right)$

Domain $= [0, \infty)$

Range $= [-1,1]$

(b) $gof(x) = g(\sin x) = 1 - \sqrt{\sin x}$

for domain

$\sin x \geq 0$

$x \in [2k\pi,(2k + 1)\pi]; k \in I$

$\because \sin x \in [-1,1]$

$\Rightarrow -\sqrt{\sin x} \in [-1,0]$

$\Rightarrow 1 - \sqrt{\sin x} \in [0,1]$

(c) $fof(x) = \sin(\sin x)$

Domain $= (-\infty, \infty)$

Range $= [-\sin 1, \sin 1]$

(d) $gog(x) = g\left(1 - \sqrt{x}\right)$

$= 1 - \sqrt{1 - \sqrt{x}}$

Domain : $x \geq 0$ & $1 - \sqrt{x} \geq 0$

$x \geq 0$ & $\sqrt{x} \leq 1$

$\Rightarrow 0 \leq x \leq 1$

Range : $[0,1]$

10. A function $f : R \to R$ is such that $f\left(\frac{1-x}{1+x}\right) = x$ for all $x \neq -1$. Prove the following :

फलन $f : R \to R$ ऐसा है कि $f\left(\frac{1-x}{1+x}\right) = x$, $x \neq -1$ निम्नलिखित को सिद्ध कीजिये :

(a) $f(f(x)) = x$ (b) $f(1/x) = -f(x),\ x \neq 0$ (c) $f(-x-2) = -f(x) - 2$

Ans.

Sol. $f : R \to R$

$$f\left(\frac{1-x}{1+x}\right) = x \qquad \forall x \in R - \{1\}$$

$$y \leftrightarrow \frac{1-x}{1+x} \qquad y = \frac{1-x}{1+x}$$

$$\Rightarrow y + xy = 1 - x$$

$$\Rightarrow x(y + 1) = 1 - y$$

$$\Rightarrow f(y) = \frac{1-y}{1+y} \qquad \Rightarrow x = \frac{1-y}{y+1}$$

$$\Rightarrow f(x) = \frac{1-x}{1+x}$$

(a) $f(f(x)) = f\left(\frac{1-x}{1+x}\right) = \frac{1-\left(\frac{1-x}{1+x}\right)}{1+\frac{1-x}{1+x}} = \frac{2x}{2} = x$

(b) $f\left(\frac{1}{x}\right) = \frac{1-\frac{1}{x}}{1+\frac{1}{x}} = \frac{x-1}{x+1} = \left(\frac{1-x}{1+x}\right)\cdot -1 = -f(x);\ x \neq 0$

(c) $f(-x-2) = \frac{1-(-x-2)}{1+(-x-2)} = \frac{x+3}{-(x+1)}$

$$= \frac{x-1}{x+1} - 2 = -f(x) - 2$$

11. $f(x)=\begin{cases}1-x & \text{if } x\le 0\\ x^2 & \text{if } x>0\end{cases}$ and $g(x)=\begin{cases}-x & \text{if } x<1\\ 1-x & \text{if } x\ge 1\end{cases}$ find (fog)(x) and (gof)(x)

$f(x)=\begin{cases}1-x & \text{if } x\le 0\\ x^2 & \text{if } x>0\end{cases}$ तथा $g(x)=\begin{cases}-x & \text{if } x<1\\ 1-x & \text{if } x\ge 1\end{cases}$ (fog)(x) तथा (gof)(x) ज्ञात कीजिये।

Ans. $(gof)(x)=\begin{cases}x & \text{if } x\le 0\\ -x^2 & \text{if } 0<x<1\\ 1-x^2 & \text{if } x\ge 1\end{cases}$; $(fog)(x)=\begin{cases}x^2 & \text{if } x<0\\ 1+x & \text{if } 0\le x<1\\ x & \text{if } x\ge 1\end{cases}$

Sol. $f(x)=\begin{cases}1-x & x\le 0\\ x^2 & x>0\end{cases}$ $\qquad$ $g(x)=\begin{cases}-x & x<1\\ 1-x & x\ge 1\end{cases}$

$$fog(x)=\begin{cases}1-g(x) & g(x)\le 0\\ (g(x))^2 & g(x)>0\end{cases}=\begin{cases}x^2 & x<0\\ 1+x & 0\le x<1\\ x & x\ge 1\end{cases}$$

$$gof(x)=\begin{cases}-f(x) & f(x)<1\\ 1-f(x) & f(x)\ge 1\end{cases}$$

$$=\begin{cases}x & x\le 0\\ -x^2 & 0<x<1\\ 1-x^2 & x\ge 1\end{cases}$$

12. f(x) and g (x) are linear function such that for all x, $f(g(x))$ and g(f(x)) are Identity functions. If f(0) = 4 and g(5) = 17, compute f (136).

f(x) व g(x) रेखीय फलन है ताकि $\forall\ x \in R$, $f(g(x))$ व g(f(x)) तत्समक फलन है। यदि f(0) = 4, g(5) = 17 तो f (136) निकालें।

Ans. 12

Sol. Let f(x) = ax + b $\qquad$ f(g(x)) = x

g(x) = cx + d $\qquad$ $\Rightarrow$ f(cx + d) = x

Given f(0) = 4 & g(5) = 17 $\qquad$ $\Rightarrow$ a(cx + d) + b = x

$\Rightarrow b = 4$ & $5c + d = 17$ $\Rightarrow (ac - 1) = 0$ & $ad + b = 0$

$gof(x) = x$

$\Rightarrow g(ax + b) = x$

$\Rightarrow c(ax + b) + d = x$

$\Rightarrow ca - 1 = 0$ & $bc + d = 0$

$\Rightarrow ac = 1, b = 4, ad = -4$ & $d = -4c$

$\Rightarrow 5c - 4c = 17$

$\Rightarrow c = 17$

$\Rightarrow a = \dfrac{1}{17}$

$\because f(x) = \dfrac{x}{17} + 4$

$\Rightarrow f(136) = \dfrac{136}{17} + 4 = 12$

EVEN AND ODD FUNCTIONS

13. Find whether the following functions are even or odd or none :

ज्ञात करे निम्न फलन सम, विषम या दोनों में से कुछ भी नही है :

(a) $f(x) = \log\left(x + \sqrt{1 + x^2}\right)$ (b) $f(x) = \dfrac{x(a^x + 1)}{a^x - 1}$

(c) $f(x) = \sin x + \cos x$ (d) $f(x) = x \sin^2 x - x^3$

(e) $f(x) = \sin x - \cos x$ (f) $f(x) = \dfrac{\left(1 + 2^x\right)^2}{2^x}$

(g) $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$ (h) $f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3}$

Ans. (a) odd (b) even (c) neither odd nor even (d) odd
(e) neither odd nor even (f) even (g) even (h) even

Sol. (a) $f(x) = \log\left(x + \sqrt{1 + x^2}\right)$

$$f(-x) = \log\left(-x + \sqrt{1 + x^2}\right) = \log\left(\frac{1}{x + \sqrt{1 + x^2}}\right)$$

$$= -\log\left(x + \sqrt{1 + x^2}\right) \rightarrow \text{odd}$$

(b) $f(x)=\dfrac{x\left(a^x+1\right)}{a^x-1}$

$$f(-x)=\frac{-x\left(1+a^x\right)}{1-a^x}=f(x)$$

$\Rightarrow$ Even

(c) $f(x)=\sin x+\cos x$

$$\underbrace{f(-x)=\cos x-\sin x}_{\text{Neither even nor odd}}$$

(d) $f(x)=x\sin^2 x-x^3=\underbrace{x\left(\sin^2 x-x^2\right)}_{\text{odd}}$

(e) $f(x)=\underbrace{\sin x-\cos x}_{\text{Neither even nor odd}}$

(f) $f(x)=\dfrac{\left(1+2^x\right)^2}{2^x}\rightarrow$ Even

(g) $f(x)=\dfrac{x}{e^x-1}+\dfrac{x}{2}+1\rightarrow$ Even

$$f(-x)=-x\left(\frac{e^x}{1-e^x}+\frac{1}{2}\right)+1$$

$$=-x\left(\frac{2e^x+1-e^x}{2\left(1-e^x\right)}\right)+1=-x\left(\frac{e^x+1}{2\left(1-e^x\right)}\right)+1$$

$$=\frac{xe^x+x}{2\left(e^x-1\right)}+1=f(x)$$

(h) $f(x)=\left[(x+1)^2\right]^{\frac{1}{3}}+\left[(x-1)^2\right]^{\frac{1}{3}}$

$f(-x)=f(x)\rightarrow$ Even function

INVERSE OF A FUNCTION

14. Compute the inverse of the functions :

निम्न का प्रतिलोम फलन ज्ञात कीजिये :

(a) $f(x) = ln\left(x + \sqrt{x^2+1}\right)$ (b) $f(x) = 2^{\frac{x}{x-1}}$

(c) $y = \dfrac{10^x - 10^{-x}}{10^x + 10^{-x}}$

Ans. (a) $\dfrac{e^x - e^{-x}}{2}$ (b) $\dfrac{\log_2 x}{\log_2 x - 1}$ (c) $\dfrac{1}{2}\log\dfrac{1+x}{1-x}$

Sol. (a) $f(x) = \ln\left(x + \sqrt{x^2+1}\right)$

$y = \ln\left(x + \sqrt{x^2+1}\right)$

$x \leftrightarrow y$

$x = \ln\left(y + \sqrt{y^2+1}\right)$

$e^x = y + \sqrt{y^2+1}$

$\Rightarrow e^{2x} + y^2 - 2ye^x = y^2 + 1$

$\Rightarrow e^{2x} - 1 = 2ye^x$

$\Rightarrow y = \dfrac{e^{2x}-1}{2e^x} = \dfrac{e^x - e^{-x}}{2} = f^{-1}(x)$

(b) $f(x) = 2^{\frac{x}{x-1}}$

$\Rightarrow y = 2^{\frac{x}{x-1}}$

$\Rightarrow x = 2^{\frac{y}{y-1}}$

$\Rightarrow \log_2 x = \dfrac{y}{y-1}$

$\Rightarrow y(\log_2 x - 1) = \log_2 x$

$\Rightarrow y = f^{-1}(x) = \dfrac{\log_2 x}{\log_2 x - 1}$

(c) $y = \dfrac{10^x - 10^{-x}}{10^x + 10^{-x}} = \dfrac{10^{2x} - 1}{10^{2x} + 1}$

$\Rightarrow x = \dfrac{10^{2y} - 1}{10^{2y} + 1}$

$\Rightarrow 10^{2y}(x - 1) + x + 1 = 0$

$\Rightarrow 10^{2y} = \dfrac{x + 1}{1 - x}$

$\Rightarrow 2y = \log_{10}\left(\dfrac{x + 1}{1 - x}\right)$

$\Rightarrow f^{-1}(x) = \dfrac{1}{2}\log_{10}\left(\dfrac{1 + x}{1 - x}\right)$

15. Function f and g are defined by $f(x) = \sin x, x \in R$; $g(x) = \tan x, x \in R - \left(K + \dfrac{1}{2}\right)\pi$ where $K \in I$.

Find (i) periods of fog and gof. (ii) range of the function fog and gof.

फलन f तथा g, $f(x) = \sin x, x \in R$; $g(x) = \tan x, x \in R - \left(K + \dfrac{1}{2}\right)\pi$ के रूप में परिभाषित किया गया है, जहाँ $K \in I$.

ज्ञात कीजिए (i) fog व gof का आवर्तकाल ज्ञात कीजिये (ii) fog व gof परिसर ज्ञात कीजिये।

Ans. (i) period of fog is π, period of gof is 2π (ii) range of fog is $[-1, 1]$, range of gof is $[-\tan 1, \tan 1]$

Sol. $fog(x) = f(\tan x) = \sin(\tan x)$

$\Rightarrow \underbrace{fog(\pi + x) = fog(x)}_{P=\pi}$

Range of fog is $[-1,1]$

$\underbrace{gof(x) = g(\sin x) = \tan(\sin x)}_{\text{period}=2\pi}$

Range = $[-\tan 1, \tan 1]$

PERIODIC FUNCTION

16. Suppose that f is an even, periodic function with period 2, and that f (x) = x for all x in the interval [0, 1]. Find the value of f (3.14).

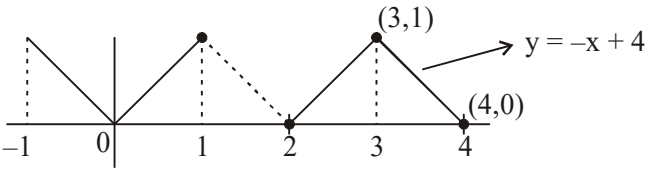
फलन f सम, आवर्ती है जिसका आवर्तकाल 2 है, f (x) = x, सभी x [0, 1] के लिये। f (3.14) का मान ज्ञात कीजिये।

Ans. 0.86

Sol. $f(-x) = f(x)$

$f(x+2) = f(x) \ \forall x \in R$

$f(3\cdot 14) = -3\cdot 14 + 4 = 0.86$



17. The fundamental period of $e^{\cos^4 \pi x + x - [x] + \cos \pi x}$ is _____. (where [ ] denotes the greatest integer function)

$e^{\cos^4 \pi x + x - [x] + \cos \pi x}$ का मूलभूत आवर्तकाल _____ है। (जहाँ [ ] महत्तम पूर्णांक फलन को प्रदर्शित करता है)

Ans. 2

Sol. $e^{\underbrace{\cos^4 \pi x}_{\text{period}=1} + \underbrace{x - [x]}_{\text{period}=1} + \underbrace{\cos \pi x}_{\text{period}=2}}$

$\Rightarrow P = \text{L.C.M.}(1,1,2) = 2$

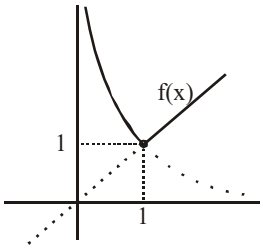
OTHERS

18. If $f(x) = \max\left(x, 1/x\right)$ for $x > 0$ where max (a, b) denotes the greater of the two real numbers a and b. Define the function $g(x) = f(x) \cdot f\left(1/x\right)$ and plot its graph.

यदि $f(x) = \max\left(x, 1/x\right), x > 0$; जहाँ max (a, b) a व b बड़ी संख्या को निरूपित करता है तो फलन $g(x) = f(x) \cdot f\left(1/x\right)$ को परिभाषित करे तथा इसका ग्राफ खीचिऐं।

Ans. $g(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } 0 < x \le 1 \\ x^2 & \text{if } x > 1 \end{cases}$

Sol. $x = \frac{1}{x} \Rightarrow x^2 = 1 \quad \Rightarrow x = 1$

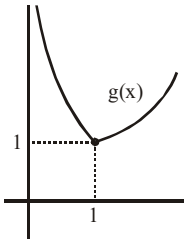


$$f(x) = \frac{1}{x}; \quad 0 < x \le 1$$

$$= x \; ; \quad x > 1$$

$$g(x) = f(x)\, f\left(\frac{1}{x}\right)$$

$$\Rightarrow \; g(x) = \begin{cases} \frac{1}{x}.\frac{1}{x} & 0 < x \le 1 \\ x.x & x > 1 \end{cases}$$



19. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where a, b, c, d $\in$ R. Suppose P(0) = 6, P(1) = 7, P(2) = 8 and P(3) = 9, then find the value of P(4).

माना $P(x) = x^4 + ax^3 + bx^2 + cx + d$, जहाँ a, b, c, d $\in$ R. यदि P(0) = 6, P(1) = 7, P(2) = 8 तथा P(3) = 9, तो P(4) का मान ज्ञात कीजिए।

Ans. 34

Sol. Let g(x) = P(x) – (x + 6)

$\Rightarrow$ g(0) = g(1) = g(2) = g(3) = 0

$\Rightarrow$ g(x) = x(x – 1) (x – 2) (x – 3)

$\Rightarrow$ P(x) = x(x – 1) (x – 2) (x – 3) + x + 6

P(4) = 4 × 3 × 2 × 1 + 10

= 24 + 10

= 34

20. Let $f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$. If f (x) is divided by $x^3 - x$ then the remainder is some function of x say g (x). Find the value of g (10).

$f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$ से विभाजित करने पर शेषफल g (x) है। g (10) का मान ज्ञात कीजिए।

Ans. 21

Sol. $f(x) = x(x - 1)(x + 1)P(x) + ax^2 + bx + c$

$$\left.\begin{array}{c} f(0) \Rightarrow c = 1 \\ f(1) = a + b + c \Rightarrow a + b + c = 3 \\ f(-1) = a - b + c \Rightarrow a - b + c = -1 \end{array}\right] \Rightarrow \begin{array}{c} a + b = 2 \\ a - b = -2 \end{array}$$

$\Rightarrow a = 0, b = 2, c = 1$

$\Rightarrow g(x) = 2x + 1$

$\Rightarrow g(10) = 21$

21. Let $f(x) = \frac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + ... + f\left(\frac{2005}{2006}\right)$.

$f(x) = \frac{9^x}{9^x + 3}$ तो योग $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + ... + f\left(\frac{2005}{2006}\right)$ का मान ज्ञात कीजिए।

Ans. 1002.5

Sol. $f(x) = \frac{9^x}{9^x + 3}$

$\Rightarrow f(x) + f(1 - x) = 1$

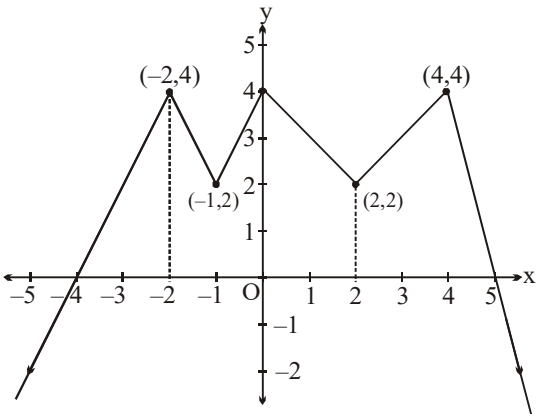
$$\Rightarrow f\left(\frac{1}{2006}\right) + f\left(\frac{2005}{2006}\right) + ... + f\left(\frac{1002}{2006}\right) + f\left(\frac{1004}{2006}\right) + f\left(\frac{1003}{2006}\right)$$

$$= 1002 + f\left(\frac{1}{2}\right) = 1002 + \frac{\sqrt{9}}{\sqrt{9} + 3} = 1002 + \frac{1}{2}$$

$= 1002.5$

22. Consider the graph of a real-valued continuous function f(x) defined on R (the set of all real numbers) as shown below.

माना R (सभी वास्तविक संख्याओं का समुच्चय) पर परिभाषित वास्तविक मानों वाले फलन f(x) का आरेख निम्न प्रकार है।



Find the number of real solutions of the equation $f(f(x)) = 4$.

समीकरण $f(f(x)) = 4$ के वास्तविक हलों की संख्या ज्ञात कीजिए।

Ans. 7

Sol. $f(f(x)) = 4$ if $\underbrace{f(x) = 0}_{2 \text{ solution}}$ or $\underbrace{-2}_{2 \text{ solution}}$ or $\underbrace{4}_{3 \text{ solution}}$

$\Rightarrow$ No. of solutions = 7

Paragraph for Question Nos. 23 to 25

Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall\, x \in R$

माना कि $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall\, x \in R$

23. Least value of 'a' for which f(x) is injective function, is :

f(x) के एकैकी फलन होने के लिए a का न्यूनतम मान है :

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

Ans. A

Sol. $f'(x) = x^2 + x + a \quad \Rightarrow \quad f'(x) \geq 0 \quad \Rightarrow \quad \left(x + \frac{1}{2}\right)^2 + \left(a - \frac{1}{4}\right) \geq 0$

Hence $a \geq \frac{1}{4}$

$f'(x) = x^2 + x + a \quad \Rightarrow \quad f'(x) \geq 0 \quad \Rightarrow \quad \left(x + \frac{1}{2}\right)^2 + \left(a - \frac{1}{4}\right) \geq 0$

अतः $a \geq \frac{1}{4}$

24. If $a = -1$ then $f(x)$ is :

(A) bijective (B) many–one and onto

(C) one–one and into (D) many– one and into

यदि $a = -1$ हो, तो $f(x)$ है :

(A) एकैकी आच्छादक (B) बहुएकैकी और आच्छादक

(C) एकैकी और अन्तर्क्षेपी (D) बहुएकैकी और अन्तर्क्षेपी

Ans. B

Sol. If $a = -1 \Rightarrow f'(x) = x^2 + x - 1$

$D = 1 + 4 > 0 \Rightarrow$ many one function and cubic polynomial have range $(-\infty, \infty)$

hence onto function

यदि $a = -1 \Rightarrow f'(x) = x^2 + x - 1$

$D = 1 + 4 > 0 \Rightarrow$ यह बहुएकैकी फलन है एवं त्रिघात बहुपद का परिसर $(-\infty, \infty)$ होता है।

अतः यह आच्छादक फलन है।

25. $f(x)$ is invertible iff :

$f(x)$ प्रतिलोमीय है, यदि और केवल यदि :

(A) $a \in \left[\frac{1}{4}, \infty\right), b \in R$ (B) $a \in \left[\frac{1}{8}, \infty\right), b \in R$ (C) $a \in \left(-\infty, \frac{1}{4}\right], b \in R$ (D) $a \in \left(-\infty, \frac{1}{4}\right), b \in R$

Ans. A

Sol. For $f(x)$ to be invertible $f(x)$ should be one-one that is $f'(x) \geq 0, \Rightarrow a \in \left[\frac{1}{4}, \infty\right)$

$f(x)$ प्रतिलोमिय हो इसके लिए आवश्यक है कि $f(x)$ एकैकी फलन होना चाहिए

अतः $f'(x) \geq 0 \Rightarrow a \in \left[\frac{1}{4}, \infty\right)$

Paragraph for Question Nos. 26 to 28

Consider the function $y = f(x) = x^2 - 6x + 5$

माना फलन $y = f(x) = x^2 - 6x + 5$

26. The domain of definition of the function $g(x) = \ln(f(x))$:

फलन $g(x) = \ln(f(x))$ का प्रांत होगा :

(A) $(4, \infty)$ (B) $(-\infty, 1)$ (C) $R - \{x \mid 1 \leq x \leq 5\}$ (D) R

Ans. C

Sol. $y = x^2 - 6x + 5 = f(x)$

$g(x) = \ln(f(x))$

For D_g $\quad f(x) > 0$

$\Rightarrow x^2 - 6x + 5 > 0$

$\Rightarrow (x - 5)(x - 1) > 0$

$\Rightarrow x \in (-\infty, 1) \cup (5, \infty)$

$\Rightarrow D_g = R - \{x | \ 1 \le x \le 5\}$

27. Range of the function $y = \ln(f(x))$ is :

फलन $y = \ln(f(x))$ का परिसर होगा :

(A) R (B) $[1, \infty)$ (C) $[0, \infty)$ (D) $[-4, \infty)$

Ans. A

Sol. $\because \ x^2 - 6x + 5 \in [-4, \infty)$

$\Rightarrow \ln(f(x)) \in (-\infty, \infty)$

$\Rightarrow$ Range of $y = R$

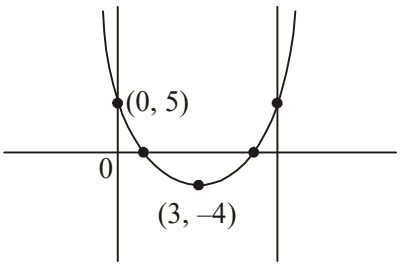
28. Number of points of intersection of a line $y = c$ (where c is a fixed constant and $0 < c < 4$) with graphs $y = \left| f(|x|) \right|$ is :

रेखा $y = c$ (जहाँ c एक स्थिर नियतांक है तथा $0 < c < 4$) तथा $y = \left| f(|x|) \right|$ के आरेख के प्रतिच्छेद बिन्दु की संख्या होगी :

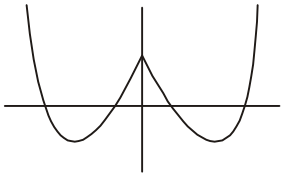
(A) 2 (B) 8 (C) 6 (D) 4

Ans. B

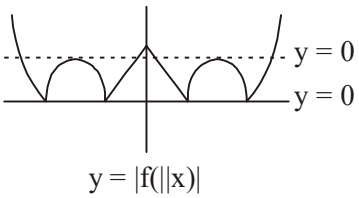
Sol. No. of points of intersection of a line $y = c; 0 < c < 4$ is 8.



$y = x^2 - 6x + 5$



$y = f(|x|)$



ONE OR MORE THAN ONE CORRECT TYPE

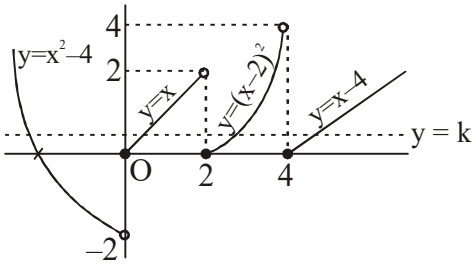
29. Consider $f(x)=\begin{cases} x^2-2 & x<0 \\ x & 0\le x<2 \\ (x-2)^2 & 2\le x<4 \\ x-4 & x\ge 4 \end{cases}$. If the equation $f(x)=k$ has four roots, then k can be equal to :

माना $f(x)=\begin{cases} x^2-2 & x<0 \\ x & 0\le x<2 \\ (x-2)^2 & 2\le x<4 \\ x-4 & x\ge 4 \end{cases}$ है। यदि समीकरण $f(x)=k$ के चार मूल है, तो k का मान हो सकता है :

(A) –2 (B) 0 (C) 1 (D) 2

Ans. BC

Sol.



So $0 \le k < 2$

Therefore $k \in \{0,1\}$

From graph obviously option (B) & (C) are correct.

30. Let $f(x) = x^2 + 4x - 1$ and $g(x) = |x|$. If $h(x) = f(g(x)) + 10$, then the range of $h(x)$, is subset of:

माना $f(x) = x^2 + 4x - 1$ तथा $g(x) = |x|$. यदि $h(x) = f(g(x)) + 10$, तब h(x) का परिसर निम्न में से किसका उपसमुच्चय है :

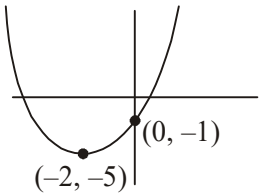
(A) $\{y : y \ge 10\}$ (B) $\{y : y \ge 0\}$ (C) $\{y : y \ge 5\}$ (D) $\{y : y \ge 9\}$

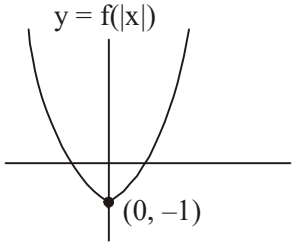
Ans. BCD

Sol. $h(x) = f(g(x)) + 10 = |x|^2 + 4|x| - 1 + 10 = |x|^2 + 4|x| + 9 = (|x| + 2)^2 + 5$

Hence range of h (x) is $[9, \infty)$

Minimum value occurs when $x = 0$





31. Which of following pairs of functions are identical :

(A) $f(x) = e^{\ell n \sec^{-1} x}$ and $g(x) = \sec^{-1} x$ (B) $f(x) = \tan(\tan^{-1} x)$ & $g(x) = \cot(\cot^{-1} x)$

(C) $f(x) = \text{sgn}(x)$ and $g(x) = \text{sgn}(\text{sgn}(x))$ (D) $f(x) = \cot^2 x . \cos^2 x$ & $g(x) = \cot^2 x - \cos^2 x$

निम्नलिखित फलनों के युग्मो में से कौनसे तत्समक है :

(A) $f(x) = e^{\ell n \sec^{-1} x}$ and $g(x) = \sec^{-1} x$ (B) $f(x) = \tan(\tan^{-1} x)$ & $g(x) = \cot(\cot^{-1} x)$

(C) $f(x) = \text{sgn}(x)$ and $g(x) = \text{sgn}(\text{sgn}(x))$ (D) $f(x) = \cot^2 x . \cos^2 x$ & $g(x) = \cot^2 x - \cos^2 x$

Ans. BCD

Sol. (A) $f(x) = e^{\ell n(\sec^{-1} x)} = \sec^{-1} x, \quad x \in (-\infty, -1] \cup (1, \infty)$

$g(x) = \sec^{-1} x, \quad x \in (-\infty, -1] \cup [1, \infty)$

non-identical functions तुल्य फलन नहीं है।

(B) $f(x) = \tan(\tan^{-1} x) = x, x \in R$

$g(x) = \cot(\cot^{-1} x) = x, x \in R$

identical functions तुल्य फलन है।

(C) $f(x) = \text{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \Rightarrow g(x) = \text{sgn}(\text{sgn}\, x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

Identical functions तुल्य फलन है।

(D) $f(x) = \cot^2 x . \cos^2 x, \quad x \in R - \{n\pi\}, n \in I$

$g(x) = \cot^2 x - \cos^2 x$

$= \cot^2 x (1 - \sin^2 x) = \cot^2 x . \cos^2 x \qquad x \in R - \{n\pi\}, n \in I$

Identical functions तुल्य फलन है।

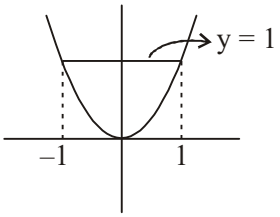
32. $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective :

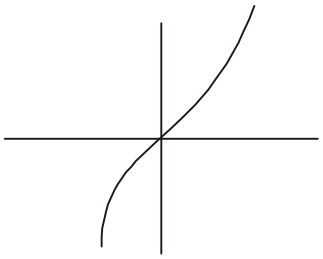
निम्नलिखित फलनों का प्रान्त $D \equiv [-1, 1]$ है, इनमें से कौनसे फलन एकैकी है :

(A) $f(x) = x^2$ (B) $g(x) = x^3$ (C) $h(x) = \sin 2x$ (D) $k(x) = \sin(\pi x/2)$

Ans. BD

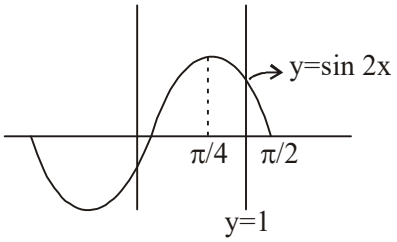
Sol. (A) $f(x) = x^2$ (B) $g(x) = x^3$





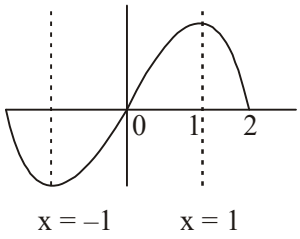
Not one – one one – one

(C)



Not one – one

(D) $k(x) = \sin\left(\frac{\pi x}{2}\right)$



one – one

33. The functions which are aperiodic are : (where [x] denotes greatest integer function)

वे फलन जो अनावर्ती है : (जहाँ [x] महत्तम पूर्णांक फलन को प्रदर्शित करता है ।)

(A) $y = [x + 1]$ (B) $y = \sin x^2$ (C) $y = \sin^2 x$ (D) $y = \sin^{-1} x$

Ans. ABD

Sol. (A) $y = [x + 1] \rightarrow$ aperiodic

(B) $y = \sin x^2 \rightarrow$ aperiodic

(C) $y = \sin^2 x \rightarrow$ periodic with period π.

(D) $y = \sin^{-1} x \rightarrow$ aperiodic

34. If $f: R \rightarrow [-1, 1]$, where $f(x) = \sin\frac{\pi}{2}[x]$, (where [ ] denotes the greatest integer function) then :

(A) $f(x)$ is onto (B) $f(x)$ is into (C) $f(x)$ is periodic (D) $f(x)$ is many one

यदि f: $R \to [-1, 1]$ जहाँ $f(x) = \sin\frac{\pi}{2}[x]$, (जहाँ [ ] महत्तम पूर्णांक फलन को प्रदर्शित करता है) तो :

(A) f(x) आच्छादक है (B) f(x) अन्तर्क्षेपी है (C) f(x) आवर्ती है (D) f(x) बहुएकैकी है

Ans. BCD

Sol. $f: R \to [-1, 1] \quad \Rightarrow \quad f(x) = \sin\left(\frac{\pi}{2}[x]\right) = \begin{cases} -1 & , \quad -1 \le x < 0 \\ 0 & , \quad 0 \le x < 1 \\ 1 & , \quad 1 \le x \le 2 \end{cases}$

Many - one function बहुएकैकी फलन है।

into function अन्तर्क्षेपी फलन

Also साथ ही $f(x+4) = \sin\left(\frac{\pi}{2}[x+4]\right) = \sin\left(2\pi + \frac{\pi}{2}[x]\right) = \sin\left(\frac{\pi}{2}[x]\right)$

$= f(x)$ and hence periodic अतः यह आवृत्ति फलन है।

35. Identify the statement(s) which is/are incorrect?

(A) the function $f(x) = \cos(\cos^{-1}x)$ is neither odd nor even

(B) the fundamental period of $f(x) = \cos(\sin x) + \cos(\cos x)$ is π

(C) the range of the function $f(x) = \cos(3\sin x)$ is $[-1, 1]$

(D) none of these

गलत कथनों की पहचान कीजिए?

(A) फलन $f(x) = \cos(\cos^{-1}x)$ न तो विषम है न ही सम

(B) $f(x) = \cos(\sin x) + \cos(\cos x)$ का मूलभूत आर्वतकाल π है

(C) फलन $f(x) = \cos(3\sin x)$ का परिसर $[-1, 1]$ है

(D) इनमें से कोई नहीं

Ans. ABC

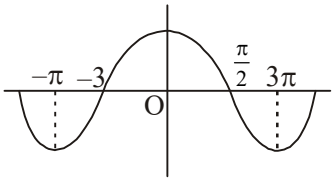
Sol. (A) $f(x) = \cos(\cos^{-1}x) = x \to$ odd

(B) $f(x) = \underbrace{\cos(\sin x)}_{\text{period}=2\pi} + \underbrace{\cos(\cos x)}_{\text{period}=2\pi}$

$\Rightarrow$ LCM $(2\pi, 2\pi) = 2\pi$

$\Rightarrow$ Period is 2π.

(C) $f(x) = \cos\left(\underbrace{3\sin x}_{\in[-3,3]}\right)$



$\Rightarrow f(x) \in [\cos 3, 1]$

36. If $F(x) = \dfrac{\sin \pi [x]}{\{x\}}$, then F(x) is :

(A) periodic with fundamental period 1

(B) even

(C) range is singleton

(D) identical to $\text{sgn}\left(\text{sgn}\ \dfrac{\{x\}}{\sqrt{\{x\}}}\right) - 1$, where {x} denotes fractional part function and [ ] denotes greatest integer function and sgn (x) is a signum function

यदि $F(x) = \dfrac{\sin \pi [x]}{\{x\}}$, हो तो F(x) :

(A) मूलभूत आवर्तकाल 1 का आवर्ती फलन है

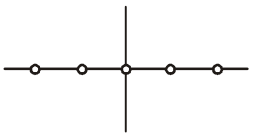
(B) सम फलन है

(C) एकल परिसर है

(D) $\text{sgn}\left(\text{sgn}\ \dfrac{\{x\}}{\sqrt{\{x\}}}\right) - 1$, के समरूप है, जहाँ {x} फलन के भिन्नात्मक भाग को प्रदर्शित करता है तथा [ ] महत्तम पूर्णांक फलन को प्रदर्शित करता है एवं sgn (x) एक सिग्नम फलन है।

Ans. ABCD

Sol. $f(x) = \dfrac{\sin(\pi[x])}{\{x\}} = 0\ ,\ x \notin I$



(A) By graph fundamental period is one

(B) $f(-x) = 0 = f(x)$ $\therefore$ even function

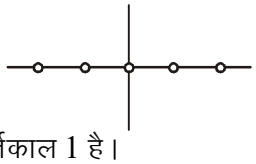
(C) Range $y \in \{0\}$

(D) $y = \text{sgn}\left(\text{sgn}\dfrac{\{x\}}{\sqrt{\{x\}}}\right) - 1, \quad x \notin I$

$y = \text{sgn}\,(1) - 1 \Rightarrow y = 1 - 1$

$y = 0, \quad x \notin I$ Identical to f(x)

Hindi $f(x) = \dfrac{\sin(\pi[x])}{\{x\}} = 0\ ,\ x \notin I$



(A) ग्राफ से यह स्पष्ट है कि मूलभूत आवर्तकाल 1 है।

(B) $f(-x) = 0 = f(x)$ $\therefore$ समफलन

(C) परिसर $y \in \{0\}$

(D) $y = \text{sgn}\left(\text{sgn}\dfrac{\{x\}}{\sqrt{\{x\}}}\right) - 1, \quad x \notin I$

$y = \text{sgn}(1) - 1 \quad \Rightarrow \quad y = 1 - 1$

$y = 0, \quad x \notin I$ $\qquad$ f(x) के समान

37. Function $f(x) = \sin x + \tan x + \text{sgn}(x^2 - 6x + 10)$ is :

(A) periodic with period 2π (B) periodic with period π

(C) Non-periodic (D) periodic with period 4π

फलन $f(x) = \sin x + \tan x + \text{sgn}(x^2 - 6x + 10)$ ह :

(A) 2π आवर्त का आवर्ती फलन (B) π आवर्त का आवर्ती फलन

(C) अनावर्ती फलन (D) 4π आवर्त का आवर्ती फलन

Ans. AD

Sol. $f(x) = \sin x + \tan x + \text{sgn}\underbrace{\left(x^2 - 6x + 10\right)}_{>0}$

$\Rightarrow f(x) = \underbrace{\sin x}_{P=\pi} + \underbrace{\tan x}_{P=\pi} + \underset{\text{No Fundamental Period}}{1}$

$\Rightarrow$ LCM $(2\pi, \pi, \pi) = 2\pi$

$\Rightarrow$ f(x) is periodic with period 2π or 4π or 6π.....

38. Let $f : R \to R - \{3\}$ is a function satisfying $f(x + \alpha) = \dfrac{f(x) - 5}{f(x) - 3}$ for all $x \in R$, ten

(A) f is periodic with period 4α (B) f is into

(C) f is constant (D) f can not take the value 1

माना कि $f : R \to R - \{3\}$ एक फलन है जो कि $f(x + \alpha) = \dfrac{f(x) - 5}{f(x) - 3}$ को x के सभी वास्तविक मानों के लिए संतुष्ट करता है, तो :

(A) f आवर्ती है तथा 4α आवर्तकाल है (B) f अन्तर्क्षेपी है

(C) f अचर है (D) f का मान 1 नहीं हो सकता

Ans. ABD

Sol. $f(x + \alpha) = \dfrac{f(x) - 5}{f(x) - 3}; f(x) \neq 3$

$$f(x+2\alpha)=\frac{f(x+\alpha)-5}{f(x+\alpha)-3}=\frac{\frac{f(x)-5}{f(x)-3}-5}{\frac{f(x)-5}{f(x)-3}-3}$$

$$\Rightarrow f(x+2\alpha)=\frac{-4f(x)+10}{-2f(x)+4}=\frac{2f(x)-5}{f(x)-2};f(x\neq 2)$$

$$f(x+3\alpha)=\frac{2\left(\frac{f(x)-5}{f(x)-3}\right)-5}{\frac{f(x)-5}{f(x)-3}-2}=\frac{-3f(x)+5}{-f(x)+1}$$

$$\Rightarrow f(x+3\alpha)=\frac{3f(x)-5}{f(x)-1};f(x)\neq 1$$

$$f(x+4\alpha)=\frac{3\left(\frac{f(x)-5}{f(x)-3}\right)-5}{\frac{f(x)-5}{f(x)-3}-1}=\frac{-2f(x)}{-2}=f(x)$$

$\therefore$ f(x) is periodic with period 4α.

f cannot take the value 1,2 & 3.

$\Rightarrow$ f is into.

f cannot be constant.

If f(x) = a then

$a^2 - 3a - a + 5 = 0$

$$\underbrace{a^2-4a+5=0}_{\text{real no. a}}$$

MATCH THE COLUMN

39. Let $f(x) = \ln x$ and $g(x) = x^2 - 1$

Column-I contains composite functions and column-II contains their domain. Match the entries of column-I with their corresponding answer is column-II :

माना $f(x) = \ln x$ तथा $g(x) = x^2 - 1$

स्तम्भ– I में फलन तथा स्तम्भ–II इनके प्रान्त है। सही स्तम्भों का मिलान कीजिए :

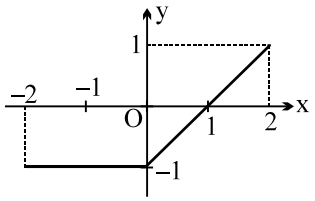
Column-I	**Column-II**
(A) fog	(P) $(1, \infty)$
(B) gof	(Q) $(-\infty, \infty)$
(C) fof	(R) $(-\infty, -1) \cup (1, \infty)$
(D) gog	(S) $(0, \infty)$

Ans. (A) R (B) S (C) P (D) Q

Sol. $f(x) = \ln x$

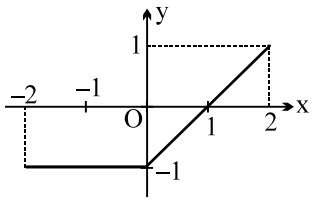
$g(x) = x^2 - 1$	Domain
$fog(x) = \ln(x^2 - 1)$	$x \in (-\infty, -1) \cup (1, \infty)$
$gof(x) = (\ln(x))^2 - 1$	$x \in (0, \infty)$
$fof(x) = \ln(\ln x)$	$x \in (1, \infty)$
$gog(x) = (x^2 - 1)^2 - 1$	$x \in (-\infty, \infty)$

40. The graph of the function $y = f(x)$ is as follows.

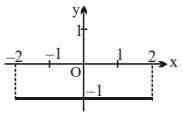
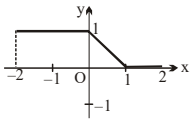
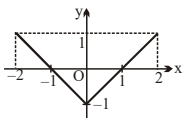
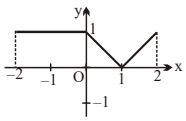


Match the *function* mentioned in Column-I with the respective *graph* given in Column-II.:

$y = f(x)$ का आरेख दिया गया है :

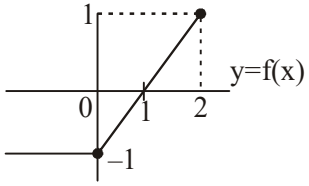


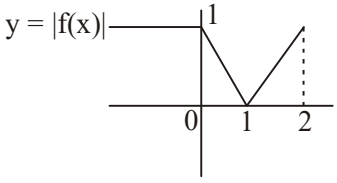
स्तम्भ– I में फलन तथा स्तम्भ–II इनके आरेख है। सही स्तम्भों का मिलान करें।

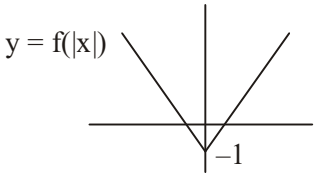
Column-I	**Column-II**
(A) $y = \lvert f(x) \rvert$	(P) 
(B) $y = f(\lvert x \rvert)$	(Q) 
(C) $y = f(-\lvert x \rvert)$	(R) 
(D) $y = \frac{1}{2}(\lvert f(x) \rvert - f(x))$	(S) 

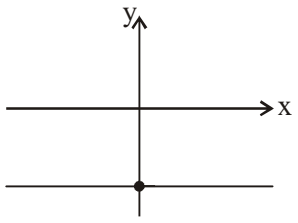
Ans. (A) S (B) R (C) P (D) Q

Sol.

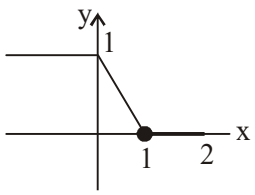








$$y = \frac{1}{2}\left(\left|f\left(x\right)\right| - f\left(x\right)\right)$$



41. Column I contains functions and column II contains their natural domains. Exactly one entry of column II matches with exactly one entry of column I :

Column I में फलन व column II में प्रान्त है। सही स्तम्भ का मिलाने कीजिए :

Column-I	**Column-II**
(A) $f(x) = \sin^{-1}\left(\frac{x+1}{x}\right)$	(P) $(1, 3) \cup (3, \infty)$
(B) $g(x) = \sqrt{ln\left(\frac{x^2+3x-2}{x+1}\right)}$	(Q) $(-\infty, 2)$
(C) $h(x) = \frac{1}{ln\left(\frac{x-1}{2}\right)}$	(R) $\left(-\infty, -\frac{1}{2}\right]$
(D) $\phi(x) = ln\left(\sqrt{x^2+12} - 2x\right)$	(S) $[-3, -1) \cup [1, \infty)$

Ans. (A) R (B) S (C) P (D) Q

Sol. (A) $f(x) = \sin^{-1}\left(\frac{x+1}{x}\right)$

$$-1 \le \frac{x+1}{x} \le 1$$

$$\frac{2x+1}{x} \ge 0 \qquad \frac{1}{x} \le 0$$

$$x \in \left(-\infty. -\frac{1}{2}\right] \cup (0, \infty) \qquad x < 0$$

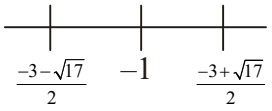
$$x \in \left(-\infty, -\frac{1}{2}\right)$$

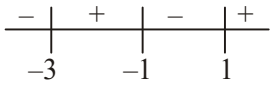
(B) $g(x)=\sqrt{\ln\left(\frac{x^2+3x-2}{x+1}\right)}$

$\ln\left(\frac{x^2+3x-2}{x+1}\right)\geq 0$ & $\frac{x^2+3x-2}{x+1}>0$

Taking antilog

$\frac{x^2+3x-2-x-1}{x+1}\geq 0$





$x\in\left(\frac{-3-\sqrt{17}}{2},-1\right)\cup\left(\frac{-3+\sqrt{17}}{2},\infty\right)$

$\Rightarrow x\in[-3,-1)\cup[1,\infty)$

$\Rightarrow$ Domain $=[-3,-1)\cup[1,\infty)$

(C) $h(x)=\frac{1}{\ln\left(\frac{x-1}{2}\right)}$

$\ln\left(\frac{x-1}{2}\right)\neq 0$ & $\frac{x-1}{2}>0$

$\Rightarrow \frac{x-1}{2}\neq 1$ & $x>1$

$\Rightarrow x\in(1,3)\cup(3,\infty)$

(D) $\phi(x)=\ln\left(\sqrt{x^2+12}-2x\right)$

$\sqrt{x^2+12}-2x>0$

$\Rightarrow \sqrt{x^2+12}>2x$

If $x\leq 0$ then it is always true. $\Rightarrow x\in(-\infty,0]$ ...(1)

If $x>0$ then $x^2+12>4x^2$

$\Rightarrow 3x^2<12$

$\Rightarrow x^2 < 4$

$\Rightarrow (x - 2)(x + 2) < 0$

$\Rightarrow x \in (0,2)$ ...(2)

$\Rightarrow$ Domain $= (-\infty,2)$

Exercise J-M

1. The equation $x^2 - 4x + [x] + 3 = x[x]$, where [x] denotes the greatest integer function, has :

[JEE(Main)Jan.2023]

(A) no solution (B) a unique solution in $(-\infty, \infty)$

(C) a unique solution in $(-\infty, 1)$ (D) exactly two solution in $(-\infty, \infty)$

समीकरण $x^2 - 4x + [x] + 3 = x[x]$, जहाँ [x] महत्तम पूर्णांक फलन है : **[JEE(Main)Jan.2023]**

(A) का कोई हल नहीं है (B) का $(-\infty, \infty)$ में ठीक एक हल है

(C) का $(-\infty, 1)$ में ठीक एक हल है (D) का $(-\infty, \infty)$ में ठीक दो हल है

Ans. B

Sol. $x^2 - 4x + [x] + 3 = x[x]$

$\Rightarrow x^2 - 4x + 3 = x[x] - [x]$

$\Rightarrow (x-1)(x-3) = [x] \cdot (x-1)$

$\Rightarrow x = 1$ or $x - 3 = [x]$

$\Rightarrow x - [x] = 3$

$\Rightarrow \{x\} = 3$ (Not Possible)

Only one solution $x = 1$ in $(-\infty, \infty)$

2. Let f(x) be a function such that $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in N$. If $f(1) = 3$ and $\sum_{k=1}^{n} f(k) = 3279$, then the value of n is : **[JEE(Main)Jan.2023]**

माना एक फलन f(x) इस प्रकार है कि सभी $x, y \in N$ के लिए $f(x+y) = f(x) \cdot f(y)$ है। यदि $f(1) = 3$ और $\sum_{k=1}^{n} f(k) = 3279$ है, तब n का मान है : **[JEE(Main)Jan.2023]**

(A) 9 (B) 8 (C) 7 (D) 6

Ans. C

Sol. $f(x+y) = f(x) \cdot f(y) \forall x, y \in N, f(1) = 3$

$f(2) = f^2(1) = 3^2$

$f(3) = f(1)f(2) = 3^3$

$f(4) = 3^4$

$f(k) = 3^k$

$\sum_{k=1}^{n} f(k) = 3279$

$f(1) + f(2) + f(3) + \ldots\ldots\ldots + f(k) = 3279$

$3 + 3^2 + 3^3 + \ldots\ldots\ldots 3^k = 3279$

$$\frac{3\left(3^k-1\right)}{3-1}=3279$$

$$\frac{3^k-1}{2}=1093$$

$$3^k-1=2186$$

$$3^k=2187$$

$$\boxed{k=7}$$

3. If $f(x)=\frac{2^{2x}}{2^{2x}+2}$, $x\in R$, then $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+.....+f\left(\frac{2022}{2023}\right)$ is equal to :

[JEE(Main)Jan.2023]

यदि $f(x)=\frac{2^{2x}}{2^{2x}+2}$, $x\in R$ है, तब $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+.....+f\left(\frac{2022}{2023}\right)$ बराबर है :

[JEE(Main)Jan.2023]

(A) 2010 (B) 1010 (C) 2011 (D) 1011

Ans. D

Sol. $f(x)=\frac{4^x}{4^x+2}$

$$f(x)+f(1-x)=\frac{4^x}{4^x+2}+\frac{4^{1-x}}{4^{1-x}+2}$$

$$=\frac{4^x}{4^x+2}+\frac{4}{4+2\left(4^x\right)}$$

$$=\frac{4^x}{4^x+2}+\frac{2}{2+4^x}$$

$= 1$

$\Rightarrow f(x) + f(1-x) = 1$

Now $f\left(\frac{1}{2023}\right)+f\left(\frac{2}{2023}\right)+f\left(\frac{3}{2023}\right)+.....+f\left(1-\frac{3}{2023}\right)+f\left(1-\frac{2}{2023}\right)+f\left(1-\frac{1}{2023}\right)$

Now sum of terms equidistant from beginning and end is 1

Sum = 1 + 1 + 1 + …….+ 1 (1011 times)

= 1011

4. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by

$f(x)=\log_{\sqrt{m}}\left\{\sqrt{2}(\sin x-\cos x)+m-2\right\}$, for some m, such that the range of f is [0,2]. Then the value of m is _____.

[JEE(Main)Jan.2023]

माना फलन $f:\mathbb{R}\to\mathbb{R}$, किसी m के लिए

$f(x) = \log_{\sqrt{m}}\left\{\sqrt{2}(\sin x - \cos x) + m - 2\right\}$ द्वारा परिभाषित है तथा f का परिसर [0, 2] है। तो m का मान है।

[JEE(Main)Jan.2023]

(A) 3 (B) 2 (C) 5 (D) 4

Ans. C

Sol. $\log_{\sqrt{5}}(\sqrt{2}(\sin x - \cos x) + m - 2) \in [0,2] \in [\log_{\sqrt{5}} 1, \log_{\sqrt{5}} 5]$

$\Rightarrow \sqrt{2}(\sin x - \cos x) + m - 2 \in [1, m]$

Range of $(\sin x - \cos x)$ is $[-\sqrt{2}, \sqrt{2}]$

So, from (i)

$[-4 + m, m] \to [1, m] \Rightarrow -4 + m = 1 \Rightarrow m = 5$

5. Let $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in N$, and $f(4) = 133, f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is : **[JEE(Main)Jan.2023]**

माना $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in N$, और $f(4) = 133, f(5) = 255$ है। तो $(f(3) - f(2))$ के सभी धनात्मक पूर्णांक भाजकों का योग है : **[JEE(Main)Jan.2023]**

(A) 60 (B) 58 (C) 59 (D) 61

Ans. A

Sol. $f(x) = 2x^n + \lambda$

$f(4) = 133$

$f(5) = 255$

$133 = 2 \times 4^n + \lambda$.... (1)

$255 = 2 \times 5^n + \lambda$.... (2)

(B) – (1)

$122 = 2(5^n - 4^n)$

$\Rightarrow 5^n - 4^n = 61$

$\therefore n = 3$ & $\lambda = 5$

Now, $f(3) - f(2) = 2(3^3 - 2^3) = 38$

divisors = 1,2,19,38

Sum = 60

6. The number of functions $f: \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$ satisfying $f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ is :

[JEE(Main)Jan.2023]

$f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1,2,3\}$ को संतुष्ट करने वाले फलनों $f: \{1,2,3,4\} \to \{a \in \mathbb{Z} : |a| \le 8\}$ की संख्या है

[JEE(Main)Jan.2023]

(A) 3 (B) 2 (C) 4 (D) 1

Ans. B

Sol. $nf(n) + f(n+1) = n$

$f(1) + f(2) = 1$

$2f(2) + f(3) = 2$

$3f(3) + f(4) = 3$

$2.(1-f(1)) + f(3) = 2$

$f(3) = 2f(1)$

$3.2f(1)+f(4) = 3$

$f(4) = 3 - 6\, f(1) \le 8$

$-8 \le f(4) \le 8$

$-8 \le 3 - 6\, f(1) \le 8$

$-11 \le -6f(1) \le 5$

$$-\frac{5}{6} \le f(1) \le \frac{11}{6}$$

$f(1) = 0, 1$

C–I: $f(1) = 0 , \Rightarrow f(2) = 1$

$f(3) = 0 , f(4) = 3$

C–II: $f(1) = 1 , \Rightarrow f(2) = 0$

$\Rightarrow f(3) = 2 , f(4) = -3$

so '2' such functions.

7. The domain of $f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in R$ is : **[JEE(Main)Jan.2023]**

$f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in R$ का प्रांत है : **[JEE(Main)Jan.2023]**

(A) $R - \{-1,3\}$ (B) $(-1,\infty) - \{3\}$ (C) $R - \{3\}$ (D) $(2,\infty) - \{3\}$

Ans. D

Sol. $\therefore f(x) = \dfrac{\log_{(x+1)}(x-2)}{e^{2\log x} - (2x+3)}$

(i) $x - 2 > 0$

$x > 2$

$x \in (2, \infty)$

(ii) $x + 1 > 0$ & $x + 1 \ne 1$

$\Rightarrow x > -1$ $\Rightarrow x \ne 0$

$x \in (-1, 0) \cup (0, \infty)$

(iii) $x > 0$

$x \in (0, \infty)$

(iv) $e^{2\ell nx} - (2x+3) \Rightarrow x^2 - 2x - 3 \neq 0$

$\Rightarrow (x-3)(x+1) \neq 0$

$\Rightarrow x \neq 3, x \neq -1$

from (i) $\cap$ (ii) $\cap$ (iii) $\cap$ (iv)

$x \in (2, \infty) - \{3\}$

8. Consider a function $f: \mathbb{N} \to \mathbb{R}$, satisfying

$f(1) + 2f(2) + 3f(3) + ... + xf(x) = x(x+1)f(x);\ x \geq 2$ with $f(1) = 1$.

Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to : **[JEE(Main)Jan.2023]**

एक फलन $f: \mathbb{N} \to \mathbb{R}$, के लिए $f(1) + 2f(2) + 3f(3) + ... + xf(x) = x(x+1)f(x);\ x \geq 2$ तथा $f(1) = 1$ है। तो $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 8000 (B) 8400 (C) 8100 (D) 8200

Ans. C

Sol. f(1) + 2f(2) + 3f(3) …… n f(n) = n (n +1) f(n) ……...(1)

replace in by n + 1

f(1) + 2f(2) + 3f(3) …… nf(n) + (n + 1) f(n+1) = (n + 1) (n +2) f(n +1) ……(2)

Subtracting (1) from (2)

(n +1) f(n +1) = (n +1) (n +2) f(n +1) – n(n +1) f(n) or (n +1)f(n+1) – nf(n) = 0 ……...(3)

put n = 2, 3, 4 …… n + 1 in eq. (3)

3f(3) – 2 f(2) = 0

4f(4) –3f(3) = 0

,,

(n +1) f(n+1) – n f(n) = 0

Add $\Rightarrow$ (n + 1) f(n + 1) = 2f(2)

$\Rightarrow \quad f(n+1) = \frac{2f(2)}{n+1}$ .....(4)

again put n = 2 in eq. (1)

f(1) + 2f(2) = 6f(2)

$\Rightarrow 1 = 4f(2) \Rightarrow f(2) = \frac{1}{4}$

Now by (4) $f(n) = \frac{1}{2n}$

$$\Rightarrow \frac{1}{f(2022)}+\frac{1}{f(2028)}=4044+4056=8100$$

9. The range of the function $f(x)=\sqrt{3-x}+\sqrt{2+x}$ is : **[JEE(Main)Jan.2023]**

फलन $f(x)=\sqrt{3-x}+\sqrt{2+x}$ का परिसर है : **[JEE(Main)Jan.2023]**

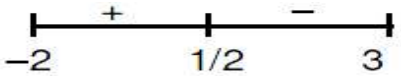
(A) $\left[2\sqrt{2},\sqrt{11}\right]$ (B) $\left[\sqrt{5},\sqrt{13}\right]$ (C) $\left[\sqrt{2},\sqrt{7}\right]$ (D) $\left[\sqrt{5},\sqrt{10}\right]$

Ans. D

Sol. $f'(x)=\frac{1}{2\sqrt{2}+x}-\frac{1}{2\sqrt{3}-x}$

$$\Rightarrow f'(x)=\frac{\sqrt{3-x}-\sqrt{2+x}}{2\sqrt{2+x}\sqrt{3-x}}=0$$

$$\Rightarrow x=\frac{1}{2} \text{ (domain is } -2\le x\le 3)$$



So at $x=\frac{1}{2}$, maxima

$$f(-2)=\sqrt{5}, f\left(\frac{1}{2}\right)=\sqrt{10}, f(3)=\sqrt{5}$$

range $\in \left[\sqrt{5},\sqrt{10}\right]$

10. If the domain of the function $f(x)=\frac{[x]}{1+x^2}$, where [x] is greatest integer $\le x$, is [2,6), then its range is : **[JEE(Main)Jan.2023]**

यदि फलन $f(x)=\frac{[x]}{1+x^2}$ जहाँ [x] महत्तम पूर्णांक $\le x$ है का प्रांत [2,6) है, तो इसका परिसर है **[JEE(Main)Jan.2023]**

(A) $\left(\frac{5}{26},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$ (B) $\left(\frac{5}{26},\frac{2}{5}\right]$

(C) $\left(\frac{5}{37},\frac{2}{5}\right]$ (D) $\left(\frac{5}{37},\frac{2}{5}\right]-\left\{\frac{9}{29},\frac{27}{109},\frac{18}{89},\frac{9}{53}\right\}$

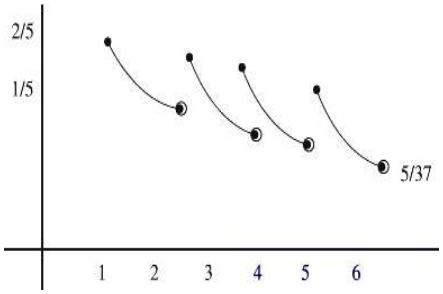
Ans. C

Sol. $f(x)=\frac{2}{1+x^2}$ $x\in[2,3)$

$f(x)=\frac{3}{1+x^2}$ $x\in[3,4)$

$$f(x)=\frac{4}{1+x^2} \qquad x \in [4, 5)$$

$$f(x)=\frac{5}{1+x^2} \qquad x \in [5, 6)$$



$$\left(\frac{5}{37}, \frac{2}{5}\right]$$

11. The absolute minimum value, of the function $f(x) = |x^2 - x + 1| + [x^2 - x + 1]$, where [t] denotes the greatest integer function, in the interval $[-1,2]$, is : **[JEE(Main)Jan.2023]**

अंतराल $[-1, 2]$ में फलन $f(x) = |x^2 - x + 1| + [x^2 - x + 1]$ जहाँ [t] महत्तम पूर्णांक फलन है, का निरपेक्ष न्यूनतम मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Ans. B

Sol. $f(x)=\left|x^2-x+1\right|+\left[x^2-x+1\right]; x \in [-1,2]$

Let $g(x) = x^2 - x + 1$

$$=\left(x-\frac{1}{2}\right)^2+\frac{3}{4}$$

$\because \left|x^2-x+1\right|$ and $\left[x^2-x+2\right]$

Both have minimum value at $x=\frac{1}{2}$

$\Rightarrow$ Minimum $f(x) = \frac{3}{4}+0$

$$=\frac{3}{4}$$

12. Let $f : R - \{0,1\} \to R$ be a function such that $f(x)+f\left(\frac{1}{1-x}\right)=1+x$. Then $f(2)$ is equal to : **[JEE(Main)Jan.2023]**

माना फलन $f:R-\{0,1\}\to R$ इस प्रकार है कि $f(x)+f\left(\frac{1}{1-x}\right)=1+x$ है। तो $f(2)$ बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{9}{4}$ (B) $\frac{7}{4}$ (C) $\frac{9}{2}$ (D) $\frac{7}{3}$

Ans. A

Sol. , $x=2\Rightarrow f(2)+f(-1)=3$ ...(1)

, $x=\frac{1}{2}\Rightarrow f\left(\frac{1}{2}\right)+f(2)=\frac{3}{2}$ ...(2)

, $x=-1\Rightarrow f(-1)+f\left(\frac{1}{2}\right)=0$ ...(3)

(B) ..... (3) $\Rightarrow f(2)-f(-1)=\frac{3}{2}$ ...(4)

$(1)+(4)\Rightarrow 2f(2)=\frac{9}{2}\Rightarrow f(2)=\frac{9}{4}$

13. Let $S=\{1,2,3,4,5,6\}$. Then the number of one-one function $f: S\to P(S)$, where P(S) denote the power set of S, such that $f(n)\subset f(m)$ where $n<m$ is _____ . **[JEE(Main)Jan.2023]**

माना $S=\{1,2,3,4,5,6\}$ है तो ऐसे एकैकी फलनों $f: S\to P(S)$, जहाँ P(S) समुच्चय S का घात समुच्चय $f(n)\subset f(m)$ है जब कि $n<m$ है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 3240

Sol. Let $S=\{1, 2, 3, 4, 5, 6\}$, then the number of one-one functions, $f: S\bullet P(S)$, where P(S) denotes the power set of S, such that $f(n)<f(m)$ where $n<m$ is

$n(S)=6$

$P(S)=\{\phi,\{1\},.....\{6\},\{1,2\},....,\{5.6\},.....,\{1,2,3,4,5,6,\}\}$

–64 elements

case – 1

$f(6)$ = S i.e. 1 option,

$f(5)$ = any 5 element subset A of S i.e. 6 options,

$f(4)$ = any 4 element subset B of A i.e. 5 options,

$f(3)$ = any 3 element subset C of B i.e. 4 options,

$f(2)$ = any 2 element subset D of C i.e. 3 options,

$f(1)$ = any 1 element subset E of D or empty subset i.e. 3 options,

Total functions = 1080

Case – 2

$f(6)$ = any 5 element subset A of S i.e. 6 options,

$f(5)$ = any 4 element subset B of A i.e. 5 options,

f? (4) = any 3 element subset C of B i.e. 4 options,

$f(3)$ = any 2 element subset D of C i.e. 3 options,

f? (2) = any 1 element subset E of D i.e. 2 options,
f (1) = empty subset i.e. 1 option
Total functions = 720
Case – 3
f(6)= S
f(5) = any 4 element subset A of ' S i.e. 15 options,
f(4) = any 3 element subset B of A i.e. 4 options,
f(3) = any 2 element subset C of B i.e. 3 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 4
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 3 element subset B of A i.e. 10 options,
f(3) = any 2 element subset C of B i.e. 3 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 5
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 4 element subset B of A i.e. 5 options,
f(3) = any 2 element subset C of B i.e. 6 options,
f(2) = any 1 element subset D of C i.e. 2 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
Case – 6
f(6) = S
f(5) = any 5 element subset A of S i.e. 6 options,
f(4) = any 4 element subset B of A i.e. 5 options,
f(3) = any 3 element subset C of B i.e. 4 options,
f(2) = any 1 element subset D of C i.e. 3 options,
f(1) = empty subset i.e. 1 option
Total functions = 360
$\therefore$ Number of such functions = 3240

14. For some a, b, c $\in$ N, let f(x) = ax – 3 and g(x) = x^b + c, x $\in$ R. If $\left(fog\right)^{-1}\left(x\right)=\left(\frac{x-7}{2}\right)^{\frac{1}{3}}$, then (fog) (ac) + (gof) (b) is equal to _____. **[JEE(Main)Jan.2023]**

किसी a, b, c $\in$ N के लिए, माना f(x) = ax – 3 तथा g(x) = x^b + c, x $\in$ R है। यदि $\left(fog\right)^{-1}\left(x\right)=\left(\frac{x-7}{2}\right)^{\frac{1}{3}}$ है, तो (fog) (ac) + (gof) (b) बराबर है। **[JEE(Main)Jan.2023]**

Ans. 2039

Sol. Let $fog(x) = h(x)$

$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$

$\Rightarrow h(x) = fog(x) = 2x^3 + 7$

$fog(x) = a(xb + c) - 3$

$\Rightarrow a = 2, b = 3, c = 5$

$\Rightarrow fog(ac) = fog(10) = 2007$

$g(f(x) = (2x - 3)^3 + 5$

$\Rightarrow gof(b) = gof(3) = 32$

$\Rightarrow sum = 2039$

15. Suppose f is a function satisfying $f(x+y) = f(x) + f(y)$ for all, $x, y \in \mathbb{N}$ and $f(1) = \frac{1}{5}$. If $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$, then m is equal to _____. **[JEE(Main)Jan.2023]**

माना f एक फलन है जो सभी $x, y \in \mathbb{N}$ के लिए $f(x + y) = f(x) + f(y)$ को संतुष्ट करता है एवं $f(1) = \frac{1}{5}$ है। यदि $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$ है, तब m बराबर है। **[JEE(Main)Jan.2023]**

Ans. 10

Sol. $\because f(1) = \frac{1}{5} \therefore f(2) = f(1) + f(1) = \frac{2}{5}$

$f(2) = \frac{2}{5}$ $f(3) = f(2) + f(1) = \frac{3}{5}$

$f(3) = \frac{3}{5}$

$\therefore \sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)}$

$= \frac{1}{5} \sum_{n=1}^{m} \left(\frac{1}{n+1} - \frac{1}{n+2}\right)$

$= \frac{1}{5}\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + ... + \frac{1}{m+1} - \frac{1}{m+2}\right)$

$= \frac{1}{5}\left(\frac{1}{2} - \frac{1}{m+2}\right) = \frac{m}{10(m+2)} = \frac{1}{12}$

$\therefore m = 10$

16. Let $f^1(x) = \frac{3x+2}{2x+3}$, $x \in R - \left\{\frac{-3}{2}\right\}$; For $n \geq 2$, define $f^n(x) = f^1 o f^{n-1}(x)$. If $f^5(x) = \frac{ax+b}{bx+a}$, gcd(a,b) = 1, then

a + b is equal to _____ . **[JEE(Main)Jan.2023]**

माना $f^1(x)=\frac{3x+2}{2x+3}, x \in R-\left\{\frac{-3}{2}\right\}$ है। $n \geq 2$, के लिए $f^n(x)=f^1 o f^{n-1}(x)$ द्वारा परिभाषित कीजिए। यदि $f^5(x)=\frac{ax+b}{bx+a}$, gcd(a,b) = 1 है, तो a + b बराबर है। **[JEE(Main)Jan.2023]**

Ans. 3125

Sol. $f^1(x)=\frac{3x+2}{2x+3}$

$\Rightarrow f^2(x)=\frac{13x+12}{12x+13}$

$\Rightarrow f^3(x)=\frac{63x+62}{62x+63}$

$\therefore f^5(x)=\frac{1563x+1562}{1562x+1563}$

a + b = 3125

17. Let A= {1,2,3,5,8,9}. Then the number of possible functions $f: A \to A$ such that $f(m \cdot n)=f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to _____ . **[JEE(Main)Jan.2023]**

माना A= {1,2,3,5,8,9} है। तब संभव फलनों $f: A \to A$, जब कि $m, n \in A$ तथा $m \cdot n \in A$ को संतुष्ट करने वाले सभी m, n के लिए $f(m \cdot n)=f(m) \cdot f(n)$ है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 432

Sol. f(1) = 1; f(9) = f(3) × f(3)

i.e., f(3) = 1 or 3

Total function = 1 × 6 × 2 × 6 × 6 × 1 = 432

18. Let $A=\{x \in \mathbb{R}:[x+3]+[x+4] \leq 3\}$, $B=\left\{x \in \mathbb{R}: 3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$, where [t] denotes greatest integer function. Then : **[JEE(Main)April,2023]**

माना $A=\{x \in \mathbb{R}:[x+3]+[x+4] \leq 3\}$, $B=\left\{x \in \mathbb{R}: 3^x\left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3}<3^{-3x}\right\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है। तब: **[JEE(Main)April,2023]**

(A) A = B (B) $B \subset C, A \neq B$ (C) $A \cap B=\phi$ (D) $A \subset B, \quad A \neq B$

Ans. A

Sol. $[x]+3+[x]+4 \leq 3$

$2[x] \leq -4$

$[x] \leq -2 \Rightarrow x \in (-\infty,-1)$. _____A

$$3^x\left(\frac{3\cdot\frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3} < 3^{-3x}$$

$27 < 3^{-3x}$

$-3x > +3$

$x < -1$ _____B

$A = B$

19. Let the sets A and B denote the domain and range respectively of the function $f(x)=\frac{1}{\sqrt{[x]-x}}$, where [x] denotes the smallest integer greater than or equal to x. Then among the statements :

(S1) : $A \cap B = (1,\infty) - \mathbb{N}$ and

(S2) : $A \cup B = (1,\infty)$ **[JEE(Main)April,2023]**

(A) only (S2) is true (B) neither (S1) nor (S2) is true

(C) only (S1) is true (D) both (S1) and (S2) are true

माना फलन $f(x)=\frac{1}{\sqrt{[x]-x}}$, जहाँ [x] न्यूनतम पूर्णांक $\geq x$ है, के प्रांत तथा परिसर क्रमशः समुच्चय A तथा B है। तो कथनों

(S1) : $A \cap B = (1,\infty) - \mathbb{N}$ तथा

(S2) : $A \cup B = (1,\infty)$ **[JEE(Main)April,2023]**

(A) केवल (S2) सही है (B) न तो (S1) न ही (S2) सही है

(C) केवल (S1) सही है (D) दोनों (S1) तथा (S2) सही है

Ans. C

Sol. $[x] - x > 0 \Rightarrow [x] > x$

$\Rightarrow x \notin I$ So $A = R - I$

Also $0 < [x] - x < 1$ when $x \in I$

So $f(x) \in (1,\infty) \Rightarrow B = (1,\infty)$

$A \cap B = (1,\infty) - N$

$A \cup B = (1,\infty) \cup \{R - I\}$

So S_1 is true & S_2 is false

20. If $f(x)=\frac{\left(\tan 1^\circ\right)x + \log_e(123)}{x\log_e(1234) - \left(\tan 1^\circ\right)}, x > 0$, then the least value of $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ is :

[JEE(Main)April,2023]

यदि $f(x)=\frac{\left(\tan 1^{\circ}\right)x+\log_e(123)}{x\log_e(1234)-\left(\tan 1^{\circ}\right)}, x>0$ है, तो $f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$ का निम्नतम मान है :

[JEE(Main)April,2023]

(A) 2 (B) 8 (C) 4 (D) 0

Ans. C

Sol. $f(x)=\frac{(\tan 1)x+\log 123}{x\log 1234-\tan 1}$

Let $A=\tan 1, B=\log 123, C=\log 1234$

$f(x)=\frac{Ax+B}{xC-A}$

$f(f(x))=\frac{A\left(\frac{Ax+B}{xC-A}\right)+B}{C\left(\frac{Ax+B}{CX-A}\right)-A}$

$=\frac{A^2x+AB+xBC-AB}{ACx+BC-ACx+A^2}$

$=\frac{x\left(A^2+BC\right)}{\left(A^2+BC\right)}=x$

$f(f(x))=x$

$f\left(f\left(\frac{4}{x}\right)\right)=\frac{4}{x}$

$f(f(x))+f\left(f\left(\frac{4}{x}\right)\right)$

$AM \geq GM$

$x+\frac{4}{x}\geq 4$

21. The domain of the function $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$ is : (where [x] denotes the greatest integer less than or equal to x) :

[JEE(Main)April,2023]

फलन $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$, जहाँ [x] महत्तम पूर्णांक $\leq x$ है, का प्रांत है :

[JEE(Main)April,2023]

(A) $(-\infty,-3]\bigcup(5,\infty)$ (B) $(-\infty,-2)\cup[6,\infty)$ (C) $(-\infty,-2)\cup(5,\infty)$ (D) $(-\infty,-3]\cup[6,\infty)$

Ans. B

Sol. $f(x)=\frac{1}{\sqrt{[x]^2-3[x]-10}}$

For Domain $[x]^2-3[x]-10>0$

$\Rightarrow \quad ([x]-5)([x]+2)>0$

$\Rightarrow \quad [x]\in(-\infty,-2)\cup(5,\infty)$

$\therefore \quad x\in(-\infty,-2)\cup[6,\infty)$

22. For $x\in\mathbb{R}$, two real valued functions f(x) and g(x) are such that, $g(x)=\sqrt{x}+1$ and $fog(x)=x+3-\sqrt{x}$. Then f(0) is equal to : **[JEE(Main)April,2023]**

$x\in\mathbb{R}$ के लिए दो वास्तविक फलन f(x) तथा g(x) इस प्रकार है कि $g(x)=\sqrt{x}+1$ तथा $fog(x)=x+3-\sqrt{x}$ हैं। तो f(0) बराबर है : **[JEE(Main)April,2023]**

(A) 5 (B) 0 (C) –3 (D) 1

Ans. (Bonus)

Sol. $g(x)=\sqrt{x}+1$

$fog(x)=x+3-\sqrt{x}$

$=\left(\sqrt{x}+1\right)^2-3\left(\sqrt{x}+1\right)+5$

$= g^2(x) - 3g(x) + 5$

$\Rightarrow f(x) = x^2 - 3x + 5$

$\therefore f(0) = 5$

But, if we consider the domain of the composite function fog(x) then in that case f(0) will not be defined as g(x) cannot be equal to zero.

23. Let R = {a, b, c, d, e} and S = {1, 2, 3, 4}. Total number of onto functions f: R → S such that f(a) ≠ 1, is equal to _____. **[JEE(Main)April,2023]**

माना R = {a, b, c, d, e} तथा S = {1, 2, 3, 4} हैं। आच्छादक फलनों f : R → S जिनके लिए f(a) ≠ 1 है, की कुल संख्या है : **[JEE(Main)April,2023]**

Ans. 384

Ans. Matrix 180

Sol. Total onto function

$\frac{\lfloor 5}{\lfloor 3\lfloor 2}\times\lfloor 4=240$

Now when f(a) = 1

$\lfloor 4+\frac{\lfloor 4}{\lfloor 2\lfloor 2}\times\lfloor 3=24+36=60$

so required $f^n = 240 - 60 = 180$

24. Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where [x] denotes the greatest integer $\le x$. Then $m^2 + mn + n^2$ is equal to _____. **[JEE(Main)April,2023]**

माना समीकरणों $x^2 - 12x + [x] + 31 = 0$ तथा $x^2 - 5|x + 2| - 4 = 0$ के वास्तविक मूलों की संख्या क्रमशः m तथा n है, जहाँ [x] महत्तम पूर्णांक $\le x$ है। तो $m^2 + mn + n^2$ बराबर है : **[JEE(Main)April,2023]**

Ans. 9

Sol. $x^2 - 12x + [x] + 31 = 0$

$\underbrace{x^2 - 12x + 31}_{\ge -5} + [x] = 0$

It could have its solution in [5, 6] but it does not exist as at x = 5 and 6

LHS = 1

m = 0

$x^2 - 5|x + 2| - 4 = 0$

Case 1

$x \ge -2$

$x^2 - 5(x + 2) - 4 = 0$

$x^2 - 5x - 14 = 0 \Rightarrow x = 7, -2$

Case 2

$x < -2$

$x^2 + 5x + 10 - 4 = 0$

$x = -2, -3$

3 solution i.e., $x = -3, -2, 7$

$n = 3$

25. Let A = {1, 2, 3, 4, 5} and B = {1, 2, 3, 4, 5, 6}. Then the number of functions $f : A \to B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to _____. **[JEE(Main)April,2023]**

माना A = {1, 2, 3, 4, 5} तथा B = {1, 2, 3, 4, 5, 6} हैं। तो $f(1) + f(2) = f(4) - 1$ को संतुष्ट करने वाले फलनों $f : A \to B$ की संख्या है _____ **[JEE(Main)April,2023]**

Ans. 360

Sol. $f(1) + f(2) + 1 = f(4)$

$\Rightarrow f(1) + f(2) \le 5$

Let $f(4) = 6$, then $f(1) + f(2) = 5$

So (f_1, f_2) may be (1, 4), (4, 1), (2, 3), (3, 2)

Total ways = 4 . 1 . 6 . 6 = 144

Let f(4) = 5, $f(1) + f(2) = 4$

so (f_1, f_2) may be (1, 3), (3, 1), (2, 2)

Total ways = 3 . 1 . 6 . 6 = 108

Let f(4) = 4, $f(1) + f(2) = 3$

so (f_1, f_2) may be (1, 2), (2, 1)

Total ways = 2 . 1 . 6 . 6 = 72

Let f(4) = 3, f(1) + f(2) = 2

Only way (1, 1)

Total ways = 1 . 1 . 36 = 36

26. Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ is equal to _____.

[JEE(Main)April,2023]

माना $[\alpha]$ महत्तम पूर्णांक $\leq \alpha$ है। तो $\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+\ldots+\left[\sqrt{120}\right]$ बराबर है _____

[JEE(Main)April,2023]

Ans. 825

Sol. $S=\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+....+\left[\sqrt{120}\right]$

$\left[\sqrt{1}\right]\rightarrow\left[\sqrt{3}\right]=1\times3$

$\left[\sqrt{4}\right]\rightarrow\left[\sqrt{8}\right]=2\times5$

$\left[\sqrt{9}\right]\rightarrow\left[\sqrt{15}\right]=3\times7$

$\vdots$

$\left[\sqrt{100}\right]\rightarrow\left[\sqrt{120}\right]=10\times21$

$S=1\times3+2\times5+3\times7+....+10\times21$

$=\sum_{r=1}^{10} r(2r+1)$

$=2\sum_{r=1}^{10} r^2+\sum_{r=1}^{10} r$

$=\frac{2\times10\times11\times21}{6}+\frac{10\times11}{2}$

$= 770 + 55$

$= 825$

27. Let $f(x)=\sum_{k=1}^{10} k\,x^k, x\in\mathbb{R}$. If $2f(2)+f'(2)=119(2)^n+1$ then n is equal to _____.

[JEE(Main)April,2023]

माना $f(x)=\sum_{k=1}^{10} k\,x^k, x\in\mathbb{R}$ है। यदि $2f(2)+f'(2)=119(2)^n+1$ है, तो n बराबर है _____

[JEE(Main)April,2023]

Ans. 10

Sol. $f(x)=\sum_{k=1}^{10} kx^k$

$f(x) = x + 2x^2 + ... + 10\,x^{10}$

$f(x) \cdot x = x^2 + 2x^3 + ... + 9x^{10} + 10x^{11}$

$f(x)\,(1 - x) = x + x^2 + x^3 + ... + x^{10} - 10\,x^{11}$

$$f(x)=\frac{x\left(1-x^{10}\right)}{(1-x)^2}-\frac{10x^{11}}{(1-x)}$$

$$f(x)=\frac{x-x^{11}-10x^{11}+10x^{12}}{(1-x)^2} \Rightarrow \frac{10x^{12}-11x^{11}+x}{(1-x)^2}$$

Hence $2f(2) + f'(2) = 119.2^{10} + 1$

$\Rightarrow$ So, n = 10

28. The function $f : N - \{1\} \to N$; is defined by f(n) = the highest prime factor of n, is :

[JEE(Main)Jan.2024]

(A) neither one-one nor onto (B) onto only

(C) one-one only (D) both one-one and onto

फलन $f : N - \{1\} \to N$; f(n) = पूर्णांक n का सबसे बड़ा अभाज्य गुणनखंड, द्वारा परिभाषित है। तो f :

[JEE(Main)Jan.2024]

(A) न तो एकैकी है न ही आच्छादक है (B) केवल आच्छादक है

(C) केवल एकैकी है (D) एकैकी और आच्छादक दोनों है

Ans. A

Sol. Largest prime factors of two different natural numbers can be same so it is many one.

For example f(6) = 3, f(12) = 3

f(10) = 5, f(15) = 5

As range of f(x) is set of prime number so range of f(x) is proper subset of co-domain so f is into.

29. Let $f : R - \left\{\frac{-1}{2}\right\} \to R$ and $g : R - \left\{\frac{-5}{2}\right\} \to R$ be defined as $f(x)=\frac{2x+3}{2x+1}$ and $g(x)=\frac{|x|+1}{2x+5}$. Then, the domain of the function fog is : **[JEE(Main)Jan.2024]**

माना $f : R - \left\{\frac{-1}{2}\right\} \to R$ तथा $g : R - \left\{\frac{-5}{2}\right\} \to R$, $f(x)=\frac{2x+3}{2x+1}$ तथा $g(x)=\frac{|x|+1}{2x+5}$ द्वारा परिभाषित हैं। तो फलन fog का प्रांत है : **[JEE(Main)Jan.2024]**

(A) $R - \left\{-\frac{7}{4}\right\}$ (B) R (C) $R - \left\{-\frac{5}{2}\right\}$ (D) $R - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

Ans. C

Sol. $$y=\frac{\frac{2(|x|+1)}{2x+5}+3}{2\left(\frac{|x|+1}{2x+5}\right)+1}$$

$$=\frac{2|x|+2+6x+15}{2|x|+2+2x+5}=\frac{(2|x|+6x+17)}{2|x|+2x+7}$$

And $x \neq -\frac{5}{2}$ for defining g(x)

30. If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$; $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of (fog) (x) is :

[JEE(Main)Jan.2024]

यदि $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}$; $g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, तो (fog) (x) का परिसर है :

[JEE(Main)Jan.2024]

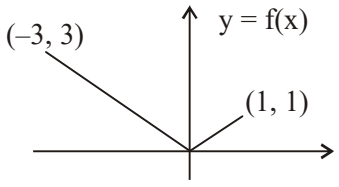
(A) [0, 1) (B) [0, 3) (C) (0, 1] (D) [0, 1]

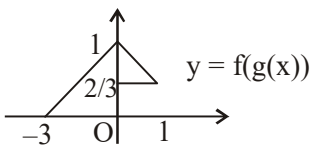
Ans. D

Sol. $$f(g(x))=\begin{cases}2+2g(x) & , \quad -3\le g(x)<0 \quad ____(1)\\ 1-\frac{g(x)}{3} & , \quad 0\le g(x)\le 3 \quad ____(2)\end{cases}$$

By (1) $x \in \phi$

And by (2) $x \in [-3, 0]$ and $x \in [0, 1]$





Range of f(g(x)) is [0, 1]

31. If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha, \beta]$ then the value of $5\beta-4\alpha$ is equal to : **[JEE(Main)Jan.2024]**

यदि फलन $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ का प्रांत $(\alpha, \beta]$ है, तो $5\beta - 4\alpha$ का मान बराबर है :

[JEE(Main)Jan.2024]

(A) 10 (B) 9 (C) 12 (D) 11

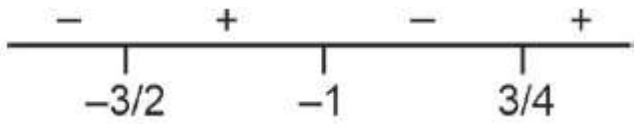
Ans. C

Sol. Given function,

$$f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$$

Clearly, $\frac{2x+3}{4x^2+x-3}>0$

$$\Rightarrow \frac{2x+3}{(4x-3)(x+1)}>0$$

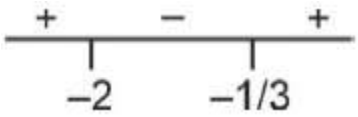


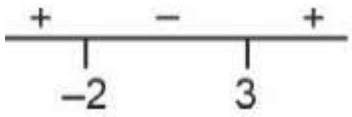
$$\therefore x\in\left(\frac{-3}{2},-1\right)\cup\left(\frac{3}{4},\infty\right)$$

Also, $-1\le\frac{2x-1}{x+2}\le 1$

$$\Rightarrow \frac{2x-1}{x+2}+1\ge 0 \qquad \frac{2x-1}{x+2}-1\le 0$$

$$\Rightarrow \frac{3x+1}{x+2}\ge 0 \qquad \frac{x-3}{x+2}\le 0$$





$$\therefore x\in(-\infty,-2)\cup\left[-\frac{1}{3},\infty\right) \text{ and } x\in(-2,3]$$

$$\therefore x\in\left(\frac{3}{4},3\right]\equiv(\alpha,\beta]$$

$\therefore 5\beta - 4\alpha = 15 - 3 = 12$

32. If $f(x)=\frac{4x+3}{6x-4}$, $x\ne\frac{2}{3}$ and $(fof)(x)=g(x)$, where $g:R-\left\{\frac{2}{3}\right\}\to R-\left\{\frac{2}{3}\right\}$, then $(gogog)(4)$ is equal to :

[JEE(Main)Jan.2024]

यदि $f(x)=\frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ तथा (fof) (x) = g(x) है, जहाँ $g: R-\left\{\frac{2}{3}\right\} \to R-\left\{\frac{2}{3}\right\}$ है, तो (gogog) (4) बराबर है:

[JEE(Main)Jan.2024]

(A) –4 (B) $-\frac{19}{20}$ (C) $\frac{19}{20}$ (D) 4

Ans. D

Sol. $g(x)=f\left(\frac{4x+3}{6x-4}\right)=\frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}=\frac{16x+12+18x-12}{24x+18-24x+16}=\frac{34x}{34}=x$

$gogog(4)=g(g(4))=g(4)=4$

33. Let $f: R \to R$ and $g: R \to R$ be defined as $f(x)=\begin{cases}\log_e x & , \quad x>0 \\ e^{-x} & , \quad x \leq 0\end{cases}$ and $g(x)=\begin{cases}x & , \quad x \geq 0 \\ e^{x} & , \quad x<0\end{cases}$. Then $gof: R \to R$ is:

[JEE(Main)Jan.2024]

(A) both one-one and onto (B) onto but not one-one

(C) one-one but not onto (D) neither one-one nor onto

माना $f: R \to R$ तथा $g: R \to R$, $f(x)=\begin{cases}\log_e x & , \quad x>0 \\ e^{-x} & , \quad x \leq 0\end{cases}$ तथा $g(x)=\begin{cases}x & , \quad x \geq 0 \\ e^{x} & , \quad x<0\end{cases}$ द्वारा परिभाषित है। तो $gof: R \to R$:

[JEE(Main)Jan.2024]

(A) एकैकी एवं आच्छादक दोनों है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) न तो एकैकी है, न ही आच्छादक है

Ans. D

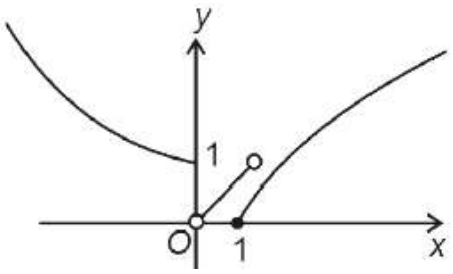
Sol. Given $f(x)=\begin{cases}e^{-x} & ; \quad x \leq 0 \\ \ell nx & ; \quad x>0\end{cases}$

and $g(x)=\begin{cases}x & ; \quad x \geq 0 \\ e^{x} & ; \quad x<0\end{cases}$

then gof(x)
= g(f(x))

$=\begin{cases}f(x) & ; \quad f(x) \geq 0 \\ e^{f(x)} & ; \quad f(x)<0\end{cases}$

$$= \begin{cases} e^{-x} & ; \quad x \le 0 \\ x & ; \quad 0 < x < 1 \\ \ell nx & ; \quad x \ge 1 \end{cases}$$



gof(x)

$\therefore$ f(x) is neither one-one nor onto

34. If the domain of the function $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2+2x-15)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^3$ is equal to : **[JEE(Main)Jan.2024]**

यदि फलन $f(x) = \frac{\sqrt{x^2-25}}{(4-x^2)} + \log_{10}(x^2+2x-15)$ का प्राँत $(-\infty,\alpha)\cup[\beta,\infty)$ है, तो $\alpha^2+\beta^3$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 175 (B) 140 (C) 150 (D) 125

Ans. C

Sol. $x^2-25 \ge 0 \quad \Rightarrow x \in (-\infty,-5]\cup[5,\infty)$

$4-x^2 \ne 0 \quad \Rightarrow x \notin \{-2,2\}$

$x^2+2x-15 > 0 \quad \Rightarrow x \in (-\infty,-5)\cup(3,\infty)$

$\Rightarrow x \in (-\infty,-5)\cup[5,\infty)$

$\Rightarrow \alpha = -5, \beta = 5$

35. Let A = {1, 2, 3, ..., 7} and let P(A) denote the power set of A. If the number of functions $f: A \to P(A)$ such that $a \in f(a)$, $\forall a \in A$, is m^n, m and $n \in N$ and m is least, then m + n is equal to _____. **[JEE(Main)Jan.2024]**

माना A = {1, 2, 3, ..., 7} है तथा माना समुच्चय A का घात (Power) समुच्चय P(A) है। यदि फलनों $f: A \to P(A)$, जिनके लिए $a \in f(a)$, $\forall a \in A$ है, की संख्या m^n, m, $n \in N$ हैं तथा m न्यूनतम है, तो m + n बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 44

Sol. $n(P(A)) = 2^7$

$a \in f(a)$

$\Rightarrow$ f(a) will have (2^6) different subsets having a in them as choice.

$\Rightarrow (2^6)^7 = 2^{42}$ function

$m^n = 2^{42}$

$\Rightarrow m + n = 44$

36. Let A = {1, 3, 7, 9, 11} and B = {2, 4, 5, 7, 8, 10, 12}. Then the total number of one-one maps f: A $\rightarrow$ B, such that f(1) + f(3) = 14, is : **[JEE(Main)April,2024]**

माना A = {1, 3, 7, 9, 11} तथा B = {2, 4, 5, 7, 8, 10, 12} हैं। तो एकैकी प्रतिचित्रों (maps) f: A $\rightarrow$ B, जिनके लिए f(1) + f(3) = 14 है, की कुल संख्या है : **[JEE(Main)April,2024]**

(A) 180 (B) 120 (C) 480 (D) 240

Ans. D

Sol. A = {1, 3, 7, 9, 11}

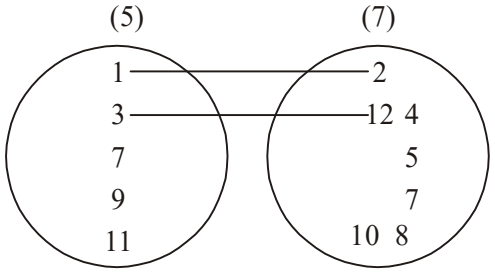
B = {2, 4, 5, 7, 8, 10, 12}

f(1) + f(3) = 14

(i) 2 + 12

(ii) 4 + 10

$2 \times (2 \times 5 \times 4 \times 3) = 240$



37. Let f, g : R $\rightarrow$ R be defined as :

$$f(x) = |x-1| \text{ and } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

Then the function f(g(x)) is : **[JEE(Main)April,2024]**

(A) Neither one-one nor onto (B) Onto but not one-one

(C) One-one but not onto (D) Both one-one and onto

माना f, g : R $\rightarrow$ R,

$$f(x) = |x-1| \text{ तथा } g(x) = \begin{cases} e^x, & x \geq 0 \\ x+1, & x \leq 0 \end{cases}$$

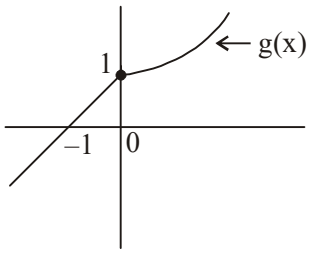
द्वारा परिभाषित हैं। तो फलन f(g(x)) : **[JEE(Main)April,2024]**

(A) न तो एकैकी न ही आच्छादक है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी है परन्तु आच्छादक नहीं है (D) एकैकी तथा आच्छादक दोनों है

Ans. A

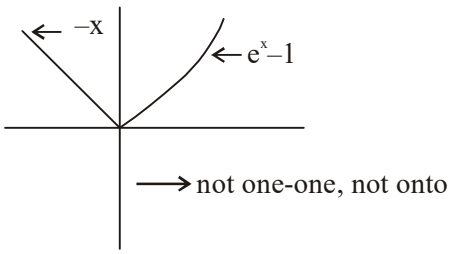
Sol. $f(x)=\begin{cases}x-1 & x\geq 1\\ 1-x & x<1\end{cases}$



$$g(x)=\begin{cases}e^x & ; \quad x\geq 0\\ x+1 & ; \quad x\leq 0\end{cases}$$

$$f(g(x))=\begin{cases}g(x)-1, & g(x)\geq 1\\ 1-g(x) & g(x)<1\end{cases}$$

$$=\begin{cases}e^x-1 & x\geq 0\\ -x & x<0\end{cases}$$



38. The function $f(x)=\dfrac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ is: **[JEE(Main)April,2024]**

(A) both one-one and onto (B) onto but not one-one

(C) neither one-one nor onto (D) one-one but not onto

फलन $f(x)=\dfrac{x^2+2x-15}{x^2-4x+9}, x\in\mathbb{R}$ है : **[JEE(Main)April,2024]**

(A) एकैकी तथा आच्छादक दोनों (B) आच्छादक किंतु एकैकी नहीं

(C) न तो एकैकी और न आच्छादक (D) एकैकी किंतु आच्छादक नहीं

Ans. C

Sol. $f'(x)=\dfrac{\left(x^2-4x+9\right)(2x+2)-\left(x^2+2x-15\right)(2x-4)}{\left(x^2-4x+9\right)^2}$

$=\dfrac{2x^3-6x^2+10x+18-\left[2x^3-38x+60\right]}{\left(x^2-4x+9\right)^2}$

$=\dfrac{-6x^2+48x-42}{\left(x^2-4x+19\right)^2}=\dfrac{-6(x-7)(x-1)}{\left(x^2-4x+9\right)^2}$

So f'(x) = positive as well as negative so many one and into.

Since codomain is not given so will be considered it R

39. Let $f(x)=\dfrac{1}{7-\sin 5x}$ be a function defined on R. Then the range of the function f(x) is equal to :

[JEE(Main)April,2024]

माना फलन $f(x)=\dfrac{1}{7-\sin 5x}$, R पर परिभाषित है। तो फलन f(x) का परिसर बराबर है : **[JEE(Main)April,2024]**

(A) $\left[\dfrac{1}{8},\dfrac{1}{6}\right]$ (B) $\left[\dfrac{1}{7},\dfrac{1}{5}\right]$ (C) $\left[\dfrac{1}{8},\dfrac{1}{5}\right]$ (D) $\left[\dfrac{1}{7},\dfrac{1}{6}\right]$

Ans. A

Sol. $f(x)=\dfrac{1}{7-\sin x}, x\in R$

$-1 \le -\sin 5x \le 1$

$-1 \le -\sin -5x \le 1$

$6 \le 7-\sin 5x \le 8$

$\dfrac{1}{8} \le \dfrac{1}{7-\sin 5x} \le \dfrac{1}{6}$

$\dfrac{1}{8} \le f(x) \le \dfrac{1}{6}$

40. Let [t] be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f: A \to Z$ be the function $f(x)=\left[\log_2\left(x^2+\left[\dfrac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is :

[JEE(Main)April,2024]

माना [t], t से कम या बराबर एक महत्तम पूर्णांक फलन है। माना A संख्या 2310 के सभी अभाज्य गुणनखण्डों को समुच्चय है और $f: A \to Z$ एक फलन $f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$ तो समुच्चय A से फलन f के परिसर में एकैकी फलनों की संख्या होगी :

[JEE(Main)April,2024]

(A) 25 (B) 20 (C) 120 (D) 24

Ans. C

Sol. $N = 2310 = 231 \times 10$

$= 3 \times 11 \times 7 \times 2 \times 5$

$A = \{2,3,5,7,11\}$

$$f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$$

$f(2) = [\log_2(5)] = 2$
$f(3) = [\log_2(25 + 25)] = 5$
$f(7) = [\log_2(117)] = 6$
$f(11) = [\log_2 387] = 8$
Range of f: $B = \{2,3,5,6,8\}$

No. of one-one functions = 5! = 120

41. Let $f(x)=\begin{cases} -a & \text{if } -a \le x \le 0 \\ x+a & \text{if } 0 < x \le a \end{cases}$ where $a > 0$ and $g(x)=\frac{\left(f(|x|)-|f(x)|\right)}{2}$. Then the function $g : [-a, a] \to [-a, a]$ is :

[JEE(Main)April,2024]

(A) One-one (B) Neither one-one nor onto

(C) Onto (D) Both one-one and onto

माना $f(x)=\begin{cases} -a & \text{if } -a \le x \le 0 \\ x+a & \text{if } 0 < x \le a \end{cases}$ है, जहाँ $a > 0$ तथा $g(x)=\frac{\left(f(|x|)-|f(x)|\right)}{2}$ है। तो फलन $g : [-a, a] \to [-a, a]$ होगा :

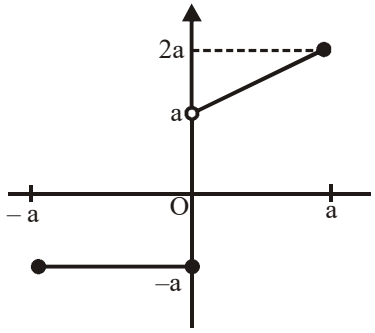
[JEE(Main)April,2024]

(A) एकैकी है (B) न तो एकैकी है न ही आच्छादक है

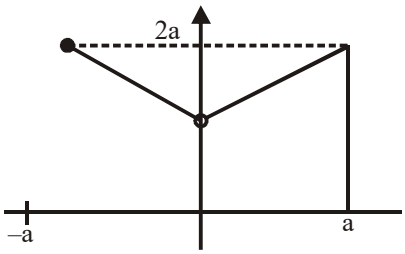
(C) आच्छादक है (D) ऐकैकी तथा आच्छादक दोनों है

Ans. B

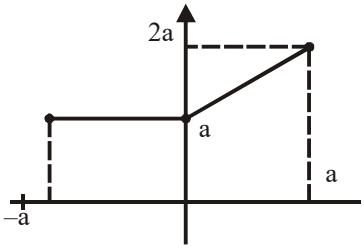
Sol. $y = f(x)$



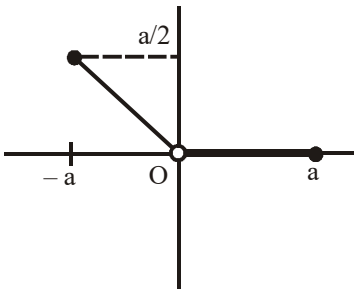
$y = f|x|$



$y = |f(x)|$



$$g(x) = \frac{f(|x|) - |f(x)|}{2}$$



42. Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = fog(10)$, $b = gof(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + l^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=x^2+9,\ g(x)=\frac{x}{x-9}$ है और $a=fog(10), b=gof(3)$ है। यदि दीर्घवृत्त $\frac{x^2}{a}+\frac{y^2}{b}=1$ की उत्केन्द्रता तथा नाभिलंब जीवा की लंबाई क्रमशः e तथा l है। तो $8e^2+l^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 6 (B) 12 (C) 16 (D) 8

Ans. D

Sol. $f(x)=x^2+9$

$$g(x)=\frac{x}{x-9}$$

$f(g(10))=f(10)=109=a$

$g(f(3))=2=b$

$$\frac{x^2}{109}+\frac{y^2}{2}=1$$

$2=109\,(1-e^2)$

$$e^2=\frac{107}{109}$$

$$\ell=\frac{2\times2}{\sqrt{109}}\Rightarrow\ell^2=\frac{16}{109}$$

$$8e^2+\ell^2=\frac{856}{109}+\frac{16}{109}=8$$

43. Let the range of the function $f(x)=\frac{1}{2+\sin 3x+\cos 3x}$, $x\in R$ be [a, b]. If α and β are respectively the A.M. and the G.M. of a and b, then $\frac{\alpha}{\beta}$ is equal to : **[JEE(Main)April,2024]**

माना फलन $f(x)=\frac{1}{2+\sin 3x+\cos 3x}$, $x\in R$ का परिसर [a, b] है। यदि a तथा b के A.M. तथा G.M. क्रमशः α तथा β हैं, तो $\frac{\alpha}{\beta}$ बराबर है : **[JEE(Main)April,2024]**

(A) $\sqrt{\pi}$ (B) π (C) 2 (D) $\sqrt{2}$

Ans. D

Sol. $$f(x)=\frac{1}{2+\sin 3x+\cos 3x}$$

we know that

$$-\sqrt{2}\le\sin 3x+\cos 3x\le\sqrt{2}$$

$$2-\sqrt{2} \le 2+\sin 3x+\cos 3x \le 2+\sqrt{2}$$

$$\frac{1}{2+\sqrt{2}} \le \frac{1}{2+\sin 3x+\cos 3x} \le \frac{1}{2-\sqrt{2}}$$

$$\therefore a = \frac{1}{2+\sqrt{2}} \text{ or } \frac{2-\sqrt{2}}{2}$$

$$b = \frac{1}{2-\sqrt{2}} \text{ or } \frac{2+\sqrt{2}}{2}$$

$$\alpha = \text{A. M of a and b} = \frac{\frac{2-\sqrt{2}}{2}+\frac{2+\sqrt{2}}{2}}{2} = 1$$

$$\beta = \text{G. M of a and b} = \sqrt{\frac{2-\sqrt{2}}{2}\times\frac{2+\sqrt{2}}{2}} = \frac{1}{\sqrt{2}}$$

$$\therefore \frac{\alpha}{\beta} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$

44. Consider the function f : R $\to$ R defined by $f(x)=\frac{2x}{\sqrt{1+9x^2}}$. If the composition of $f,\underbrace{(fofofo.....f)}_{10 \text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the vlaue of $\sqrt{3\alpha+1}$ is equal to _____.

[JEE(Main)April,2024]

विचार कीजिए फलन $f: R \to R$, $f(x)=\frac{2x}{\sqrt{1+9x^2}}$ द्वारा परिभाषित है। यदि f का संयोजन $f,\underbrace{(fofofo.....f)}_{10 \text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$ है, तो $\sqrt{3\alpha+1}$ का मान बराबर है _____

[JEE(Main)April,2024]

Ans. 1024

Sol. $f(x)=\frac{2x}{\sqrt{1+9x^2}}$

$$(fof)(x)=\frac{2f(x)}{\sqrt{1+9(f(x))^2}}=\frac{\frac{4x}{\sqrt{1+9x^2}}}{\sqrt{1+9\times\frac{4x^2}{1+9x^2}}}=\frac{4x}{\sqrt{1+45x^2}}$$

$$(\text{fofof})(x)=\frac{4\times\frac{2x}{\sqrt{1+9x^2}}}{\sqrt{1+45\times\frac{4x^2}{1+9x^2}}}=\frac{8x}{\sqrt{1+21\times9x^2}}$$

$$(\text{fofofof})(x)=\frac{16x}{\sqrt{1+85\times9x^2}}$$

$\Rightarrow \alpha$ is 10th term of 1, 5, 21, 85, ...

α is 10th term of $\frac{\left(2^1\right)^2-1}{3},\frac{\left(2^2\right)^2-1}{3},\frac{\left(2^3\right)^2-1}{3},\frac{\left(2^4\right)^2-1}{3},\ldots$

$$\Rightarrow \alpha=\frac{\left(2^{10}\right)^2-1}{3}$$

$$\Rightarrow \sqrt{3\alpha+1}=2^{10}=1024$$

45. If $S=\left\{a\in R:|2a-1|=3[a]+2\{a\}\right\}$, where [t] deonts the greatest integer less than or equal to t and {t} represents the fractional part t, then $72\sum_{a\in S}$ a is equal to _____. **[JEE(Main)April,2024]**

यदि $S=\left\{a\in R:|2a-1|=3[a]+2\{a\}\right\}$ है, जहाँ [t] महत्तम पूर्णांक फलन है तथा {t}, t के भिन्नात्मक भाग को दर्शाता है, तो $72\sum_{a\in S}$ a बराबर है _____ **[JEE(Main)April,2024]**

Ans. 18

Sol. $|2a-1|=3[a]+2\{a\}$

Case-I : $a\geq\frac{1}{2}$

$2a-1=[a]+2a \qquad (\because\ a=[a]+\{a\})$

$[a]=-1$

$a\in[-1,0)$ but $a\geq\frac{1}{2}$, $\Rightarrow$ No value of a

Case-II: $a<\frac{1}{2}$

$1-2a=[a]+2a$

$1=[a]+4a$

$1=5[a]+4\{a\}$

$4\{a\}=1-5[a]$

$0\leq 4\{a\}<4$

$0 \le 1 - 5[a] < 4$

$\frac{1}{3} \ge [a] \ge -\frac{3}{5} \quad \Rightarrow \quad [a] = 0 \quad \Rightarrow \quad \{a\} = \frac{1}{4}$

$\Rightarrow \quad a = \frac{1}{4}$

Now, $72\sum a = 72 \times \frac{1}{4} = 18$

46. If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in N$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ is equal to _____. **[JEE(Main)April,2024]**

यदि सभी $m, n \in N$ के लिए, $f(m + n) = f(m) + f(n)$ है तथा $f(1) = 1$ है, के अधिकतम धन पूर्णांक λ जिसके लिए $\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$ है, है _____ **[JEE(Main)April,2024]**

Ans. 1010

Sol. $f(m+n) = f(m) + f(n)$

$\Rightarrow f(x) = kx$

$\Rightarrow f(1) = 1$

$\Rightarrow k = 1$

$f(x) = x$

Now

$\sum_{k=1}^{2022} f(\lambda + k) \le (2022)^2$

$\Rightarrow \sum_{k=1}^{2022} (\lambda + k) \le (2022)^2$

$\Rightarrow 2022\lambda + \frac{2022 \times 2023}{2} \le (2022)^2$

$\Rightarrow \lambda \le 2022 - \frac{2023}{2}$

$\Rightarrow \lambda \le 1010.5$

47. Let $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ and $B = \{x : (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____. **[JEE(Main)April,2024]**

माना $A = \{(x, y) : 2x + 3y = 23, x, y \in N\}$ तथा $B = \{x : (x, y) \in A\}$ हैं। तो A से B के एकैकी फलनों की संख्या है

_____.

Ans. 24

Sol. $2x + 3y = 23$

$x = 1 \quad y = 7$

$x = 4 \quad y = 5$

$x = 7 \quad y = 3$

$x = 10 \quad y = 1$

A	B
(1, 7)	1
(4, 5)	4
(7, 3)	7
(10, 1)	10

The number of one-one functions from A to B is equal to 4!

Exercise J - A

1. The function f : [0, 3] → [1, 29], defined by f(x) = $2x^3 - 15x^2 + 36x + 1$, is : **[JEE-2012]**

(A) one-one and onto (B) onto but not one - one

(C) one-one but not onto (D) neither one- one nor onto

फलन f : [0, 3] → [1, 29] परिभाषित है, f(x) = $2x^3 - 15x^2 + 36x + 1$ तो f होगा : **[JEE-2012]**

(A) एकैकी लेकिन आच्छादक नहीं है (B) आच्छादक लेकिन एकैकी नहीं है

(C) एकैकी एवं आच्छादक दोनों है (D) न एकैकी न आच्छादक है

Ans. B

Sol. $f(x) = 2x^3 - 15x^2 + 36x + 1$

$\Rightarrow f'(x) = 6(x^2 - 5x + 6)$

$= 6(x - 2)(x - 3)$

$\because f(x)$ is non monotonic in $x \in [0,3]$

$\Rightarrow f(x)$ is not one-one

$f(x)$ is increasing in $x \in [0,2)$ and decreasing in $x \in (2,3]$

$f(0) = 1,\ f(2) = 29$ & $f(3) = 28$

$\therefore$ Range of $f(x)$ is $[1,29]$

$\Rightarrow f(x)$ is onto.

2. Let f : (–1, 1) → R be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then the value(s) of $f\left(\frac{1}{3}\right)$ is (are) : **[JEE-2012]**

माना f: (–1, 1) → R इस प्रकार की $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$, $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ तो $f\left(\frac{1}{3}\right)$ का मान होगा : **[JEE-2012]**

(A) $1 - \sqrt{\frac{3}{2}}$ (B) $1 + \sqrt{\frac{3}{2}}$ (C) $1 - \sqrt{\frac{2}{3}}$ (D) $1 + \sqrt{\frac{2}{3}}$

Ans. AB

Sol. $\because \theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \Rightarrow 2\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$

Now $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$

$= \frac{1 + \cos 2\theta}{\cos 2\theta} = 1 + \frac{1}{\cos 2\theta}$ .....(i)

Let $\cos 4\theta = \frac{1}{3}$ $\Rightarrow$ $2\cos^2 2\theta - 1 = \frac{1}{3}$

$\Rightarrow$ $\cos 2\theta = \pm\sqrt{\frac{2}{3}}$

$\Rightarrow$ From (i), $f\left(\frac{1}{3}\right) = 1 \pm \sqrt{\frac{3}{2}}$

$\Rightarrow$ (A, B) are correct

3. Let $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then : **[JEE(Adv.)-2014]**

(A) f(x) is an odd function
(B) f(x) is a one-one function
(C) f(x) is an onto function
(D) f(x) is an even function

माना कि $f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R$ जहाँ $f(x) = (\log(\sec x + \tan x))^3$ के द्वारा परिभाषित किया गया है, तो : **[JEE(Adv.)-2014]**

(A) f(x) एक विषम फलन होगा
(B) f(x) एक ऐकैकी फलन होगा
(C) f(x) एक आच्छादक फलन होगा
(D) f(x) एक सम फलन होगा

Ans. ABC

Sol. (i) $f(-x) = -f(x)$ so it is odd function

(ii) $f'(x) = 3(\log(\sec x + \tan x))^2 \frac{1}{(\sec x + \tan x)} (\sec x \tan x + \sec^2 x) > 0$

(iii) Range of f(x) is R as $f\left(-\frac{\pi}{2}\right) \Rightarrow -\infty$ $\Rightarrow f\left(\frac{\pi}{2}\right) \Rightarrow \infty$

4. Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta - \alpha)$ is _____ **[JEE(Adv.)-2018]**

माना कि समुच्चय X में ठीक 5 अवयव हैं और समुच्चय Y में ठीक 7 अवयव हैं। यदि X से Y में एकैकी फलनों की संख्या α है और Y से X में आच्छादक फलनों की संख्या β है, तब $\frac{1}{5!}(\beta - \alpha)$ का मान है _____ **[JEE(Adv.)-2018]**

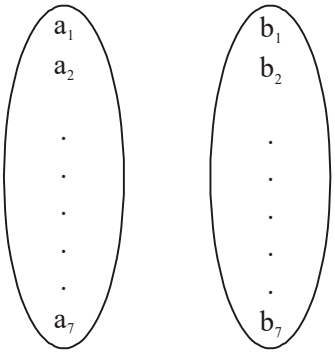
Ans. 119

Sol. $n(X) = 5$

$n(Y) = 7$

$\alpha \rightarrow$ Number of one-one function $= {}^7C_5 \times 5! = 21 \times 120 = 2520$

$\beta \rightarrow$ Number of onto function Y to X



1, 1, 1, 1, 3 1, 1, 1, 2, 2

$$\frac{7!}{3!4!} \times 5! + \frac{7!}{(2!)^3\, 3!} \times 5! = \left({}^7C_3 + 3.{}^7C_3\right)5! = 4 \times {}^7C_3 \times 5!$$

$$\frac{\beta - \alpha}{5!} = 4 \times {}^7C_3 - {}^7C_5 = 4 \times 35 - 21 = 119$$

5. If the function $f: R \rightarrow R$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is TRUE?

[JEE(Adv.)-2020]

(A) f is one-one, but NOT onto (B) f is onto, but NOT one-one

(C) f is BOTH one-one and onto (D) f is NEITHER one-one NOR onto

यदि फलन $f: R \rightarrow R$ को $f(x) = |x|(x - \sin x)$ से परिभाषित किया जाता है, तब निम्न में से कौनसा कथन सत्य है?

[JEE(Adv.)-2020]

(A) f एकैकी है, लेकिन आच्छादक नही है (B) f आच्छादक है, लेकिन एकैकी नहीं है

(C) f एकैकी एवं आच्छादक दोनों है (D) f एकैकी भी नहीं है एवं आच्छादक भी नहीं है

Ans. C

Sol. $f(x) \begin{cases} \rightarrow x(x - \sin x) & x \geq 0 \\ \rightarrow -x\,(x - \sin x) & x < 0 \end{cases}$

$\forall\ x \geq 0$

$f'(x) = 2x - (\sin x + x \cos x)$

$= (x - \sin x) + x\,(1 - \cos x) \geq 0$

$f(x)$ is increasing

$f(x))_{min} = f(0) = 0$

$x \rightarrow \infty \Rightarrow f(x) \rightarrow \infty$

$\Rightarrow \forall\ x \geq 0 \Rightarrow f(x) \in [0, \infty)$

$\forall\ x < 0$

$\because$ f is odd $\Rightarrow$ f(x) is also increasing

rangs of (x) for $x < 0$

$f(x) \in (-\infty\ , 0)$

Rangs of f(x) $=(-\infty, \infty) \Rightarrow$ f(x) is into

f(x) is increasing $\forall\ x \in R \Rightarrow$ f is one – one

6. Let the function $f: [0, 1] \to \mathbb{R}$ be defined by $f(x) = \frac{4^x}{4^x + 2}$. Then the value of

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) \text{ is.}$$ **[JEE(Adv.)-2020]**

माना कि फलन $f: [0, 1] \to \mathbb{R}$ को $f(x) = \frac{4^x}{4^x + 2}$ से परिभाषित किया जाता है। तब

$$f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + f\left(\frac{3}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right)$$ का मान है। **[JEE(Adv.)-2020]**

Ans. 19

Sol. $f(x) = \frac{4^x}{4^x + 2}$; Replace $x \to 1 - x$

$$f(1-x) = \frac{4^{1-x}}{4^{1-x} + 2} = \frac{4 / 4^x}{4 / 4^x + 2} = \frac{2}{4^x + 2}$$

$$f(x) + f(1-x) = \frac{4^x}{4^x + 2} + \frac{2}{4^x + 2} = 1\ \forall x.$$

So, $f\left(\frac{1}{40}\right) + f\left(\frac{39}{40}\right) = f\left(\frac{2}{40}\right) + f\left(\frac{38}{40}\right) = ... = f\left(\frac{19}{40}\right) + f\left(\frac{21}{40}\right) = 1$

$$\Rightarrow f\left(\frac{1}{40}\right) + f\left(\frac{2}{40}\right) + ...f\left(\frac{19}{40}\right) + f\left(\frac{20}{40}\right) + f\left(\frac{21}{40}\right) + ... + f\left(\frac{39}{40}\right) - f\left(\frac{1}{2}\right) = 19 + f\left(\frac{20}{40}\right) - f\left(\frac{1}{2}\right)$$

= 19

7. Let |M| denote the determinant of a square matrix. M. Let $g: \left[0, \frac{\pi}{2}\right] \to R$ be the function defined by

$$g(\theta) = \sqrt{f(\theta) - 1} + \sqrt{f\left(\frac{\pi}{2} - \theta\right) - 1}$$

$$\text{where } f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}.$$

Let p(x) be a quadratic polynomial whose roots are the maximum and minimum values of the function g(θ), and $p(2)=2-\sqrt{2}$. Then, which of the following is/are TRUE? **[JEE(Adv.)2022]**

माना |M|, एक वर्ग आव्यूह M के सारणिक को दर्शाता है। माना $g:\left[0,\frac{\pi}{2}\right]\to R$ एक फलन है जो

$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$ द्वारा परिभाषित है, जहाँ

$$\text{where } f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}$$ है।

माना p(x) एक द्विघात बहुपद है जिसके मूल, फलन g(θ) के अधिकतम तथा न्यूनतम मान हैं और $p(2)=2-\sqrt{2}$ है। तब निम्नलिखित में से कौनसा/कौनसे विकल्प सत्य है/हैं? **[JEE(Adv.)2022]**

(A) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$ (B) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$ (C) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$ (D) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

Ans. AC

Sol. $$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\frac{\pi}{4} & \tan\pi\end{vmatrix}$$

$$f(\theta)=\frac{1}{2}\begin{vmatrix}2 & \sin\theta & 1\\ 0 & 1 & \sin\theta\\ 0 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}0 & -\sin\left(\theta-\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & 0 & \log_e\left(\frac{4}{\pi}\right)\\ -\tan\left(\theta-\frac{\pi}{4}\right) & -\log_e\left(\frac{4}{\pi}\right) & 0\end{vmatrix}$$

$f(\theta)=(1+\sin^2\theta)+0$ (skew symmetric)

$$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$$

$=|\sin\theta|+|\cos\theta|$ for $\theta\in\left[0,\frac{\pi}{2}\right]$

$g(\theta)\in\left[1,\sqrt{2}\right]$

Again let $P(x)=k(x-\sqrt{2})(x-1)$

$2-\sqrt{2}=k(2-\sqrt{2})(2-1)$

$\Rightarrow k=1$ ($P(2)=2-\sqrt{2}$ given)

$\therefore P(x)=\left(x-\sqrt{2}\right)(x-1)$

for option (A) $P\left(\frac{3+\sqrt{2}}{4}\right)<0$ correct

option (B) $P\left(\frac{1+3\sqrt{2}}{4}\right)<0$ incorrect

option (C) $P\left(\frac{5\sqrt{2}-1}{4}\right)>0$ correct

option (D) $P\left(\frac{5-\sqrt{2}}{4}\right)>0$ incorrect

8. Let $f:R\to R$ be a function defined by $f(x)=\begin{cases}x^2\sin\left(\frac{\pi}{x^2}\right), & \text{if } x\neq 0,\\ 0, & \text{if } x=0\end{cases}$. Then which of the following statements is TRUE? **[JEE(Adv.)2024]**

(A) $f(x)=0$ has infinitely many solutions in the interval $\left[\frac{1}{10^{10}},\infty\right)$

(B) f(x) = 0 has no solutions in the interval $\left[\frac{1}{\pi},\infty\right)$

(C) The set of solutions of f(x) = 0 in the interval $\left(0,\frac{1}{10^{10}}\right)$ is finite

(D) f(x) = 0 has more than 25 solutions in the interval $\left(\frac{1}{\pi^2},\frac{1}{\pi}\right)$

माना कि $f : R \to R$ एक फलन (function) है, जो $f(x)=\begin{cases} x^2\sin\left(\frac{\pi}{x^2}\right), & \text{यदि } x \neq 0, \\ 0, & \text{यदि } x = 0 \end{cases}$ द्वारा परिभाषित है, तब निम्नलिखित कथनों में से कौनसा सत्य है? **[JEE(Adv.)2024]**

(A) अन्तराल (interval) $\left[\frac{1}{10^{10}},\infty\right)$ में f(x) = 0 के अनंत (infinitely many) हल (solutions) हैं

(B) अंतराल $\left[\frac{1}{\pi},\infty\right)$ में f(x) = 0 का कोई हल (solution) नहीं है

(C) अन्तराल $\left(0,\frac{1}{10^{10}}\right)$ में f(x) = 0 के हलों (solutions) का समुच्चय (set) परिमित (finite) है

(D) अंतराल $\left(\frac{1}{\pi^2},\frac{1}{\pi}\right)$ में f(x) = 0 के 25 से अधिक हल (solutions) हैं

Ans. D

Sol. for x > 0

$f(x) = 0 \Rightarrow x^2\sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \sin\left(\frac{\pi}{x^2}\right) = 0$

$\Rightarrow \frac{\pi}{x^2} = n\pi$

$\Rightarrow x = \pm\frac{1}{\sqrt{n}}$

$x \in \left(\frac{1}{\pi^2},\frac{1}{\pi}\right) \Rightarrow \frac{1}{\pi^2} < \frac{1}{\sqrt{n}} < \frac{1}{\pi}$

$\pi < \sqrt{n} < \pi^2$

$\pi^2 < n < \pi^4$

f(x) = 0 has more than 25 solutions

9. Let f : R $\to$ R be a fuction such that f(x + y) = f(x) + f(y) for all x, y $\in$ R, and g : R $\to$ (0, ∞) be a function such that g(x + y) = g(x) g(y) for all x, y $\in$ R. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is _____. **[JEE(Adv.)2024]**

माना कि f : R $\to$ R एक ऐसा फलन (fuction) है कि सभी x, y $\in$ R के लिए f(x + y) = f(x) + f(y) है, और g : R $\to$ (0, ∞) एक ऐसा फलन है कि सभी x, y $\in$ R के लिए g(x + y) = g(x) g(y) है। यदि $f\left(\frac{-3}{5}\right) = 12$ और $g\left(\frac{-1}{3}\right) = 2$ हैं, तब $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 51

Sol. f : R $\to$ R

f(x + y) = f(x) + f(y) for all x, y $\in$ R

$\Rightarrow$ f(x) = ax

Now $f\left(\frac{-3}{5}\right) = 12$

a = – 20

$\therefore$ f(x) = – 20x

Again, g (x + y) = g(x). g(y) x, y $\in$ R.

$\Rightarrow f(x) = k^x$

Now $g\left(\frac{-1}{3}\right) = 2$

$2 = k^{\frac{-1}{3}}$

$\Rightarrow \left(2^{-1}\right)^3 = k$

$$k = \frac{1}{8}$$

$$\therefore\ g(x) = \left(\frac{1}{8}\right)^x$$

Now, $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right) g(0)$

$\Rightarrow [-5 + 64 - 8] \cdot 1 \quad \Rightarrow 64 - 13 \qquad \Rightarrow 51$

PARAGRAPH

Paragraph for Q. No 10 to 11

Let S = {1, 2, 3, 4, 5, 6} and X be the set of all relations R from S to S that satisfy both the following properties:

i. R has exactly 6 elements.

ii. For each (a, b) $\in$ R, we have $|a - b| \geq 2$.

Let Y = {R $\in$ X : The range of R has exactly one element } and

Z = {R $\in$ X : R is a function from S to S}.

Let n(A) denote the number of elements in a set A.

(There are two questions based on PARAGRAPH)

माना कि S = {1, 2, 3, 4, 5, 6} है, और X, S से S में उन सभी संबंधों (relations) R का समुच्चय (set) है जो निम्नलिखित दोनों गुणधर्मों (properties) को संतुष्ट करते हैं :

i. R में ठीक (exactly) 6 अवयव (elements) हैं।

ii. प्रत्येक (a, b) $\in$ R के लिए $|a - b| \geq 2$ है।

माना कि Y = {R $\in$ X : R के परिसर (range) में ठीक (exactly) एक अवयव (element) है} और

Z = {R $\in$ X : R, S से S में एक फलन (function) है।}

माना कि n(A), समुच्चय A में अवयवों की संख्या (number of elements) को दर्शाता है।

(PARAGRAPH पर आधारित दो प्रश्न हैं, नीचे दिया गया प्रश्न उनमें से एक है)

10. If $n(X) = {}^mC_6$, then the value of m is _____. **[JEE(Adv.)2024]**

यदि $n(X) = {}^mC_6$ है, तब m का मान _____ है। **[JEE(Adv.)2024]**

Ans. 20

11. If the value of $n(Y) + n(Z)$ is k^2, then $|k|$ is _____. **[JEE(Adv.)2024]**

यदि $n(Y) + n(Z)$ का मान k^2 है, तब $|k|$ _____ है। **[JEE(Adv.)2024]**

Ans. 36

Sol. All possible ordered pair (a, b) s.t. $|a - b| \geq 2$

(1, 3) (2, 4) (3, 1) (4, 1) (5, 1) (6, 1)

(1, 4) (2, 5) (3, 5) (4, 2) (5, 2) (6, 2)

(1, 5) (2, 6) (3, 6) (4, 6) (5, 3) (6, 3)

(1, 6) (6, 4)

10. $n(X) = {}^{20}C_6 \Rightarrow m = 20$

11. $n(Y) = 0$ as maximum 4 ordered pairs are possible having same image

$n(Z) = 4 \times 3 \times 3 \times 3 \times 3 \times 4 = 1296$

$n(Y) + n(Z) = 1296 = (36)^2$

$\Rightarrow |k| = 36$

12. Let R denote the set of all real numbers. Let $a_i, b_i \in R$ for $i \in \{1,2,3\}$.

Define the functions $f : R \to R$, $g : R \to R$, and $h : R \to R$ by

$f(x) = a_1 + 10x + a_2x^2 + a_3x^3 + x^4$,

$g(x) = b_1 + 3x + b_2x^2 + b_3x^3 + x^4$,

$h(x) = f(x + 1) - g(x + 2)$.

If $f(x) \neq g(x)$ for every $x \in R$, then the coefficient of x^3 in $h(x)$ is : **[JEE(Adv.)2025]**

मान लीजिए कि R सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है। मान लीजिए कि $i \in \{1,2,3\}$ के लिए $a_i, b_i \in R$ हैं। फलनों (functions) $f : R \to R$, $g : R \to R$, और $h : R \to R$ को

$f(x) = a_1 + 10x + a_2x^2 + a_3x^3 + x^4$,

$g(x) = b_1 + 3x + b_2x^2 + b_3x^3 + x^4$,

$h(x) = f(x + 1) - g(x + 2)$.

द्वारा परिभाषित कीजिए। यदि प्रत्येक $x \in R$ के लिए $f(x) \neq g(x)$ है, तब $h(x)$ में x^3 का गुणांक (coefficient) है :

[JEE(Adv.)2025]

(A) 8 (B) 2 (C) –4 (D) –6

Ans. C

Sol. Equation $f(x) = g(x)$ will not have any real solution.

$(a_3 - b_3)x^3 + (a_2 - b_2)x^2 + 7x + (a_1 - b_1) = 0$

$a_3 = b_3$ & $49 - 4(a_2 - b_2)(a_1 - b_1) < 0$

$h(x) = a_1 + 10(x + 1) + a_2(x + 1)^2 + a_3(x + 1)^3 + (x + 1)^4$

$-b_1 - 3(x + 2) - b_2(x + 2)^2 - b_3(x + 2)^3 - (x + 2)^4$

On RHS only $(x + 1)^4 - (x + 2)^4$ will have term containing x^3

So, coefficient of $x^3 = {}^4C_1 - {}^4C_1(2)$

$= 4 - 8 = -4$

13. Let N denote the set of all natural numbers, and Z denote the set of all integers. Consider the functions $f: N \to Z$ and $g: Z \to N$ defined by

$$f(n) = \begin{cases} \frac{(n+1)}{2} & \text{if n is odd,} \\ \frac{(4-n)}{2} & \text{if n is even,} \end{cases}$$

and

$$f(n) = \begin{cases} 3+2n & \text{if} \quad n \geq 0 \\ -2n & \text{if} \quad n < 0 \end{cases}$$

Define $(g \cdot f)(n) = g(f(n))$ for all $n \in N$, and $(f \cdot g)(n) = f(g(n))$ for all $n \in Z$.

Then which of the following statements is (are) TRUE ? **[JEE(Adv.)2025]**

(A) $g \cdot f$ is **NOT** one-one and $g \cdot f$ is **NOT** onto

(B) $f \cdot g$ is **NOT** one-one but $f \cdot g$ is onto

(C) g is one-one and g is onto

(D) f is **NOT** but f is onto

मान लीजिए कि N सभी प्राकृत संख्याओं के समुच्चय को दर्शाता है, और Z सभी पूर्णांकों के समुच्चय को दर्शाता है। फलनों $f: N \to Z$ और $g: Z \to N$ जो

$$f(n) = \begin{cases} \frac{(n+1)}{2} & \text{य दि n विषम है,} \\ \frac{(4-n)}{2} & \text{य दि n सम है,} \end{cases}$$

और

$$f(n) = \begin{cases} 3+2n & \text{य दि} \quad n \geq 0 \text{ है} \\ -2n & \text{य दि} \quad n < 0 \text{ है} \end{cases}$$

द्वारा परिभाषित हैं, पर विचार कीजिए। सभी $n \in N$ के लिए $(g \cdot f)(n) = g(f(n))$, और सभी $n \in Z$ के लिए $(f \cdot g)(n) = f(g(n))$ परिभाषित कीजिये।

तब निम्नलिखित कथनों में से कौन सा (से) सत्य है (हैं) ? **[JEE(Adv.)2025]**

(A) $g \cdot f$ कएकैकी **नहीं** है और $g \cdot f$ आच्छादक भी **नहीं** है

(B) $f \cdot g$ एकैकी **नहीं** है परंतु $f \cdot g$ आच्छादक है

(C) g एकैकी है और g आच्छादक भी है

(D) f एकैकी **नहीं** है परंतु f आच्छादक है

Ans. AD

Sol. $f(n)=\begin{cases}\dfrac{n+1}{2} & n \to \text{odd} \\ \dfrac{4-n}{2} & n \to \text{even}\end{cases}$ $\qquad N \to Z$

$f(1) = 1$ $\qquad$ $f(5) = 3$

$f(2) = 1$ $\qquad$ $f(6) = -1$ $\qquad$ not one-one and onto

$f(3) = 2$ $\qquad$ $f(7) = 4$

$f(4) = 0$ $\qquad$ $f(8) = -2$

$g(n)=\begin{cases}3+2n, & n \geq 0 \\ -2n, & n < 0\end{cases}$

$g(0) = 3$ $\qquad$ $g(-1) = 2$

$g(1) = 5$ $\qquad$ $g(-2) = 4$ $\qquad$ not one-one and into

$g(2) = 7$ $\qquad$ $g(-3) = 6$

$g(3) = 9$ $\qquad$ $g(-4) = 8$

fog(x)

$-4 \Rightarrow -2$

$-3 \Rightarrow -1$

$-2 \Rightarrow 0$

$-1 \Rightarrow 1$ $\qquad$ one-one and onto

$0 \Rightarrow 2$

$1 \Rightarrow 3$

$2 \Rightarrow 4$

$3 \Rightarrow 5$

gof(x)

$1 \Rightarrow 5$

$2 \Rightarrow 5$

$3 \Rightarrow 7$

$4 \Rightarrow 3$

$5 \Rightarrow 9$

$6 \Rightarrow 2$

$7 \Rightarrow 11$

$8 \Rightarrow 4$

not one-one and into

14. Let R denote the set of all real numbers. Let $f : R \to R$ be a function such that $f(x) > 0$ for all $x \in R$, and $f(x + y) = f(x) f(y)$ for all $x, y \in R$.

Let the real numbers $a_1, a_2, ..., a_{50}$ be in an arithmetic progression. If $f(a_{31}) = 64f(a_{25})$, and $\sum_{i=1}^{50} f(a_i) = 3\left(2^{25}+1\right)$,

then the value of $\sum_{i=6}^{30} f(a_i)$ is _____. **[JEE(Adv.)2025]**

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है। मान लीजिए कि $f : R \to R$ एक इस प्रकार का फलन है कि सभी $x \in R$ के लिए $f(x) > 0$ है, और सभी $x, y \in R$ के लिए $f(x + y) = f(x) f(y)$ है।

मान लीजिए कि वास्तविक संख्याएँ $a_1, a_2, ..., a_{50}$ एक समांतर श्रेणी में हैं। यदि $f(a_{31}) = 64f(a_{25})$ है, और $\sum_{i=1}^{50} f(a_i) = 3\left(2^{25}+1\right)$,

है, तब $\sum_{i=6}^{30} f(a_i)$ का मान _____ है। **[JEE(Adv.)2025]**

Ans. 96

Sol. $f : R \to R$

$f(x) > 0 \quad \forall x \in R$

$f(x + y) = f(x).f(y) \ \forall x, y \in R$

$\Rightarrow f(x) = a^{\lambda x}$; $\lambda \in R$

Now $a_1, a_2, \ldots\ldots\ldots a_{50} \rightarrow$ A.P.

$f(a_{31}) = 64\ f(a_{25})$

$a^{\lambda(A + 30D)} = 64\ a^{\lambda(A + 24\ D)}$

$a^{\lambda A} . .a^{30\lambda D} = 64\ a^{\lambda A} . a^{\lambda 24D}$

$a^{6\lambda D} = 64$ .....(1)

Again $6a^{6\lambda D} = 2^6 \Rightarrow a = 2$; $\lambda D = 1$

$f(a_1) + f(a_2) + f(a_3) + \ldots\ldots\ldots + f(a_{50}) = 3(2^{25} + 1)$

$a^{\lambda A} + a^{\lambda(A + D)} + a^{\lambda(A + 2D)} + \ldots\ldots\ldots + a^{\lambda(A + 49D)} = 3(2^{25} + 1)$

$a^{\lambda A} (1 + a^{\lambda D} + a^{2\lambda D} + \ldots\ldots\ldots + a^{49\lambda D}) = 3(2^{25} + 1)$

$$a^{\lambda A}\left\{\frac{1-\left(a^{\lambda D}\right)^{50}}{1-a^{\lambda D}}\right\} = 3\left(2^{25}+1\right)$$

$$2^{\lambda A}\left\{\frac{1-(2)^{50}}{1-2}\right\} = 3\left(2^{25}+1\right)$$

$$2^{\lambda A}\left\{\left(2^{25}-1\right)\left(2^{25}+1\right)\right\} = 3\left(2^{25}+1\right)$$

$2^{\lambda A} (2^{25} - 1) = 3$

$$2^{\lambda A} = \frac{3}{2^{25}-1}$$

$$\sum_{i=6}^{30} f\left(a_i\right) = f\left(a_6\right) + f\left(a_7\right) + \ldots + f\left(a_{30}\right)$$

$= a^{\lambda(A + 5D)} + a^{\lambda(A + 6D)} + \ldots\ldots\ldots + a^{\lambda(A + 29D)}$

$= a^{\lambda A}\ \{a^{5\lambda D} + a^{6\lambda D} + \ldots\ldots\ldots + a^{29\lambda D}\}$

$= 2^{\lambda A}\ \{2^5 + 2^6 + \ldots\ldots\ldots + 2^{29}\}$

$$\Rightarrow 2^{\lambda A}\left\{\frac{2^5\left(1-2^{25}\right)}{1-2}\right\}$$

$$\frac{3}{2^{25}-1} \times \frac{2^5\left(2^{25}-1\right)}{1}$$

$= 32 \times 3$

$= 96$

15. Let S = {1, 2, 3, ..., 10}. Consider the set X = {R : R is an equivalence relation on the set S such that R has exactly 42 elements}. Then the number of elements in X is _____. **[JEE(Adv.)2026]**

मान लीजिए कि S = {1, 2, 3, ..., 10} है। समुच्चय X = {R : R, समुच्चय S पर एक ऐसा तुल्यता संबंध है कि R में ठीक 42 अवयव हैं } पर विचार कीजिए। तब X में अवयवों की संख्या _____है। **[JEE(Adv.)2026]**

Ans. 2520

Sol. A relation R on set S = {1, 2, ..., 10} has exactly 42 ordered pairs (a, b).

1. Reflexivity: For every element $a \in S$, the pair $(a, a) \in R$.

   Since there are 10 elements in S, all 10 diagonal pairs– (1,1), (2,2), ..., (10,10) must be in R.

   Remaining elements to choose = 42 – 10 = 32

2. Symmetry: If $(a, b) \in R$ (where $a \neq b$), then (b, a) must also be in R.

   This means the remaining 32 elements must come in pairs.

   Number of symmetric pairs (a, b) with $a \neq b = \frac{32}{2} = 16$

3. Transitivity: If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

Concurrently with symmetry, this means if any set of elements are related to one another, every single element in that subset must be related to every other element in that same subset. They form closed, mutually related groups.

If a mutually related group has x elements, it will contribute exactly x^2 ordered pairs to the relation R (since every element pairs with all x elements, including itself).

Let the 10 distinct elements of S be divided into k mutually related groups, with sizes $x_1, x_2, ..., x_k$.

1. Total number of elements: $x_1 + x_2 + ... + x_k = 10$

2. Total number of ordered pairs: $x_1^2 + x_2^2 + \ldots + x_k^2 = 42$

Case I: Group sizes of 6, 2, 1, 1 $\Rightarrow (6^2 + 2^2 + 1^2 + 1^2 = 36 + 4 + 1 + 1 = 42)$

Case II: Group sizes of 5, 4, 1 $\Rightarrow (5^2 + 4^2 + 1^2 = 25 + 16 + 1 = 42)$

the number of ways to divide n distinct objects into groups of sizes p, q, r, ... is given by the formula:

$$\text{Number of ways} = \frac{n!}{(p! \cdot q! \cdot r! \ldots) \times a! \cdot b! \ldots}$$

(where a, b, ... represent the number of times a group size repeats).

Calculation for Case I: Group sizes (6, 2, 1, 1)

$$\text{Ways}_1 = \frac{10!}{6!\cdot 2!\cdot 1!\cdot 1!\times 2!} \Rightarrow \quad \text{Ways}_1 = \frac{10\times 9\times 8\times 7\times 6!}{6!\times 2\times 1\times 2} = \frac{5040}{4} = 1260$$

Calculation for Case II: Group sizes (5, 4, 1)

$$\text{Ways}_2 = \frac{10!}{5!\cdot 4!\cdot 1!} \Rightarrow \quad \text{Ways}_2 = \frac{10\times 9\times 8\times 7\times 6\times 5!}{5!\times 24} = \frac{30240}{24} = 1260$$

Total number of relations = $\text{Ways}_1 + \text{Ways}_2$

Total = 1260 + 1260 = 2520

16. Let N denote the set of all positive integers. Consider the sets

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{1, 2, 3, 4, 5, 6, 7\}$$

Let S be the set of all functions $f: A \to B$ such that $f(2) \neq 2$ and $f(4) \neq 4$. Consider the set $T = \{f \in S$: there exists a function $g: B \to N$ such that $g(f(x)) = 2^x$ for all $x \in A\}$. Then the number of elements in the set T is _____.

[JEE(Adv.)2026]

मान लीजिए कि N सभी धन पूर्णांकों के समुच्चय को दर्शाता है। समुच्चयों

$$A = \{1, 2, 3, 4, 5\} \text{ और } B = \{1, 2, 3, 4, 5, 6, 7\}$$

पर विचार कीजिए। मान लीजिए कि S उन सभी फलनों $f: A \to B$, जिनके लिए $f(2) \neq 2$ और $f(4) \neq 4$ हैं, का समुच्चय है। समुच्चय $T = \{f \in S$: एक ऐसे फलन $g: B \to N$ का अस्तित्व है कि सभी $x \in A$ के लिए $g(f(x)) = 2^x$ है$\}$ पर विचार कीजिए।

तब समुच्चय T में अवयवों की संख्या _____ है। **[JEE(Adv.)2026]**

Ans. 1860

Sol. $g(f(x)) = 2^x \ \forall\ x \in A$

because 2^x is one-one function.

So, f(x) must be one-one function.

So, total no. of one-one function = $7 \times 6 \times 5 \times 4 \times 3 = 2520$

but $f(2) \neq 2$ and $f(4) \neq 4$

So, required function = Total – (when f(2) = 2 OR f(4) = 4) + (when f(2) = 2 and f(4) = 4)

$= 2520 - ({}^6C_4 \times 4! + {}^6C_4 \times 4!) + {}^5C_3 \times 3!$

$= 2520 - (360 + 360) + 60$

$= 1860$

Exercise C - I

1. The mapping $f : R \to R$ given by $f(x) = x^3 + ax^2 + bx + c$ is a bijection if :

प्रतिचित्रण $f : R \to R$, $f(x) = x^3 + ax^2 + bx + c$ द्वारा परिभाषित फलन एकैकी आच्छादक है यदि :

(A) $b^2 \le 3a$ (B) $a^2 \le 3b$ (C) $a^2 \ge 3b$ (D) $b^2 \ge 3a$

Ans. B

Sol. $f : R \to R$, $f(x) = x^3 + ax^2 + bx + c$

$f'(x) = 3x^2 + 2ax + b$

$D \le 0$

$4a^2 - 12b \le 0$

$a^2 \le 3b$

2. If $f : [0, \infty) \to [0, \infty)$ and $f(x) = \frac{x}{1+x}$, then f is :

(A) one-one and onto (B) one-one but not onto

(C) onto but not one-one (D) neither one-one nor onto

यदि $f : [0, \infty) \to [0, \infty)$ एवं $f(x) = \frac{x}{1+x}$ हो, तो f :

(A) एकैकी और आच्छादक है (B) एकैकी लेकिन आच्छादक नहीं है

(C) आच्छादक लेकिन एकैकी नहीं है (D) न तो एकैकी न ही आच्छादक है

Ans. B

Sol. $f : [0, \infty) \to [0, \infty) \quad \Rightarrow \quad f(x) = \frac{x}{1+x}$

$\frac{x_1}{1+x_1} = \frac{x_2}{1+x_2} \Rightarrow \quad x_1 = x_2$ only

for given domain

$f(x) < 1 \quad \therefore \quad$ function is into

3. Suppose $f(x) = (x + 1)^2$ for $x \ge -1$. If $g(x)$ is the function whose graph is the reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals :

माना $f(x) = (x + 1)^2$ for $x \ge -1$ यदि फलन $g(x)$ का आरेख $f(x)$ के आरेख का रेखा $y = x$ के सापेक्ष प्रतिबिम्ब है, तो $g(x)$ होगा:

(A) $-\sqrt{x} - 1, x \ge 0$ (B) $\frac{1}{(x+1)^2}, x > -1$ (C) $\sqrt{x+1}, x > -1$ (D) $\sqrt{x} - 1, x \ge 0$

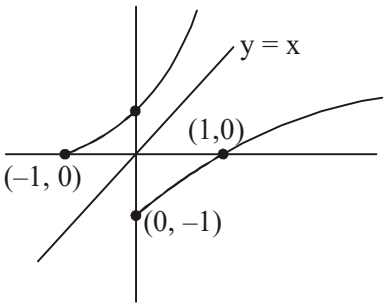
Ans. D

Sol. $f(x) = (x+1)^2 \qquad x \geq -1$

$x \leftrightarrow y$

$x = (y+1)^2$

$\Rightarrow (y+1) = \sqrt{x}$ or $\underbrace{y+1 = -\sqrt{x}}_{\text{Not possible}}$



$\Rightarrow \underbrace{y = \sqrt{x} - 1}_{x \geq 0}$

4. Let function $f : R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is :

(A) one-one and onto (B) one-one but not onto

(C) onto but not one-one (D) neither one-one nor onto

माना फलन $f : R \to R$ इस प्रकार कि $f(x) = 2x + \sin x$ for $x \in R$ तो f होगा :

(A) एकैकी तथा आच्छादक (B) एकैकी परन्तु आच्छादक नहीं

(C) आच्छादक परन्तु एकैकी नही (D) ना तो एकैकी ना ही आच्छादक

Ans. A

Sol. $f(x) = 2x + \sin x$

For one-one

$f'(x) = \underbrace{2 + \cos x}_{>0 \quad \forall x \in R}$

$\Rightarrow$ f is one-one.

$\because \sin x \in [-1, 1]$ & $x \in (-\infty, \infty)$

& $2x + \sin x \in (-\infty, \infty)$

$\Rightarrow$ f is onto.

5. Let $g(x) = 1 + x - [x]$ & $f(x) = \begin{cases} -1 & , \quad x < 0 \\ 0 & , \quad x = 0 \\ 1 & , \quad x > 0 \end{cases}$. Then for all x, f(g (x)) is equal to :

माना $g(x) = 1 + x - [x]$ एवं $f(x) = \begin{cases} -1 & , x < 0 \\ 0 & , x = 0 \\ 1 & , x > 0 \end{cases}$. हो तो x के लिए, f(g (x)) =

(A) x (B) 1 (C) f(x) (D) g(x)

Ans. B

Sol. $g(x) = 1 + \{x\}$

$0 \le \{x\} < 1$

$\Rightarrow 1 + \{x\} \in [1, 2)$

$\Rightarrow f(g(x)) = 1$

6. If f: $[1, \infty) \to [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals :

यदि $f : [1, \infty) \to [2, \infty)$, $f(x) = x + \frac{1}{x}$, द्वारा परिभाषित हो तो $f^{-1}(x) =$

(A) $\frac{x + \sqrt{x^2 - 4}}{2}$ (B) $\frac{x}{1 + x^2}$ (C) $\frac{x - \sqrt{x^2 - 4}}{2}$ (D) $1 - \sqrt{x^2 - 4}$

Ans. A

Sol. $f : [1, \infty) \to [2, \infty)$

$f(x) = x + \frac{1}{x}$

Clearly f is one-one & onto.

$y = x + \frac{1}{x}$

$\Rightarrow y \leftrightarrow x$

$x = y + \frac{1}{y}$

$\Rightarrow y^2 - xy + 1 = 0$

$y = \frac{x \pm \sqrt{x^2 - 4}}{2}$

$y = \frac{x + \sqrt{x^2 - 4}}{2}$ & $\underbrace{y = \frac{x - \sqrt{x^2 - 4}}{2}}_{\text{wrong}}$

$\Rightarrow f^{-1}(x)=\dfrac{x+\sqrt{x^2-4}}{2}$ at $x=\dfrac{5}{2}$

$$y=\frac{\frac{5}{2}-\frac{3}{2}}{2}=\frac{1}{2}\notin[1,\infty)$$

7. Let $f(x)=\dfrac{\alpha x}{x+1}$, $x\neq -1$. Then for what value of α is $f(f(x))=x$?

माना $f(x)=\dfrac{\alpha x}{x+1}$, $x\neq -1$ है तो $f(f(x))=x$ के लिए α का मान है?

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) -1

Ans. D

Sol. $f(x)=\dfrac{\alpha x}{x+1}$, $x\neq -1$

Now $f(f(x))=x \Rightarrow f(x)=f^{-1}(x)$

Let $y=\dfrac{\alpha x}{x+1} \Rightarrow xy+y=\alpha x$

$\Rightarrow \quad x(y-\alpha)=-y \quad \Rightarrow \quad x=\dfrac{-y}{y-\alpha}$

$f^{-1}(x)=\dfrac{-x}{x-\alpha}$

Now $\dfrac{\alpha x}{x+1}=\dfrac{-x}{x-\alpha}$

on solving we get $\alpha=-1$

8. If the function $f: [1,\infty)\to[1,\infty)$ is defined by $f(x)=2^{x(x-1)}$, then $f^{-1}(x)$ is :

(A) $\left(\dfrac{1}{2}\right)^{x(x-1)}$ (B) $\dfrac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$

(C) $\dfrac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (D) not defined

यदि फलन f: $[1, \infty) \to [1, \infty)$, $f(x) = 2^{x(x-1)}$ द्वारा परिभाषित हो तो $f^{-1}(x) =$:

(A) $\left(\frac{1}{2}\right)^{x(x-1)}$ (B) $\frac{1}{2}\left(1+\sqrt{1 + 4 \log_2 x}\right)$

(C) $\frac{1}{2}\left(1-\sqrt{1+4 \log_2 x}\right)$ (D) परिभाषित नहीं

Ans. B

Sol. $f(x) = 2^{x(x-1)}$

It is one-one onto function

$\log_2 y = x(x-1)$

$\Rightarrow \quad x^2 - x - \log_2 y = 0$

$$x = \frac{1 \pm \sqrt{1+4\log_2 y}}{2}$$

$$f^{-1}(x) = \frac{1+\sqrt{1+4\log_2 x}}{2}$$

9. Let $f(x) = (x + 1)^2 - 1, (x \geq -1)$. Then the set $S = \{x : f(x) = f^{-1}(x)\}$ is, if f is onto :

(A) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$ (B) $\{0, 1, -1\}$

(C) $\{0, -1\}$ (D) empty

माना $f(x) = (x + 1)^2 - 1, (x \geq -1)$ तो समुच्चय $S = \{x : f(x) = f^{-1}(x)\}$ होगा, (यदि फलन f आच्छादक है) :

(A) $\left\{0, -1, \frac{-3+i\sqrt{3}}{2}, \frac{-3-i\sqrt{3}}{2}\right\}$ (B) $\{0, 1, -1\}$

(C) $\{0, -1\}$ (D) रिक्त

Ans. C

Sol. $f(x) = (x + 1)^2 - 1 \qquad (x \geq -1)$

$f: [-1,\infty) \to [-1, \infty)$

$y = (x + 1)^2 - 1$

For inverse

$x = (y + 1)^2 - 1$

$\Rightarrow (y + 1)^2 = x + 1$

$\Rightarrow y + 1 = \sqrt{x+1}$

$\Rightarrow f^{-1}(x) = -1 + \sqrt{x+1}$

$\Rightarrow (x+1)^2 - 1 = -1 + \sqrt{x+1}$

$\Rightarrow (x+1)^2 = \sqrt{x+1}$

$\Rightarrow (x+1)^4 = x+1$

let $x + 1 = t$

$t^4 - t = 0$

$t = 0$ or $t^3 = 1$

$\Rightarrow t = 0$ or $t = 1$

$\Rightarrow x = -1$ or $x = 0$

10. Let $f(x) = \sin x$ and $g(x) = |\ell n\, x|$ if composite functions $fog(x)$ and $gof(x)$ are defined and have ranges R_1 and R_2 respectively then :

(A) $R_1 = \{u: -1 < u < 1\}$ $R_2 = \{v: 0 < v < \infty\}$

(B) $R_1 = \{u: -\infty < u \le 0\}$ $R_2 = \{v: -1 \le v \le 1\}$

(C) $R_1 = \{u: 0 \le u < \infty\}$ $R_2 = \{v: -1 < v < 1; v \ne 0\}$

(D) $R_1 = \{u: -1 \le u \le 1\}$ $R_2 = \{v: 0 \le v < \infty\}$

माना $f(x) = \sin x$ एवं $g(x) = |\ell n\, x|$, यदि संयुक्त फलन $fog(x)$ एवं $gof(x)$ परिभाषित हो तथा इनके परिसर क्रमशः R_1 एवं R_2 हो, तो :

(A) $R_1 = \{u: -1 < u < 1\}$ $R_2 = \{v: 0 < v < \infty\}$

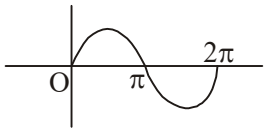
(B) $R_1 = \{u: -\infty < u \le 0\}$ $R_2 = \{v: -1 \le v \le 1\}$

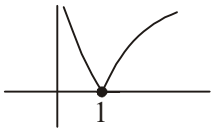
(C) $R_1 = \{u: 0 \le u < \infty\}$ $R_2 = \{v: -1 < v < 1; v \ne 0\}$

(D) $R_1 = \{u: -1 \le u \le 1\}$ $R_2 = \{v: 0 \le v < \infty\}$

Ans. D

Sol. $f(x) = \sin x$ $g(x) = \sin x$





$fog(x) = f(|\ln x|) = \sin(|\ln x|) \in [-1,1]$ as $|\ln x| \in [0, \infty)$

$gof(x) = g(\sin x) = |\ln(\sin x)|$

For $\log_e(\sin x)$; $\sin x > 0$

$\Rightarrow 0 < \sin x \le 1$

$-\infty < \ln(\sin x) \le 0$

$\Rightarrow gof(x) = |\ln(\sin x)| \in [0, \infty)$

11. If the functions f (x) and g (x) are defined on R $\to$ R such that f (x) = $\begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{irrational} \end{cases}$,

g (x) = $\begin{cases} 0, & x \in \text{irrational} \\ x, & x \in \text{rational} \end{cases}$ then (f – g)(x) is :

(A) one-one and onto (B) neither one-one nor onto

(C) one-one but not onto (D) onto but not one-one

f(x) व g(x) R $\to$ R इस प्रकार परिभाषित है कि f (x) = $\begin{cases} 0, & x \in \textit{परिमेय} \\ x, & x \in \textit{अपरिमेय} \end{cases}$, g (x) = $\begin{cases} 0, & x \in \textit{अपरिमेय} \\ x, & x \in \textit{परिमेय} \end{cases}$ तो (f – g)(x) है:

(A) एकैकी, आच्छादक (B) बहुएकैकी, अन्तःक्षेपी

(C) एकैकी व अन्तःक्षेपी (D) आच्छादक व बहुएकैकी

Ans. A

Sol. $\phi(x) = f(x) - g(x)$

$= \begin{cases} -x & x \in Q \\ x & x \notin Q \end{cases}$

It is one-one onto function

12. Which of the following statements are incorrect?

I If f(x) and g(x) are one-one then f(x) + g(x) is also one-one.

II If f(x) and g(x) are one-one then f(x) · g(x) is also one-one.

III If f(x) is odd then it is necessarily one-one.

(A) I and II only (B) II and III only (C) III and I only (D) I, II and III

निम्न में से कौनसे कथन असत्य है?

I यदि f (x) तथा g (x) एकैकी फलन है तो f (x) + g(x) भी एकैकी फलन होगा।

II यदि f (x) तथा g (x) एकैकी फलन है तो f (x) · g(x) भी एकैकी फलन होगा।

III यदि f (x) विषम फलन है तो यह निश्चित रूप से एकैकी फलन होगा।

(A) केवल I तथा II (B) केवल III तथा I (C) केवल III तथा I (D) I, II तथा III

Ans. D

Sol. (I) $\underbrace{f(x)=x}_{\text{one-one}}$ $\underbrace{g(x)=-x}_{\text{one-one}}$

$\Rightarrow f(x)+g(x)=0 \rightarrow$ Not one-one

(II) $f: R \rightarrow R$ $g: R-\{0\} \rightarrow R$

$f(x)=x \rightarrow$ one-one $g(x)=\frac{1}{x} \rightarrow$ one-one

$\Rightarrow f(x) \cdot g(x)=1 \rightarrow$ Not one-one

(III) Let $f(x)=\sin x \rightarrow$ odd function but not one-one

13. Which of the following equations have the same graphs?

I. $y=x-2$ II. $y=\frac{(x^2-4)}{(x+2)}$ III. $(x+2)y=x^2-4$

(A) I and II only (B) I and III only

(C) II and III only (D) All the equations have different graphs

निम्न में से कौनसी समीकरणों का आरेख समान होगा?

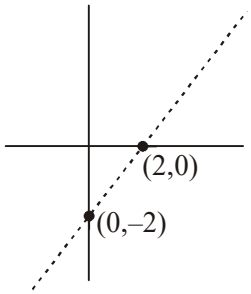
I. $y=x-2$ II. $y=\frac{(x^2-4)}{(x+2)}$ III. $(x+2)y=x^2-4$

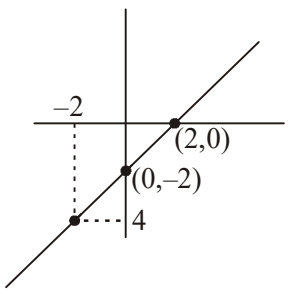
(A) केवल I तथा II (B) केवल I तथा III

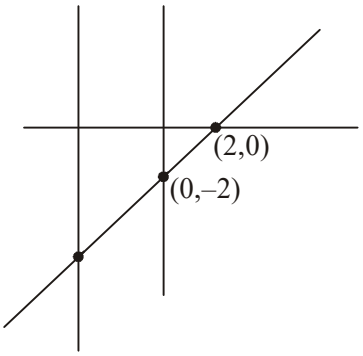
(C) केवल II तथा III (D) सभी समीकरणों के आरेख भिन्न है

Ans. D

Sol. $y=x-2 \rightarrow$ Range $=R$ $\underbrace{y=\frac{x^2-4}{x+2}}_{y \in R-\{-4\}}$ $(x+2)\,y=x^2-4$



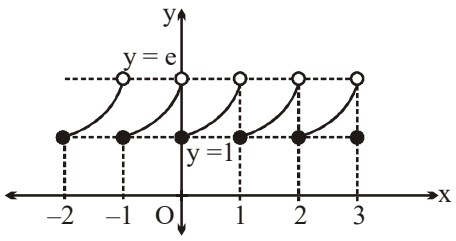


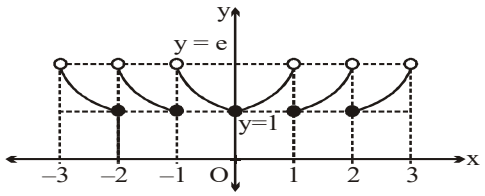


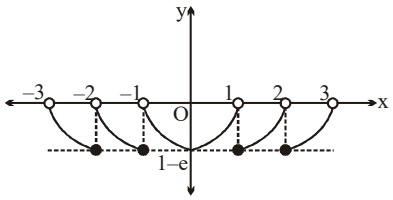
g(x)= –2

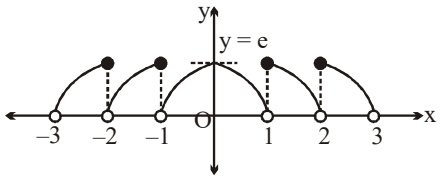
14. Which one of the following best represent the graph of function $f(x) = e^{\{|x|\}}$: [Note: $\{\alpha\}$ denotes the fractional part of α.]

निम्न में से कौनसा आरेख फलन $f(x) = e^{\{|x|\}}$ का सबसे उतम निरूपण है : [Note : $\{\alpha\}$, α के आंशिक भाग को निरूपित करता है]

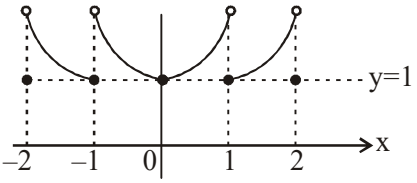
(A) 

(B) 

(C) 

(D) 

Ans. B

Sol. 

$f(x) = e^{\{|x|\}}$

$0 \le \{|x|\} < 1$

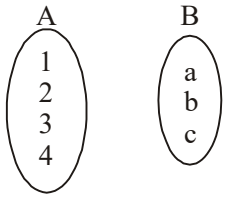
$1 \le e^{\{|x|\}} < e$

15. Set A consists of 4 distinct and set-B consists of 3 distinct elements. Number of functions that can be defined from a set $A \to B$ which are not onto is :

समुच्चय A में 4 भिन्न तथा समुच्चय B में 3 भिन्न अवयव है। समुच्चय $A \to B$ में परिभाषित उन फलनों की संख्या ज्ञात करो जो आच्छादक नहीं है :

(A) 35 (B) 45 (C) 63 (D) 9

Ans. B

Sol. 

No. of function = $3^4 = 81$

No. of onto function

$$= \frac{\lfloor 4}{(\lfloor 1)^2 \times \lfloor 2 \cdot \lfloor 2} \times \lfloor 3 = \lfloor 3 \times \lfloor 3 = 36$$

$\Rightarrow$ No. of into function = 81 – 36 = 45

ONE OR MORE THAN ONE CORRECT TYPE

16. Let $f : R \to R$ be defined as $f(x) = \sqrt[3]{x} + \sin 2x$ and $g : R \to R$ is defined by $g(x) = \frac{1-x^2}{1+x^2}$, then which of the following statement(s) is (are) correct?

(A) f is surjective
(B) f(x) + g(x) is neither even nor odd
(C) Range of g(x) is [–1, 1]
(D) f is bounded and g is unbounded

माना $f : R \to R$ इस प्रकार परिभाषित है कि $f(x) = \sqrt[3]{x} + \sin 2x$ तथा $g : R \to R$ इस प्रकार परिभाषित है कि $g(x) = \frac{1-x^2}{1+x^2}$, तो निम्न में से कौनसा/कौनसे कथन सत्य है?

(A) f आच्छादक है
(B) f(x) + g(x) न सम फलन है और न विषम फलन है
(C) g(x) का परिसर [–1, 1] है
(D) f परिबद्ध है तथा g अपरिबद्ध है

Ans. AB

Sol. $f : R \to R$

$$f(x) = x^{\frac{1}{3}} + \sin 2x$$

odd & many-one

$g : R \to R$

$$g(x) = \frac{1-x^2}{1+x^2}$$

even

$\Rightarrow$ f(x) + g(x) is neither even nor odd

$\Rightarrow f(x) \in (-\infty,\infty)$ as $x^{\frac{1}{3}} \in (-\infty,\infty)$ and $\sin 2x \in [-1,1]$

$\Rightarrow$ f(x) is surjective.

$\therefore$ f is not bounded.

$$g(x) = \frac{1-x^2}{1+x^2} = \frac{2}{\underbrace{1+x^2}_{\in(0,2]}} - 1$$

$$\Rightarrow g(x) \in (-1,1]$$

17. Let $f(x) = [x]^2 + [x+1] - 3$, where $[x]$ denotes greatest integer less than or equal to x, then which of the following statement(s) is/are CORRECT?

(A) f(x) is many one function
(B) f(x) vanishes for atleast three values of x
(C) f(x) is neither even nor odd function
(D) f(x) is aperiodic

माना $f(x) = [x]^2 + [x+1] - 3$, जहाँ $[x]$ महतम पूर्णांक x से छोटा या बराबर है, तो निम्न में से कौनसा/कौनसे कथन सत्य है?

(A) f(x) बहुएकैकी फलन है
(B) f(x) कम से कम x के तीन मानों के लिए शून्य है
(C) f(x) न सम फलन है और न विषम फलन है
(D) f(x) अनावृति फलन है

Ans. ABCD

Sol. We have $f(x) = [x]^2 + [x] - 2 \Rightarrow f(x) = ([x]+2)([x]-1) = \begin{cases} -2, & 0 \le x < 1 \\ 0, & 1 \le x < 2 \end{cases}$ and so on.

$\Rightarrow$ f(x) is many one function.

Also $f(x) = 0 \Rightarrow x \in [-2,-1) \cup [1,2) \Rightarrow f(x) = 0$ has infinite solutions.

As $f(-x) \neq f(x)$ and $f(-x) \neq -f(x) \Rightarrow$ f(x) is neither even nor odd function.

Now $f(x) = [x]^2 + [x] - 2$, where $[x]^2 + [x]$ is aperiodic.

$\Rightarrow$ f(x) is aperiodic.

18. Let $f: [-1, 1]$ onto $[3, 5]$ be a linear polynomial. Which of the following can be true?

माना एक आच्छादक फलन $f: [-1, 1] \to [3, 5]$ एक रैखिक बहुपद है। निम्न में से कौनसा कथन सत्य हो सकता है?

(A) $f\left(\frac{-1}{2}\right) = \frac{7}{2}$ (B) $f^{-1}\left(\frac{15}{4}\right) = \frac{1}{4}$ (C) $f(0) \neq 4$ (D) $f\left(\frac{1}{2}\right) + f\left(\frac{-1}{2}\right) = 8$

Ans. ABD

Sol. $f: [-1,1] \xrightarrow{\text{onto}} [3,5]$

Let $f(x) = ax + b$

$\because x \in [-1,1]$

$\Rightarrow ax \in [-a, a]$ if $a > 0$

$\Rightarrow ax + b \in [b-a, a+b]$

$\Rightarrow b - a = 3$ & $a + b = 5$

$\Rightarrow b = 4 \quad a = 1$

$\Rightarrow f(x) = x + 4$

$y = x + 4$

$x = y + 4$

$\Rightarrow f^{-1}(x) = x - 4$

$f^{-1}\left(\frac{15}{4}\right) = \frac{15}{4} - 4 = -\frac{1}{4}$

$$f\left(-\frac{1}{2}\right)=-\frac{1}{2}+4=\frac{7}{2}$$

$$f\left(\frac{1}{2}\right)+f\left(-\frac{1}{2}\right)=\frac{9}{2}+\frac{7}{2}=8$$

$f(0) = 4$

If $a < 0$

then $ax \in [a, -a]$

$\Rightarrow f(x) \in [a + b, b - a]$

$\Rightarrow a + b = 3$ & $b - a = 5$

$\Rightarrow b = 4$ & $a = -1$

$\therefore f(x) = - x + 4$

$\Rightarrow y = - x + 4$

$\Rightarrow x = -y + 4$

$\Rightarrow y = 4 - x = f^{-1}(x)$

$$\therefore f^{-1}\left(\frac{15}{4}\right)=\frac{16-15}{4}=\frac{1}{4}$$

19. Let $f: R \to R$ and $g: R \to R$ be mappings such that $f(g(x))$ is an injective mapping, then which of the following statement(s) is(are) correct?

(A) g(x) must be an injective mapping

(B) If g(x) is surjective then f(x) may not be injective mapping

(C) If g(x) is not surjective, then f(x) may be injective or many-one mapping

(D) f(x) may not be an injective mapping

माना $f: R \to R$ तथा $g: R \to R$ फलन इस प्रकार है कि $f(g(x))$ एक एकैकी फलन है, तो निम्न में से कौनसा/कौनसे कथन सत्य है?

(A) g(x) एकैकी फलन होगा

(B) यदि g(x) आच्छादक है तो f(x) एकैकी फलन ना होने की संभावना है

(C) यदि g(x) आच्छादक नहीं है, तो f(x) के एकैकी या बहुएकैकी फलन होने की संभावना है

(D) f(x) के एकैकी फलन ना होने की संभावना है

Ans. ACD

Sol. (A) fog is one-one

$\Rightarrow$ g(x) must be one-one.

(B) If g(x) is surjective

then $R_g = R$.

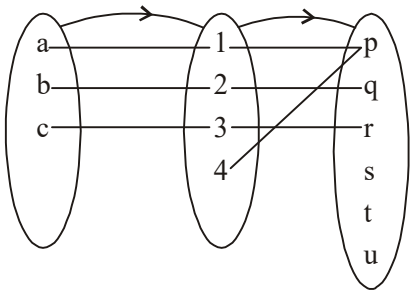
If f(x) is many-one then $f(x_1) = f(x_2) \Rightarrow x_1 \neq x_2$

$\Rightarrow$ fog is not injective

(C) If g(x) is not surjective, then $R_g \subset R$

(D) If g(x) is surjective then f is one-one.

Not surjective then f is many-one or one-one.



20. Let $f(x) = \begin{cases} x^2 - 4, & \text{if } |x| \leq 3 \\ 5\,\text{sgn}|x-3|, & \text{if } |x| > 3 \end{cases}$ and $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ for all $x \in R$, then which of the following is (are) correct? [Note : sgn(k) denotes the signum function of k.]

(A) fog(x) is an even function

(B) gof(x) is an even function

(C) gog(x) is an odd function

(D) fof(x) is an odd function

माना $f(x) = \begin{cases} x^2 - 4, & \text{यदि } |x| \leq 3 \\ 5\,\text{sgn}|x-3|, & \text{यदि } |x| > 3 \end{cases}$ तथा $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ सभी $x \in R$ के लिए, तो निम्न में से कौनसा/कौनसे कथन सत्य है? [Note : sgn(k), k के सिग्नम फलन को निरूपित करता है]

(A) fog(x) सम फलन है

(B) gof(x) सम फलन है

(C) gog(x) विषम फलन है

(D) fof(x) विषम फलन है

Ans. ABC

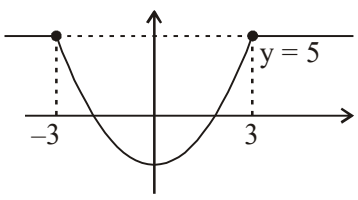
Sol. $f(x) = \begin{cases} x^2 - 4 & |x| \leq 3 \\ 5\,\text{sgn}|x-3| & |x| > 3 \end{cases} = \begin{cases} x^2 - 4 & |x| \leq 3 \\ 5 & |x| > 3 \end{cases}$

$\Rightarrow$ f(x) is an even function.

$\Rightarrow$ gof(x) is an even function & fof(x) is an even function.

$g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$

$\because e^x \in (0, \infty)$



$\Rightarrow g(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

fog(x) is an even function.

as $g(-x) = 2\tan^{-1}\left(\frac{1}{e^x}\right) - \frac{\pi}{2} = 2\cot^{-1}\left(e^x\right) - \frac{\pi}{2}$

$= 2\left(\frac{\pi}{2} - \tan^{-1}\left(e^x\right) - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2\tan^{-1}\left(e^x\right) = -g(x)$

& gog(x) is an odd function.

21. Let f(x) is a polynomial function defined R to R with leading coefficient 2 and degres 5 such that $f(1) = 2$, $f(2) = 3$, $f(3) = 4$, $f(4) = 5$ and $f(5) = 6$, then which of the following hold(s) good?

(A) $f(x)$ is many-one onto function (B) $f(x)$ is one-one onto function

(C) $f(6) = 127$ (D) $f(0) = -239$

माना R से R में परिभाषित बहुपद फलन $f(x)$ जिसका मुख्य गुणांक 2 तथा घात 5 इस प्रकार है कि $f(1) = 2$, $f(2) = 3$, $f(3) = 4$, $f(4) = 5$ तथा $f(5) = 6$ है, तो निम्न में से कौनसा/कौनसे कथन सत्य होगा/होगें?

(A) $f(x)$ बहुएकैकी आच्छादक फलन है (B) $f(x)$ एकैकी आच्छादक फलन है

(C) $f(6) = 127$ (D) $f(0) = -239$

Ans. AD

Sol. Let $g(x) = f(x) - (x + 1)$

$\Rightarrow g(1) = 0, g(2) = 0, g(3) = 0, g(4) = 0, g(5) = 0$

$\Rightarrow g(x) = 2(x - 1)(x - 2)(x - 3)(x - 4)(x - 5)$

$\Rightarrow f(x) = 2(x - 1)(x - 2)(x - 3)(x - 4)(x - 5) + (x + 1)$

$\therefore f(0) = -2 \times 2 \times 3 \times 4 \times 5 + 1 = -239$

$f(6) = 2 \times 5 \times 4 \times 3 \times 2 \times 1 + 7 = 247$

$\therefore$ f(x) is many one onto function

as $f\left(\frac{5}{2}\right) = \frac{11}{6} < 1$

Exercise C - II

1. Let f be a one–one function with domain {x,y,z} and range {1,2,3}. It is given that exactly one of the following statements is true and the remaining two are false. $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. Determine $f^{-1}(1)$

   f एकैकी फलन है। जिसका प्रान्त {x,y,z} व परिसर {1,2,3} है दिये गये कथनों में से मात्र एक सत्य है $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. $f^{-1}(1)$ ज्ञात कीजिए।

Ans. $f^{-1}(1) = y$

Sol. Case-I

S-I is true, S-II & S-III are false.

$f(x) = 1$

$f(y) = 1$

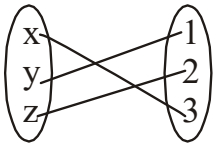
$f(z) = 2$

Not one-one.

Case-II

If S-II is true & S-I & S-III are false.

$\underbrace{f(x) \neq 1, f(y) \neq 1}_{\text{Not one-one}}$, $f(z) = 2$



Case-III

S-III is true, S-II, S-I false.

$f(x) \neq 1, f(y) = 1, f(z) = 2$

$f^{-1}(1) = y$

2. Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ and $P(6) = 6$ then find the value of $P(7)$.

   $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ बहुपद है जहाँ $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ व $P(6) = 6$ तो $P(7)$ का मान ज्ञात कीजिए।

Ans. 727

Sol. $P(x) = x \ \forall \ x = \{1,2,3,4,5,6\}$

$P(x) - x = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5)(x - 6)$

$P(7) - 7 = 720$

$P(7) = 727$

3. Let a and b be real numbers and let $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$. If $f\left(\log_{10}(\log_3 10)\right) = 5$ then find the value of $f\left(\log_{10}(\log_{10} 3)\right)$.

a व $b \in R$ तथा $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$ यदि $f\left(\log_{10}(\log_3 10)\right) = 5$ तो $f\left(\log_{10}(\log_{10} 3)\right)$ का मान ज्ञात कीजिए।

Ans. 3

Sol. $f(\log_{10}\log_3 10) = 5$

$$\Rightarrow f(-x) = -a\sin x - bx^{\frac{1}{3}} + 4$$

$$f(x) = a\sin x + bx^{\frac{1}{3}} + 4$$

$$\Rightarrow f(x) + f(-x) = 8$$

$$\Rightarrow f(\log_{10}\log_3 10) + f(-\log_{10}\log_3 10) = 8$$

$$\Rightarrow f(\log_{10}\log_{10} 3) = 3$$

4. Let f be a real valued invertible function such that $f\left(\frac{2x-3}{x-2}\right) = 5x - 2, x \neq 2$. Find $f^{-1}(13)$.

माना f कोई वास्तविक मान का व्युत्क्रमणीय फलन है जिसके लिये $f\left(\frac{2x-3}{x-2}\right) = 5x - 2, x \neq 2$ है। $f^{-1}(13)$ मान निकालिये।

Ans. 3

Sol. $f\left(\frac{2x-3}{x-2}\right) = 5x - 2$

$$f^{-1}(5x-2) = \frac{2x-3}{x-2}$$

Put $x = 3$

$$f^{-1}(13) = \frac{6-3}{3-2} = 3$$

5. Let [x] = the greatest integer less than or equal to x. If all the values of x such that the product $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_3, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$.

गुणनफल $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right]$ अभाज्य है। x का मान $[x_1, x_2) \cup [x_3, x_4)$ में है तो $x_1^2 + x_2^2 + x_3^2 + x_4^2$ का मान ज्ञात करें। [.] महत्तम पूर्णांक फलन है।

Ans. 11

Sol. $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right] =$ prime

Case-I $1 \times 2 = 2$

$\left[x+\frac{1}{2}\right]=2$ $\qquad$ $\left[x-\frac{1}{2}\right]=1$

$x+\frac{1}{2}\in[2,3)$ $\qquad$ $x-\frac{1}{2}\in[1,2)$

$x\in\left[\frac{3}{2},\frac{5}{2}\right)$ $\qquad$ $x\in\left[\frac{3}{2},\frac{5}{2}\right)$

Case-II $(-2) \times (-1) =$ prime

$\left[x+\frac{1}{2}\right]=-1$ $\qquad$ $\left[x-\frac{1}{2}\right]=-2$

$x+\frac{1}{2}\in[-1,0)$ $\qquad$ $x-\frac{1}{2}\in[-2,-1)$

$x\in\left[-\frac{3}{2},-\frac{1}{2}\right)$ $\qquad$ $x\in\left[-\frac{3}{2},-\frac{1}{2}\right)$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = \left(\frac{3}{2}\right)^2 + \left(\frac{5}{2}\right)^2 + \left(\frac{-3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 = \frac{44}{4} = 11$$

6. Suppose p(x) is a polynomial with integer coefficients. The remainder when p(x) is divided by x – 1 is 1 and the remainder when p(x) is divided by x – 4 is 10. If r (x) is the remainder when p(x) is divided by (x – 1) (x – 4), find the value of r (2006).

बहुपद p(x) के गुणांक पूर्णांक है। x – 1 से विभाजित करने पर शेषफल 1 व x – 4 से विभाजित करने पर शेषफल 10 है। (x – 1)(x – 4) से P(x) को विभाजित करने पर r(x) शेषफल है तो r (2006) ज्ञात कीजिए।

Ans. 6016

Sol. $p(x) = (x-1)\,Q_1(x) + 1 \Rightarrow P(1) = 1$

$p(x) = (x-4)\,Q_2(x) + 10 \Rightarrow P(4) = 10$

$P(x) = (x-1)(x-4)\,Q(x) + ax + b$

$P(1) = a + b = 1$

$P(4) = 4a + b = 10$

$a = 3,\ b = -2$

$r(x) = 3x - 2$

$r(2006) = 6016$

7. Prove that the function defined as, $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$ f(x) is odd as well as even. (where {x} denotes the fractional part function)

सिद्ध कीजिये कि $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$ जहाँ परिभाषित है, सम व विषम दोनों है।

Ans.

Sol. $f(x) = \begin{cases} e^{-\sqrt{|\ell n\{x\}|}} - \{x\}\sqrt{\frac{1}{|\ell n\{x\}|}} & \text{wherever it exists} \\ \{x\} & \text{otherwise} \end{cases}$

Let

$y = e^{-\sqrt{|\ell n\{x\}|}}$ $\qquad$ $z = \{x\}^{\sqrt{\frac{1}{|\ell n\{x\}|}}}$

$\ell ny = -\sqrt{|\ell n\{x\}|}$ $\qquad$ $\ell nz = \sqrt{\frac{1}{|\ell n\{x\}|}}\ell n\{x\}$

$$= -\sqrt{|\ell n\{x\}|}$$

$\Rightarrow y = z \forall x \in R - z$

$$f(x) = \begin{cases} 0 & x \in R - z \\ 0 & x \in z \end{cases}$$

f(x) is both even and odd.

8. In a function $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\left|\sqrt{2}\ \sin\left(\pi\left(x + \frac{1}{4}\right)\right)\right|\right) = 4\cos^2\frac{\pi x}{2} + x\cos\frac{\pi}{x}$

Prove that (i) $f(2) + f(1/2) = 1$ and (ii) $f(2) + f(1) = 0$

फलन $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\left|\sqrt{2}\ \sin\left(\pi\left(x + \frac{1}{4}\right)\right)\right|\right) = 4\cos^2\frac{\pi x}{2} + x\cos\frac{\pi}{x}$

के लिये सिद्ध कीजिए कि (i) $f(2) + f(1/2) = 1$ तथा (ii) $f(2) + f(1) = 0$

Ans.

Sol. put $x = 1$

$2f(1) + 1f(1) - 2f(|\sqrt{2}\ \sin\frac{5\pi}{4}|) = -1$

$\Rightarrow \quad 3f(1) - 2f(1) = -1 \quad \Rightarrow \quad f(1) = -1$

Now put $x = 2$

$$2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4\cos^2\pi + 2\cos\frac{\pi}{2}$$

$$\Rightarrow \quad 2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4$$

$$\Rightarrow \quad f(2) + f\left(\frac{1}{2}\right) = 1 \qquad .....(i)$$

Now put $x = 1/2$ we get

$$4f\left(\frac{1}{2}\right) + f(2) = 1 \quad .....(ii)$$

from (i) and (ii)

$$f\left(\frac{1}{2}\right) = 0 \text{ and } f(2) = 1$$

9. A function f, defined for all $x, y \in R$ is such that $f(1) = 2$; $f(2) = 8$ and $f(x+y) - kxy = f(x) + 2y^2$, where k is some constant. Find f(x) and show that $f(x+y)\, f\left(\frac{1}{x+y}\right) = k$ for $x + y \neq 0$.

फलन f, $x, y \in R$ इस प्रकार है कि $f(1) = 2$; $f(2) = 8$ तथा $f(x+y) - kxy = f(x) + 2y^2$, जहाँ k नियतांक है। f(x) ज्ञात करें और सिद्ध कीजिये कि : $f(x+y)\, f\left(\frac{1}{x+y}\right) = k$ for $x + y \neq 0$.

Ans. $f(x) = 2x^2$

Sol. $f(x + y) - kxy = f(x) + 2y^2$

$x = y = 1$

$8 - k = 2 + 2$

$k = 4$

$x = 0, y = 1$

$f(1) = f(0) + 2$

$f(0) = 0$

$x = 0, y = y$

$f(y) = 2y^2$

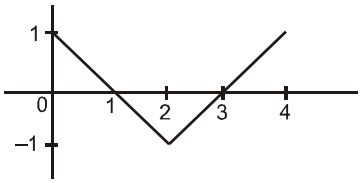
10. If $f(x) = -1 + |x-2|$, $0 \le x \le 4$
$g(x) = 2 - |x|$, $-1 \le x \le 3$
Then find fog(x) and gof(x). Draw rough sketch of the graphs of fog(x) and gof(x).

यदि $f(x) = -1 + |x-2|$, $0 \le x \le 4$
$g(x) = 2 - |x|$, $-1 \le x \le 3$
fog(x) व gof(x) ज्ञात कीजिए। ग्राफ भी खीचियें।

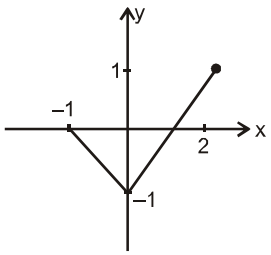
Ans. $fog(x) = \begin{matrix} -(1+x) & , & -1 \le x \le 0 \\ x-1 & , & 0 < x \le 2 \end{matrix}$; $gof(x) = \begin{matrix} x+1 & , & 0 \le x < 1 \\ 3-x & , & 1 \le x \le 2 \\ x-1 & , & 2 < x \le 3 \\ 5-x & , & 3 < x \le 4 \end{matrix}$;

$fof(x) = \begin{matrix} x & , & 0 \le x \le 1 \\ 4-x & , & 3 \le x \le 4 \end{matrix}$; $gog(x) = \begin{matrix} -x & , & -1 \le x \le 0 \\ x & , & 0 < x \le 2 \\ 4-x & , & 2 < x \le 3 \end{matrix}$

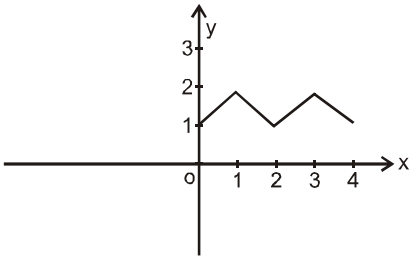
Sol. $f(x) = -1 + |x-2|$



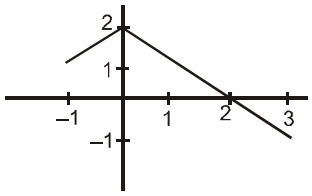
graph of fog(x) is



graph of gof(x) is



$g(x) = 2 - |x|$



$fog(x) = -1 + |x|$ $\quad -1 \le x \le 2$

$gof(x) = 2 - |-1 + |x-2|| \quad 0 \le x \le 4$

$fof(x) = -1 + |-1 + |x-2| - 2|$

$= -1 + | |x-2| - 3| \quad 0 \le x \le 1$, and $3 \le x \le 4$

$gog(x) = 2 - |2 - |x|| \quad x \in [-1, 3]$

11. If for all real values of u and v, $2f(u)\cos v = f(u+v) + f(u-v)$, prove that, for all real values of x.

(i) $f(x) + f(-x) = 2a\cos x$

(ii) $f(\pi - x) + f(-x) = 0$

(iii) $f(\pi - x) + f(x) = -2b\sin x$. Deduce that $f(x) = a\cos x - b\sin x$, a, b are arbitrary constants.

यदि u तथा v के लिए सभी वास्तविक मान, $2f(u)\cos v = f(u+v) + f(u-v)$, सिद्ध कीजिये कि x के वास्तविक मान होंगे।

(i) $f(x) + f(-x) = 2a\cos x$

(ii) $f(\pi - x) + f(-x) = 0$

(iii) $f(\pi - x) + f(x) = -2b\sin x$. दिखाइये कि $f(x) = a\cos x - b\sin x$, a, b नियतांक है।

Ans.

Sol. $2f(u)\cos v = f(u+v) + f(u-v)$

(i) put $u = 0$, $v = x$

$\Rightarrow f(x) + f(-x) = 2f(0)\cos x$

$f(0) =$ any constant a

$f(x) + f(-x) = 2a\cos x$ _____(1)

(ii) $v = \frac{\pi}{2}$, $u = \frac{\pi}{2} - x$

$$2f\left(\frac{\pi}{2}-x\right)(0)=f(\pi-x)+f(-x)$$

$$\Rightarrow f(\pi-x)+f(-x)=0 \qquad \text{_____(2)}$$

(iii) $v=\frac{\pi}{2}-x$, $u=\frac{\pi}{2}$

$$2f\left(\frac{\pi}{2}\right)\sin x=f(\pi-x)+f(x)$$

$$\Rightarrow f(\pi-x)+f(x)=-2b\sin x \text{ where } b=-f\left(\frac{\pi}{2}\right) \qquad \text{_____(3)}$$

(1) + (3) – (2)

$2f(x) = 2a\cos x - 2b\sin x$

$$\Rightarrow f(x)=a\cos x-b\sin x$$

12. A function f: $R \to R$ where R is the set of real numbers is defined by $f(x)=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$. Find the interval of values of α for which f is onto. Is the function one–to–one for $\alpha = 3$? Justify your answer.

फलन $f: R \to R$ जहाँ R सभी वास्तविक संख्याओं का समुच्चय है, $f(x)=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$. α के मानों का अन्तराल ज्ञात कीजिए, यदि f आच्छादक हो। यदि $\alpha = 3$, तो क्या फलन f एकैकी होगा।

Ans. [2, 14]; for $\alpha = 3$, f is not one–one

Sol. $y=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$

$$\Rightarrow y\alpha+6xy-8x^2y=\alpha x^2+6x-8$$

$$\Rightarrow (\alpha+8y)x^2+(6-6y)x-8-\alpha y=0$$

$x \in R$; $\quad D \geq 0$

$$(6(1-y))^2-4(\alpha+8y)(-8-\alpha y)\geq 0$$

$$4\left(9+9y^2-18y+8\alpha+\alpha^2y+64y+8\alpha y^2\right)\geq 0$$

$\Rightarrow y^2(9+8\alpha)+y(\alpha^2+46)+8\alpha+9\geq 0$

$\Rightarrow 9+8\alpha>0$ & $D\leq 0$

$\alpha>-\frac{9}{8}$ & $(\alpha^2+46)^2-4(8\alpha+9)^2<0$ _____(1)

$(\alpha^2+46+16\alpha+18)(\alpha^2+46-16\alpha-18)\leq 0$

$(\alpha^2+16\alpha+64)(\alpha^2-16\alpha+28)\leq 0$

$(\alpha+8)^2(\alpha-14)(\alpha-2)\leq 0$

$\alpha\in[2,14]\cup\{-8\}$ _____(2)

from (1) & (2)

$\alpha\in[2,14]$

for $\alpha=3$, $y=\frac{\text{Quad}}{\text{Quad}}$, many-one function.

13. The number of maps from the set {1, 2, 3} into the set {1, 2, 3, 4, 5} such that f(i) ≤ f(j) whenever i < j is _____

समुच्चय {1, 2, 3} से समुच्चय {1, 2, 3, 4, 5} में परिभाषित फलन f, इस प्रकार कि f (i) ≤ f(j) जब भी i < j की संख्या होगी _____

Ans. 35

Sol. Case-I When all the mappings are distinct

No. of functions $\Rightarrow {}^5C_3=10$

Case-II When two of the mappings are same

No. of functions $\Rightarrow {}^5C_2\times 2=20$

A factor of 2 has to be multiplied because after we have chosen two numbers, there can be two cases, aab or abb.

Case-III All mappings are same.

No. of functions = 5

Total functions = 35

14. Let the range of the function $f: \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ assumes exactly 3 distinct values. If the number of such function is N, then find the value of $\frac{N}{300}$.

माना फलन $f: \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ के परिसर में ठीक तीन भिन्न मान है। यदि ऐसे फलनों की संख्या N है, तो $\frac{N}{300}$ का मान ज्ञात करो।

Ans. 5

Sol. ${}^5C_3$ (selecting 3 distinct range values) $\times$ $(3^5 - {}^3C_2\, 2^5 + {}^3C_1)$ (No. of onto functions) $= 1500 = N$

$$\frac{N}{300} = \frac{1500}{300} = 5$$

Paragraph for Question Nos. 15 to 17

Given a function $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\pi\left(\frac{(1-2[x])}{2}\right) + \sin\left(\frac{x-[x]}{2}\right)$ whose fundamental period is p, (where {.} and [.] represent fractional part and greatest integral part functions respectively) and $y = \sqrt{2p + \frac{p}{2}[x] - [x]^2}$ the domain of y is [q, r), one another function $f_2(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$ then on the basis of above information answer the following.

मूलभूत आवर्तकाल p का एक फलन $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\pi\left(\frac{(1-2[x])}{2}\right) + \sin\left(\frac{x-[x]}{2}\right)$ दिया गया है। (जहाँ {.} और [.] क्रमशः भिन्नात्मक भाग फलन तथा महत्तम पूर्णांक फलन को बताते है।) तथा $y = \sqrt{2p + \frac{p}{2}[x] - [x]^2}$ है एवं y का प्रान्त [q, r), हैं।

एक दूसरा फलन $f_2(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$

हो, तो दी गई सूचना के आधार पर निम्न के उत्तर दीजिए।

15. The period p of $f_1(x)$ is :

(A) An irrational number
(B) Prime number
(C) A composite number
(D) Neither prime nor a composite number

$f_1(x)$ का आवर्तकाल p है :

(A) एक अपरिमेय संख्या
(B) अभाज्य संख्या
(C) एक संयुक्त संख्या
(D) न तो अभाज्य न ही संयुक्त संख्या

Ans. C

16. Value of $r - q - 1$ is equal to :

$r - q - 1$ का मान है :

(A) 6 (B) 7 (C) 8 (D) 9

Ans. A

17. Range of $f_2(f_2(x))$ in terms of p, q, r is :

p, q, r के पदों में $f_2(f_2(x))$ का परिसर है :

(A) $[p, \infty)$ (B) $[q, \infty)$ (C) $[r, \infty)$ (D) $(-\infty, -r) \cup (p, \infty)$

Ans. A

Sol. $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\left(\frac{\pi}{2} - \pi[x]\right) + \sin\left(\frac{\{x\}}{2}\right)$

$e^{\tan\left\{\frac{x}{4}\right\}} \to T = 4$

$\cos\left(\frac{\pi}{2} - \pi[x]\right) = \sin(\pi[x]) = 0$

$\sin\left(\frac{\{x\}}{2}\right) \to T = 1$

$P = 4$

$\therefore y = \sqrt{8 + 2[x] - [x]^2}$

$8 + 2[x] - [x]^2 \geq 0$

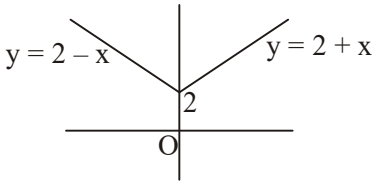
$([x] - 4)([x] + 2) \leq 0$

$[x] \in [-2, 4]$

$x \in [-2, 5)$

$q = -2, r = 5$

$\Rightarrow f_2(x) = \begin{cases} 2 + x; & x \geq 0 \\ 2 - x; & x < 0 \end{cases}$



$$f_2\left(f_2(x)\right)=\begin{cases}2+f_2(x); & f_2(x)\geq 0\\ 2-f_2(x); & f_2(x)<0\end{cases}$$

$$=\begin{cases}4-x; & x<0\\ 4+x; & x\geq 0\end{cases}$$

Range : $[4, \infty)$

Paragraph for Question Nos. 18 to 21

Let $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$. Let $f: (-\infty, a] \to [b, \infty)$, where 'a' is the largest real number for which $f(x)$ is bijective.

माना $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$ है। माना $f: (-\infty, a] \to [b, \infty)$, जहाँ 'a' अधिकतम वास्तविक संख्या जिसके लिए $f(x)$ एकैकी आच्छादक है।

18. The value of $(a + b)$ is equal to :

$(a + b)$ का मान होगा :

(A) -2 (B) -1 (C) 0 (D) 1

Ans. B

Sol. $f(x) = (x-1)^2 - 2 \qquad a = 1, b = -2$

$a + b = -1$ Ans.

19. Let $f: R \to R$, $g(x) = f(x) + 3x - 1$, then the least value of function $y = g(|x|)$ is :

माना $f: R \to R$, $g(x) = f(x) + 3x - 1$, तो फलन $y = g(|x|)$ का न्यूनतम मान होगा :

(A) $-9/4$ (B) $-5/4$ (C) -2 (D) -1

Ans. C

Sol. $g(x) = x^2 + x - 2 = \left(x + \frac{1}{2}\right)^2 - \frac{9}{4} \qquad \Rightarrow \qquad g|x| = \left(|x| + \frac{1}{2}\right)^2 - \frac{9}{4}$

$g_{min} = g(0) = -2$ Ans.

20. Let $f: [a, \infty) \to [b, \infty)$, then $f^{-1}(x)$ is given by :

माना $f: [a, \infty) \to [b, \infty)$, तो $f^{-1}(x)$ होगा :

(A) $1 + \sqrt{x+2}$ (B) $1 - \sqrt{x+3}$ (C) $1 - \sqrt{x+2}$ (D) $1 + \sqrt{x+3}$

Ans. A

Sol. $f: [1, \infty) \to [-2, \infty)$, then $f^{-1}: [-2, \infty) \to [1, \infty)$;

$f(x) = y \quad \Rightarrow \quad x^2 - 2x - (1 + y) = 0$

$x = \dfrac{2 \pm \sqrt{4 + 4(1+y)}}{2}$; $\quad x = 1 \pm \sqrt{2+y}$; $\quad f^{-1}(y) = 1 + \sqrt{2+y}$;

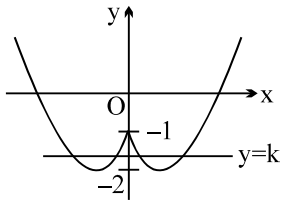
$f^{-1}(x) = 1 + \sqrt{2+x}$ Ans.

21. Let $f: R \to R$, then range of values of k for which equation $f(|x|) = k$ has 4 distinct real roots is :

माना $f: R \to R$, तो समीकरण $f(|x|) = k$ के 4 भिन्न हल होने के लिए k का परिसर होगा :

(A) $(-2, -1)$ (B) $(-2, 0)$ (C) $(-1, 0)$ (D) $(0, 1)$

Ans. A

Sol. 

; from graph $k \in (-2, -1)$ Ans.

MATCH THE COLUMN

22. Match the following columns :

Column – I	**Column – II**
(A) The period of the function $y = \sin(2\pi t + \pi/3) + 2\sin(3\pi t + \pi/4) + 3\sin 5\pi t$	(P) 1/2
(B) $y = \{\sin(\pi x)\}$ is a many one function for $x \in (0, a)$ where a may be	(Q) 8
(C) The fundamental period of the function $y = \dfrac{1}{2}\left(\dfrac{\lvert\sin(\pi/4)x\rvert}{\cos(\pi/4)x} + \dfrac{\sin(\pi/4)x}{\lvert\cos(\pi/4)x\rvert}\right)$	(R) 2
(D) If $f: [0, 2] \to [0, 2]$ is bijective function defined by $f(x) = ax^2 + bx + c$, where a, b, c are non-zero real numbers, then f(2) is equal to	(S) 0

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) निम्न फलन का आवर्तकाल $y = \sin(2\pi t + \pi/3) + 2\sin(3\pi t + \pi/4) + 3\sin 5\pi t$	(P) 1/2
(B) $x \in (0, a)$ के लिए $y = \{\sin(\pi x)\}$ एक बहुएकैकी फलन है, जहाँ a का मान हो सकता है	(Q) 8
(C) फलन $y = \dfrac{1}{2}\left(\dfrac{\lvert\sin(\pi/4)x\rvert}{\cos(\pi/4)x} + \dfrac{\sin(\pi/4)x}{\lvert\cos(\pi/4)x\rvert}\right)$ का मूलभूत आवर्तकाल	(R) 2
(D) यदि $f: [0, 2] \to [0, 2]$ एक एकैकी आच्छादक फलन हैं जो $f(x) = ax^2 + bx + c$ से परिभाषित है, जहाँ a, b, c अशून्य वास्तविक संख्याएँ हैं, तो $f(2) =$	(S) 0

Ans. (A) QR (B) QR (C) Q (D) S

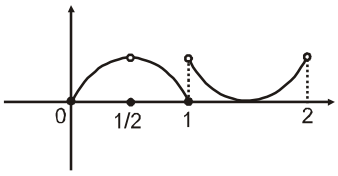
Sol. (A) $y = \sin\left(2\pi t + \frac{\pi}{3}\right) + 2\sin\left(3\pi t + \frac{\pi}{4}\right) + 3\sin(5\pi t)$

$$\text{LCM}\left(\frac{2\pi}{2\pi}, \frac{2\pi}{3\pi}, \frac{2\pi}{5\pi}\right) = \text{LCM} = \left(1, \frac{2}{3}, \frac{2}{5}\right) = 2$$

(B) $y = \{\sin \pi x\}$ period = 2

For $x \in \left(0, \frac{1}{2}\right)$ function is one-one

For $x \in (0, 2)$ and $x \in (0, 8)$
function is many-one



(C) $y = \frac{1}{2}\left(\frac{\left|\sin\frac{\pi}{4}x\right|}{\cos\frac{\pi}{4}x} + \frac{\sin\frac{\pi}{4}x}{\left|\cos\frac{\pi}{4}x\right|}\right) \Rightarrow$ F.T.P. of y is 8.

(D) Since f(x) is bijective,
$\therefore$ $f(0) = 0$ or 2 but $f(0) = 0 \Rightarrow c = 0$
(which is not true) $\therefore$ $f(0) = 2$ & $f(2) = 0$

23. Match the following columns :

Column – I	**Column – II**
(A) Domain of $f(x) = \sqrt{2^x - 3^x} + \log_{\frac{1}{2}} x$ is	(P) [0, 1]
(B) Solution set of equation $2\cos^2\frac{x}{2}\sin^2 x = x^2 + \frac{1}{x^2}$ is	(Q) $[0, \infty)$
(C) If $A = \{(x, y) ; y = \frac{1}{x}, x \in R_0\}$ and $B = \{(x, y); y = -x, x \in R\}$ then $A \cap B$ is	(R) $[1, \infty)$
(D) The functions $f(x) = \sqrt{x}\ \sqrt{x-1}$ and $g(x) = \sqrt{x^2 - x}$ are identical in	(S) ϕ

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) $f(x) = \sqrt{2^x - 3^x} + \log_{\frac{1}{2}} x$ का प्रान्त होगा	(P) [0, 1]

(B) समीकरण $2\cos^2\frac{x}{2}\sin^2 x = x^2 + \frac{1}{x^2}$ का हल होगा (Q) $[0, \infty)$

(C) यदि $A = \{(x, y);\ y = \frac{1}{x}, x \in R_0\}$ तथा $B = \{(x, y); y = -x, x \in R\}$ (R) $[1, \infty)$

तो $A \cap B$ होगा

(D) फलन $f(x) = \sqrt{x}\ \sqrt{x-1}$ तथा $g(x) = \sqrt{x^2 - x}$ किस अन्तराल में समरूप होंगे (S) ϕ

Ans. (A) S (B) S (C) S (D) R

Sol. (A) $2^x - 3^x \geq 0$

$2^x \geq 3^x$ $\qquad g(x) = \log_{\frac{1}{2}} x$

$x \in (-\infty, 0]$ $\qquad x > 0$

Domain of $f(x)$ is ϕ

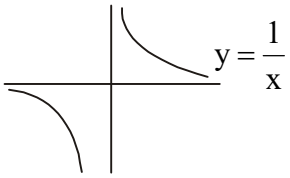
(B) $f(x) = (1 + \cos x)\sin^2 x = (1 + \cos x)(1 - \cos^2 x)$

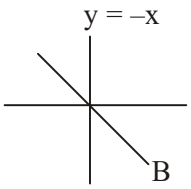
$= 1 + \cos x - \cos^2 x - \cos^3 x$

Put $\cos x = t; \Rightarrow 1 + t - t^2 - t^3$

for $t \in [0,1]$; $1 + t - t^2 - t^3 \in [0,1]$

$\therefore$ LHS $\in [0,1]$; RHS $[2, \infty]$

(C) 



$A \cap B = \phi$

(D) Domain of $f(x)$ & $g(x)$ is $[1, \infty)$

24. Match the following columns :

Column – I	**Column – II**
(A) The function $f(x) = \frac{3x-2}{x+4}$ has an inverse that can be written in the form $f^{-1}(x) = \frac{x+b}{cx+d}$. The value of $(b + c + d)$ is	(P) even

(B) Let $f(x)=\frac{x}{1+x}$ and let $g(x)=\frac{rx}{1-x}$. Let S be the set of all real numbers r such that $f(g(x))=g(f(x))$ for infinitely many real number x. The number of elements in set S is — (Q) odd

(C) If $f(x)=2x+1$ then the value of x satisfying the equation $f(x)+f(f(x))+f(f(f(x)))+f(f(f(f(x))))=116$, is — (R) prime

(S) neither prime nor composite

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) फलन $f(x)=\frac{3x-2}{x+4}$ का व्युत्क्रम $f^{-1}(x)=\frac{x+b}{cx+d}$ है, तो (b + c + d) का मान होगा	(P) सम
(B) माना $f(x)=\frac{x}{1+x}$ तथा let $g(x)=\frac{rx}{1-x}$. माना S सभी वास्तविक संख्याओं r का समुच्चय इस प्रकार है कि $f(g(x))=g(f(x))$ बहुत सारे अनन्त वास्तविक x के लिए है, तो समुच्चय S में अवयवों की संख्या होगीं	(Q) विषम
(C) यदि $f(x)=2x+1$ है, तो समीकरण $f(x)+f(f(x))+f(f(f(x)))+f(f(f(f(x))))=116$ को संतुष्ट करने वाले x का मान होगा।	(R) अभाज्य
	(S) ना अभाज्य और ना ही संयुक्त

Ans. (A) QS (B) P R (C) QR

Sol. (A) $y=\frac{3x-2}{x+4}$

$xy+4y=3x-2$

$x(y-3)=-4y-2$

$$x=\frac{-4y-2}{y-3}=\frac{y+\frac{1}{2}}{\frac{3}{4}-\frac{1}{4}y}$$

$$f^{-1}(x)=\frac{x+\frac{1}{2}}{\frac{3}{4}-\frac{1}{4}x}$$

$$b=\frac{1}{2},c=-\frac{1}{4},d=\frac{3}{4}$$

(B) $f(g(x)) = g(f(x))$

$$\frac{\frac{rx}{1-x}}{1+\frac{rx}{1-x}} = \frac{r\frac{x}{1+x}}{1-\frac{x}{1+x}}$$

$$\Rightarrow \frac{rx}{1-x+rx} = rx$$

$r = 0$ or 1

(C) $f(x) = 2x + 1$

$f(f(x)) = 2(2x + 1) + 1 = 4x + 3$

$f(f(f(x))) = 2(4x + 3) + 1 = 8x + 7$

$f(f(f(f(x)))) = 2(8x + 7) + 1 \Rightarrow 16x + 15$

$\Rightarrow (2x + 1) + (4x + 3) + (8x + 7) + (16x + 15) = 116$

$\Rightarrow 30x + 26 = 116$

$30x = 90$

$x = 3$